

Cofinality, Factorization and Scalar Extension for Entire Hahn Functions at Arbitrary Rank

A classification of the entire ring, its ideal theory,
and its behaviour under enlargement of the value group

Merged research report prepared for Vladimir Reshetnikov

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Abstract

Let Γ be a nonzero set-sized divisible ordered abelian group and $K_\Gamma = \mathbb{C}((t^\Gamma))$ the full Hahn field over the trivially valued complex numbers. We study ordinary power series $\sum a_n Z^n$ whose evaluation families are strongly Hahn-summable at every point of this one fixed field. An exact coefficient criterion holds: this happens precisely when $v(a_n)/n$ tends cofinally to $+\infty$ in Γ . Consequently nonpolynomial entire series exist exactly when $\text{cf}(\Gamma) = \aleph_0$. Support-controlled preparation, proved by well-founded recursion over a well-ordered monoid rather than by any contraction mapping, yields exact root counts on every valuation ball, shell and residue direction, a genus-zero canonical product for every ball-finite divisor with no exponential correction factors, and complete factorization by zeros at arbitrary rank.

Three sharp classifications follow. The first is ideal-theoretic. The entire ring $\mathcal{E}_\Gamma$ is always a GCD domain with units $K_\Gamma^\times$; it is the polynomial ring, hence a PID, when $\text{cf}(\Gamma) > \aleph_0$; it is a non-Noetherian Bézout domain with unrestricted Hermite interpolation when Γ has an order unit; and otherwise it is a non-Noetherian GCD domain that is *not* Bézout, witnessed by two explicit entire functions with disjoint simple zero sets generating a proper ideal. A separate order-unit criterion decides invertibility in the valuation-restricted algebra $\mathcal{T}_\Gamma$, and shows that the positive case of the trichotomy is sharp rather than merely sufficient.

The second classification is the anatomy of that third case. Grouping the nodes of a radially finite divisor into valuation shells of radii $\gamma_1 < \gamma_2 < \dots$ and normalizing each shell to valuation zero, a complete infinite Hermite prescription is realized by an entire function if and only if, on all sufficiently distant shells, the coefficients of that shell's finite Hermite interpolation polynomial lie in the valuation ring obtained by killing the Archimedean scale of the shell radius. The quotient of $\mathcal{E}_\Gamma$ by the divisor's canonical product is therefore an algebraic restricted product of finite shell algebras. This gives a necessary-and-sufficient Bézout criterion, a close-pair obstruction invisible to the sizes of the individual prescribed values, and explicit non-evaluation maximal ideals whose residue fields are ultraproducts of lower-rank Hahn fields. For one node per shell, with arbitrary multiplicities, these ideals and the evaluation ideals are all the maximal ideals containing the canonical product; the maximal spectrum of the quotient is the Stone–Čech space $\beta\mathbb{N}$ of the shell indices, and its boundary residue fields are algebraically closed. A unit-ideal criterion for finitely many generators and an exact separation threshold for paired zeros complete the ideal-theoretic picture. Apart from the statement that those residue fields are algebraically closed, these results use no algebraic closedness, and all of them hold with real Hahn coefficients, the boundary residue fields then being real closed. Beyond the maximal ideals, every prime ideal containing a product with one *simple* node per shell is given by a nonprincipal ultrafilter together with a convex subgroup of an ultraproduct of quotient value groups, or is an evaluation ideal. That quotient is zero-dimensional exactly when Γ has an order unit, and otherwise every nonprincipal fibre of its spectrum contains a strict chain of continuum many primes, although the quotient is

a reduced Bézout ring in which finite matrices have Smith forms. A cofinal enlargement of the value group leaves the product and its zeros unchanged and can nevertheless split one prime into continuum many; in the explicit surreal example even an old maximal ideal acquires continuum many nonmaximal extensions. The third classification is analytic. For every ordered-group extension $\Gamma \subseteq \Delta$, all nonpolynomial entire series over K_Γ share one exact strong evaluation domain in K_Δ : the valuation ring of v_Δ coarsened by the convex hull of Γ , whose residue field is $\mathbb{C}((t^H))$ rather than $\mathbb{C}$. Entireness survives exactly for cofinal extensions, no new zeros appear anywhere in the surviving domain, and on a noncofinal extension no alternative entire series agrees with the old one even on all ordinary complex constants. The scale boundary is intrinsic: the value group, the coefficient slopes and the negated zero valuations generate the same convex subgroup. In finitely many ordinary variables the uniform domain has an exact replacement, proved without divisibility and over any trivially valued coefficient field: the monomial family of an entire series is strongly summable at a point of the larger field exactly when the image of its ordinary support under the quotient weights of the point is well ordered. For the separate convention of homogeneous-block summation, the ordinary complex directions along which a new dominating scale can be crossed are exactly the proper countable unions of projective algebraic sets.

This report is the union of six manuscripts. Two were written independently and proved the same engine; a third, written against their merge, answered the question about the interpolation image that the merge had left open; a fourth, written in parallel against a snapshot in which that question was still open, proved the same image theorem independently by a different construction and added the ideal-theoretic refinements above; a fifth answered the scalar-extension clause of the several-variable question; a sixth described the prime ideals above a product with one simple node per shell. The arbitrary-rank classifications, the explicit non-Bézout construction, the exact shellwise image and its ideal theory, the universal extension domain and its several-variable replacement, and the analytic prime fibres with their splitting under cofinal extension are proposed original contributions, not a claimed solution of any named published conjecture. Classical Hahn-field inputs and rank-one antecedents are identified and credited, and so is the classical ultraproduct and valuation-domain mechanism behind the prime fibres. Complete proofs are supplied; they have not been independently refereed or verified in a proof assistant.

Contents

1	The question, the answer, and its scope	4
2	Hahn supports, the surreal interpretation, and the algebra chain	10
3	An exact coefficient criterion and the cofinality dichotomy	16
4	Restricted series: the sharp order-unit criterion	19
5	Support-controlled preparation	21
6	Exact root counting at every scale	22
7	Radial divisors, canonical products and factorization	25
8	The higher-rank obstruction and the exact image of the jet map	28
9	The positive case: an order unit restores interpolation	50
10	Change of workspace: the universal scalar-extension domain	53
11	Several ordinary variables: exact extension domains and exceptional directions	57
12	Prime spectra above a one-node-per-shell product	69
13	Assembly of the classifications	85
14	Explicit higher-rank and surreal examples	85
15	Relation to other reports in this collection	89
16	What the results do and do not establish	92
17	Conclusion	99
A	Proof-dependency and hypothesis audit	100
B	Reproducible finite checks	102
C	Source and novelty ledger	104
	References	108

1 The question, the answer, and its scope

1.1 Two questions that turn out to have one engine

Over an algebraically closed field, finite polynomials with disjoint zero sets generate the unit ideal. In familiar one-variable analytic theories, interpolation extends this fact to entire functions, and it is tempting to infer the same conclusion as soon as an infinite product theorem is available. The inference is false in the higher-rank setting studied here. We obtain a ring in which every admissible zero divisor is realized, every function factors completely according to its zeros, and greatest common divisors exist, yet two functions without a common zero need not satisfy a Bézout identity.

The second question is what the word *entire* names. A formal series with coefficients in one Hahn field may be evaluable at every point of that field and lose even strong summability at a single point of a larger one. Both questions are decided by the same object: the ordered hierarchy of scales above the coefficient slopes of the series. The first is decided by whether those scales have a greatest Archimedean class; the second by whether they remain cofinal after the enlargement.

The obstruction behind the first question is genuinely infinite. At increasingly large arguments, any one entire function is restricted by a common lower bound on the valuations of its coefficients. We place a second sequence of zeros so close to the first that the reciprocal values required by a Bézout identity escape those bounds. Every finite interpolation problem remains solvable by a polynomial. It is the simultaneous infinite problem that fails.

1.2 The analytic category

Throughout, except in [section 11](#), which states its own weaker hypotheses, Γ is a *set-sized divisible ordered abelian group*, written additively. Except where the degenerate case $\Gamma = 0$ is explicitly discussed, assume $\Gamma \neq 0$. Put

$$K_\Gamma = \mathbb{C}((t^\Gamma)), \quad v\left(\sum_{\eta} c_{\eta} t^{\eta}\right) = \min\{\eta : c_{\eta} \neq 0\}.$$

A Hahn series has well-ordered support. We put $v(0) = +\infty$. This is the *full* Hahn field: all well-ordered supports are admitted, not a support-restricted subfield. The complex coefficient field carries the *trivial* valuation, so every nonzero complex coefficient has valuation zero and an infinite ordinary complex sum at one exponent is not an allowed Hahn coefficient operation. The valuation ring and its maximal ideal are

$$\mathcal{O}_\Gamma = \{x \in K_\Gamma : v(x) \geq 0\}, \quad \mathfrak{m}_\Gamma = \{x \in K_\Gamma : v(x) > 0\},$$

with residue field exactly $\mathbb{C}$.

Definition 1.1. The ring of *strongly entire Hahn functions* is

$$\mathcal{E}_\Gamma = \left\{ F = \sum_{n \geq 0} a_n Z^n \in K_\Gamma[[Z]] : (a_n z^n)_{n \geq 0} \text{ is strongly Hahn-summable for every } z \in K_\Gamma \right\}.$$

Here strong summability means that the union of the term supports is well ordered and each exponent belongs to only finitely many term supports. That this set is a ring is proved below.

We use $\mathbb{N} = \{0, 1, 2, \dots\}$. The variable Z is an ordinary single indeterminate; its exponents are nonnegative *integers*. The Hahn exponents occur only in the coefficients, and the two infinities must not be confused: a finite index n in a power series is not a Hahn exponent, and integer multiples of a positive Hahn exponent need not be cofinal. Nothing is asserted about arbitrary generalized series in Z with ordinal support order types, about arbitrary local functions for the fine surreal topology, or about the Berarducci–Mantova derivation. All derivatives in this report hold K_Γ constant.

Convention 1.2 (Ring discipline). Every numbered statement below is asserted over one named coefficient ring, and no statement is transferred to another. The rings used here are

$$K_\Gamma[Z] \subseteq \mathcal{E}_\Gamma \subseteq \mathcal{T}_\Gamma \subseteq \mathcal{A}_\Gamma \subseteq K_\Gamma[[Z]],$$

defined in [sections 2](#) and [4](#), all of which have *Hahn series over $\mathbb{C}$* as coefficients. In particular no theorem here is a theorem about $\mathcal{O}(U)((t^\Gamma))$ or $\mathbb{C}\{z\}((t^\Gamma))$, whose coefficients are ordinary holomorphic functions on a common complex domain or convergent germs. That distinction is not a formality: the companion report *Analytic Geometry over Surcomplex Hahn Fields* adopts the same discipline by name for its own four coefficient rings, and records there an explicit element of the radius-free algebra lying in no common-domain algebra. We invoke that convention here so that no reader transports a statement of this report into those categories. See [section 15](#).

1.3 The two classifications

A subset of Γ is cofinal if every element of Γ is at most some member of that subset; $\text{cf}(\Gamma)$ is the least cardinality of a cofinal subset, and it is at least $\aleph_0$ for $\Gamma \neq 0$, since an ordered group has no greatest element.

Definition 1.3 (Order unit). An element $\delta > 0$ of Γ is an *order unit* if for every $\gamma \in \Gamma$ some positive integer m satisfies $m\delta > \gamma$; that is, $\{m\delta : m \geq 1\}$ is cofinal in Γ .

Remark 1.4 (One name for one invariant). This report uses exactly two order-theoretic invariants of Γ , and they must not be conflated.

- *Cofinality* always means $\text{cf}(\Gamma)$, the least cardinality of a cofinal subset of the underlying ordered set of the value group. It is never the cofinality of an ordinal index, of a diagram, or of the image of some displacement or spectral map; other reports in this collection use the word for such other invariants, and nothing here transfers to them.
- *Order unit* is the condition of [definition 1.3](#). The first two source manuscripts merged here named it twice: one called δ an order unit, the other called it a *cofinal-cyclic element*, meaning exactly that the cyclic subgroup it generates is cofinal. These are the same condition, stated twice; we keep the single name *order unit* and record the synonym here once. For a fixed $\delta > 0$, it is equivalent to each of the following: δ represents the greatest positive Archimedean class of Γ ; $m\delta \rightarrow +\infty$; the monomial t^δ is topologically nilpotent in the intrinsic valuation topology of K_Γ . Thus an order unit exists exactly when the positive Archimedean classes have a greatest member; a positive element in a smaller class is not an order unit.

An order unit forces $\text{cf}(\Gamma) = \aleph_0$, but not conversely: the group Γ_∞ of [section 14](#) has countable cofinality and no order unit. That gap is exactly the third case of the trichotomy below, and also the difference between the two existence questions of [theorems 1.5](#) and [4.1](#). The report `docs/surcomplex/expanding-polynomial-dynamics/` uses the same order unit and renamed its delivered phrase

“cofinal scale” accordingly (`epd:rem:one-name`). The report on transcendence over bounded support uses `cf(G)` in exactly this sense (`bst:subsec:terms`); its ordinal-index use of “cofinal” is confined to `bst:lem:coding`. The same cofinality dichotomy governs completion of the algebraic scalar extension in Hahn vector spaces (`docs/surcomplex/three-duals-of-hahn-vector-spaces/`, `duals:thm:cofinality-completion`, `duals:thm:basechange`).

Theorem 1.5 (Cofinality–rank trichotomy). *Let $\Gamma \neq 0$ be a divisible ordered abelian group. Then $\mathcal{E}_\Gamma$ is a GCD domain, its units are exactly $K_\Gamma^\times$, and exactly one of the following cases occurs.*

- (i) *If $\text{cf}(\Gamma) > \aleph_0$, then $\mathcal{E}_\Gamma = K_\Gamma[Z]$, a principal ideal domain.*
- (ii) *If Γ has an order unit, then $\mathcal{E}_\Gamma$ contains nonpolynomial functions and is a non-Noetherian Bézout domain. Arbitrary finite-order jets on every radially finite set can be interpolated by an entire function.*
- (iii) *If $\text{cf}(\Gamma) = \aleph_0$ and Γ has no order unit, then $\mathcal{E}_\Gamma$ contains nonpolynomial functions and is a non-Noetherian GCD domain that is not Bézout. More strongly, there are $f, g \in \mathcal{E}_\Gamma$ with disjoint simple zero sets and $(f, g) \neq \mathcal{E}_\Gamma$.*

For $\Gamma = 0$, the strong-summation convention gives $\mathcal{E}_0 = \mathbb{C}[Z]$.

A GCD domain is an integral domain in which every pair has a greatest common divisor, defined up to a unit. A Bézout domain is one in which every finitely generated ideal is principal. Case (iii) therefore separates two properties that coincide for finite polynomial algebra. This is not a distinction about algebraic closedness: every K_Γ here is algebraically closed.

The trichotomy says that case (iii) occurs. The next theorem is the anatomy of that third branch: it measures exactly how far interpolation fails, for every radially finite divisor and every group, and it specializes to the other two branches. It was proved after the first two source manuscripts, against the gap their merged report itself recorded as open ([question 16.1](#)), and is merged in here. A fourth manuscript, written in parallel against a snapshot in which the gap was still open, proved it independently by a different construction; its proof is kept as a marked second route ([section 8.5](#)).

Theorem 1.6 (Exact image of the infinite jet map). *Let D be an infinite radially finite divisor. Group its nodes into valuation shells of radii $0 < \gamma_1 < \gamma_2 < \dots$, cofinal in Γ , normalize shell n by $\tau_n = t^{-\gamma_n}$, and let R_n be the unique finite Hermite interpolation polynomial of that shell’s prescribed jets in the normalized coordinate. Let*

$$V_n = \{x \in K_\Gamma : v(x) \geq -m\gamma_n \text{ for some } m \in \mathbb{N}\}$$

be the valuation ring obtained by killing the Archimedean scale of γ_n . Then:

- (a) *an entire function realizing all the prescribed jets exists if and only if $R_n \in V_n[T]$ for all sufficiently large n , with no condition on finitely many exceptional shells and no uniform bound on the number of nodes per shell, on the jet orders, or on the admitted scale exponents;*
- (b) *consequently $\mathcal{E}_\Gamma/(P_D)$ is the algebraic restricted product $\Lambda_0 \times \{(r_n) : r_n \in \Lambda_n^\circ \text{ eventually}\}$ of the finite shell algebras $\Lambda_n = K_\Gamma[T]/(P_n)$ and $\Lambda_n^\circ = V_n[T]/(P_n)$;*
- (c) *a Bézout identity $AP_D + BF = 1$ with $F \in \mathcal{E}_\Gamma$ exists if and only if F is nonzero at every node of D and, on all sufficiently large shells, $v(F(\rho_{n,j})) \leq c_n\gamma_n$ for some finite integer c_n ;*

- (d) *for one simple node per shell, every nonprincipal ultrafilter on the positive integers yields an explicit maximal ideal of $\mathcal{E}_\Gamma$ that is not an evaluation ideal, with residue field the ultraproduct of the lower-rank Hahn fields $\mathbb{C}((t^{H_n}))$, H_n the convex subgroup generated by γ_n .*

These statements also hold over the real Hahn field $\mathbb{R}((t^\Gamma))$, with no use of algebraic closedness.

Part (a) is [theorem 8.6](#), (b) is [theorem 8.7](#), (c) is [theorem 8.26](#), (d) is [theorem 8.33](#), and the real form is [proposition 8.42](#); all are proved in [section 8](#). The restricted product here is an algebraic union, not a topological or adelic object; no norm, no local compactness and no completion is asserted. Case (ii) of [theorem 1.5](#) is the degenerate case $V_n = K_\Gamma$, and case (i) is the case where no infinite radially finite divisor exists at all.

The fourth source adds to parts (c) and (d). It decides when finitely many entire functions generate the unit ideal together with P_D ([theorem 8.27](#)), gives the exact threshold at which a pair of canonical products with paired simple zeros stops generating the unit ideal ([theorem 8.30](#)), and shows that for one node per shell, with arbitrary multiplicities, the ideals of part (d) and the evaluation ideals are *all* the maximal ideals containing P_D , forming a copy of the Stone–Čech space $\beta\mathbb{N}$ ([theorems 8.36](#) and [8.37](#)). None of these classifies the whole maximal spectrum of $\mathcal{E}_\Gamma$; see [question 16.4](#).

The sixth source goes below the maximal ideals, for one *simple* node per shell ([section 12](#)). Starting from the quotient of part (b), which it credits, it shows that every prime ideal of $\mathcal{E}_\Gamma$ containing such a product is an evaluation ideal or is given by a nonprincipal ultrafilter $\mathcal{U}$ and a convex subgroup of the ultraproduct $\prod_{\mathcal{U}}(\Gamma/H_n)$, the fibre over $\mathcal{U}$ being the spectrum of an explicit ultraproduct valuation domain ([theorem 12.9](#)). The quotient is zero-dimensional exactly when Γ has an order unit, and otherwise each nonprincipal fibre contains a chain of continuum many primes ([theorem 12.17](#)); it is nevertheless a reduced semihereditary Bézout ring with Smith forms ([theorems 12.21](#) and [12.23](#)). Under a cofinal enlargement $\Gamma \subseteq \Delta$ the contraction of these primes is intersection of convex subgroups, with all extensions of a prime described by an interval of convex subgroups ([theorem 12.29](#)); for Γ_∞ inside its rational-indexed enlargement one prime splits into continuum many while the product and its zeros do not change ([theorem 12.33](#) and [corollary 12.35](#)). Bounded multiplicities do not change the spectrum, and unbounded ones produce primes that the reduced description misses ([theorem 12.40](#)). This answers the “prime ideals” clause of [question 16.4](#) for one simple node per shell. The ultraproduct and valuation-domain mechanism is classical [[15](#), [19](#)]; what is proposed is its exact analytic realization, the dichotomy and the splitting construction.

Theorem 1.7 (Universal scalar-extension package). *Let $\Gamma \subseteq \Delta$ be nonzero divisible ordered abelian groups and let $H = \text{conv}_\Delta(\Gamma)$ be the convex hull of Γ in Δ . Write $\mathcal{D}_{\Gamma,\Delta}$ for the valuation ring of v_Δ coarsened by H . Then for every nonpolynomial $f \in \mathcal{E}_\Gamma$:*

- (a) *the family $(a_n z^n)_n$ is strongly summable in K_Δ exactly for $z \in \mathcal{D}_{\Gamma,\Delta}$, so the domain does not depend on f , and $f(\mathcal{D}_{\Gamma,\Delta}) \subseteq \mathcal{D}_{\Gamma,\Delta}$;*
- (b) *$\mathcal{D}_{\Gamma,\Delta} = K_\Delta$, equivalently $f \in \mathcal{E}_\Delta$, exactly when Γ is cofinal in Δ ;*
- (c) *every zero of the extended function on $\mathcal{D}_{\Gamma,\Delta}$ already lies in K_Γ with unchanged multiplicity, even when f has infinitely many zeros;*
- (d) *if Γ is not cofinal in Δ , there is no $g \in \mathcal{E}_\Delta$ agreeing with f on $\mathbb{C}$, let alone on K_Γ ;*
- (e) *Γ , the slope set $\{v(a_n)/n : n \geq 1, a_n \neq 0\}$ and the negated zero valuations $\{-v(r) : r \neq 0, f(r) = 0\}$ generate the same convex subgroup of any larger ordered group, so $\mathcal{D}_{\Gamma,\Delta}$ is recoverable from the coefficients alone or from the zeros alone.*

The residue field of $\mathcal{D}_{\Gamma,\Delta}$ is canonically $\mathbb{C}((t^H))$, not $\mathbb{C}$.

Remark 1.8 (The two extension statements are one statement). Part (b) is often phrased negatively — in the noncofinal case the series is simply *not entire over* K_Δ , and indeed $\mathcal{E}_\Delta \cap K_\Gamma[[Z]] = K_\Gamma[Z]$ there ([theorem 10.1](#)) — while parts (a) and (c) say that the very same series remains strongly summable on all of $\mathcal{D}_{\Gamma,\Delta}$ and keeps every one of its zeros there. Read side by side these look like a disagreement about whether the function survives an enlargement. They are not in conflict: they are the coarse and the exact form of one fact. “Not entire over K_Δ ” means precisely that the strong evaluation domain is a *proper* subring of K_Δ , and [theorem 10.4](#) computes exactly which subring it is, namely $\mathcal{D}_{\Gamma,\Delta}$, on which the function is unchanged. The negative statement locates the boundary; the positive statement describes everything inside it. Throughout this report “entire over K ” is reserved for strong summability at *every* point of K , and any weaker survival is described by naming its domain.

The proof of [theorem 1.5](#) is distributed through the article. The coefficient criterion and cofinality dichotomy are in [section 3](#); the order-unit criterion for the restricted algebra, which is what makes case (ii) sharp rather than merely sufficient, is in [section 4](#); preparation and root counting are in [sections 5](#) and [6](#); factorization and GCDs are in [section 7](#); the negative and positive Bézout results are in [sections 8](#) and [9](#). [Section 13](#) assembles them. [Theorem 1.7](#) is proved in [section 10](#). [Theorem 1.6](#) is proved in [section 8](#), whose shell notation is fixed in [sections 2.5](#) and [8.2](#).

Several ordinary variables. Part (a) of [theorem 1.7](#) has no uniform analogue in several variables: a series independent of one coordinate imposes no constraint in that coordinate. [Section 11](#), the fifth source’s contribution, gives the exact replacement. For a whole-family entire series $F = \sum_\alpha a_\alpha X^\alpha$ in finitely many variables and a point z of the larger field with nonzero coordinates, the family $(a_\alpha z^\alpha)_\alpha$ is strongly summable if and only if

$$\left\{ \sum_j \alpha_j (v(z_j) + H) : a_\alpha \neq 0 \right\} \text{ is well ordered in } \Delta/H$$

([theorem 11.8](#)). That section works under weaker hypotheses than the rest of the report — no divisibility, any trivially valued coefficient field — and keeps two summation conventions apart; only the whole-family one is called entire. It answers the “In particular” clause of [question 16.5](#) and no other part of that question.

1.4 Relationship to the repository and the literature

A note on how this report grew. Its first two source manuscripts were independently written against pinned snapshots of this repository, proved the same engine, and carried complementary payloads; their merge is described in [section C](#). A third manuscript was written later, against the merged report itself at a pinned snapshot, and answers the question that the merged report had recorded as open in [question 16.1](#). Its contribution is [theorem 1.6](#) and the material of [sections 8.2](#) to [8.4](#), [8.7](#) and [8.8](#); it credits, rather than claims, the coefficient criterion ([theorem 3.1](#)), the canonical product ([theorem 7.3](#)), the order-unit interpolation theorem ([theorem 9.2](#)) and the explicit non-Bézout pair ([theorem 8.29](#)) as already present here. Where it re-proved one of those for dependency control, both proofs are printed and marked; see [proposition 8.8](#) and the two proofs of [theorems 9.2](#) and [9.3](#). Its symbols ρ_n and $a_{n,j}$ were the exact inverses of this report’s, and the single renormalizing pass that fixed that is recorded in [section 2.5](#).

Three further manuscripts were merged later. The *fourth* was written against a snapshot of the merged report taken after the third source’s pin but before the third source was merged in,

and it too set out to identify the missing image of the infinite jet map. It did so independently and at the same generality, with a different sufficiency construction, so the image, quotient and Bézout theorems of [section 8](#) now have two independent sources; they are printed once, and its construction is kept as a marked second route ([section 8.5](#)). Its framing as *the* resolution of that gap is accurate for its snapshot and stale for this tree, where the third source answered the question first; the ledger in [section C](#) records this. Its genuinely additional results are the explicit truncation bound of the second route, the finite-generator and membership criteria, the exact threshold for paired zeros, and the classification of all maximal ideals above a one-node-per-shell product with their topology, residue-field type and boundary coordinate ([section 8.9](#)). Its letters collide with this report's more seriously than the third source's did — two of them are swapped with each other — and [section 2.5](#) records the translation. The *fifth* was written against a later snapshot, after the third source had been merged, and answers the scalar-extension clause of [question 16.5](#); it is [section 11](#), which absorbs the finite-variable criteria that earlier versions kept in an appendix. The *sixth* was written against a still later snapshot, after the fifth source had been merged; this report's directory is identical at that pin and at the base of the sixth merge. It takes [theorem 8.7](#) and [corollary 8.13](#) and the maximal-ideal results of the fourth source as its starting point, credits them, and describes all prime ideals above a product with one simple node per shell, their behaviour under cofinal enlargement of the value group, and the multiplicity threshold; it is [section 12](#). It reproves the simple-node interpolation theorem for dependency control, by the argument of the first route ([remark 12.2](#)), so that proof is printed once. Its sequence ring, convex subgroups and several other letters collide with this report's and are renamed ([section 2.5](#)).

The repository supplied with the question already contains a rank-one report over $\mathbb{C}((t^{\mathbb{R}}))$, including preparation, zero counting, zero-free rigidity and a partial-theta canonical product [\[25\]](#). None of the source manuscripts claims to originate those points, and neither do we; the present report is their arbitrary-rank generalization, and the rank-one report itself proves the obstruction that makes rank one non-generic. The repository also contains a global-divisor report in a different ring, $\mathcal{O}(U)((t^{\Gamma}))$, with ordinary holomorphic coefficient functions on a common complex domain [\[26\]](#); its global support obstructions belong to a different analytic category and our argument is not a restatement of its Picard-group or moving-divisor results. [Section 15](#) states the cross-references in full.

Classical non-Archimedean factorization is an essential antecedent. Lazard's work is a foundational reference for one-variable zeros [\[5\]](#); Cherry records the one-variable product factorization and develops GCD and factorization results under a complete real-valued non-Archimedean absolute value [\[6, 7\]](#), and himself calls those factorization results well known. Hahn-field algebraic closedness is standard, used here in the form of Poonen's Corollary 4 [\[4\]](#); the support calculus is Neumann's [\[3\]](#); support-based operator methods are developed extensively by van der Hoeven [\[9\]](#). The distinction between positive valuation and topological nilpotence in rank two is established, and we follow Conrad's discussion of it [\[8\]](#). We claim none of these facts as new.

Novelty statement. The proposed contributions are the arbitrary-rank coefficient, unit and workspace criteria, the explicit non-Bézout construction, the exact shellwise interpolation image with its restricted-product quotient, sharp Bézout criterion and explicit residue fields, the universal scalar-extension domain with zero conservation, and their combination into [theorems 1.5 to 1.7](#). The literature comparison for the image theorem was targeted in the same bounded sense as the others and located no matching statement of the shellwise restricted-product theorem; that is a reason to present it as proposed new work, not a proof that no such statement occurs anywhere. The ultrafilter construction of [theorem 8.33](#) is standard algebra; what is proposed there is the

analytic surjection onto the restricted product and the consequent identification of the residue fields. The support-controlled preparation and product arguments provide a self-contained route to those results; their rank-one specializations are not new. The literature checks performed for the first two source manuscripts did not locate the stated arbitrary-rank classifications or the explicit obstruction in the primary sources consulted. This is a bounded literature statement, not a certification of priority. The repository review was targeted at the catalogue, relevant READMEs and stated scope, not a line-by-line audit of every manuscript. No named published open problem is claimed solved.

The fourth source proposes as new the exact block restricted-product quotient, its cardinal construction with cofinal truncation bounds independent of block dimension and multiplicity, the unit-ideal criterion and paired-root threshold, and the scale-dependent boundary residue fields with the full ultrafilter locus of the paired example. The first and third of these were already in this tree from the third source when it was merged, so here they count as an independent second derivation. It credits as antecedents Henriksen’s free maximal ideals and transcendental boundary coordinate for the classical entire ring [13], Bruno’s non-Archimedean maximal-ideal classifications under a real value group [16], Helmer’s divisibility theory [12], and Cherry; it does not present the existence of ultrafilter maximal ideals, or of a transcendental boundary coordinate, as new. The fifth source proposes [theorem 11.8](#) with its repeated-weight proof, and the combination of the extension geometry with the exceptional-direction classification and realization. It credits the Ma–Neelon realization of countable algebraic unions as convergence sets of divergent power series [17], and does not reclaim the finite-variable growth criterion, the one-variable extension domain or the whole-class rigidity. The sixth source proposes the description of all prime fibres above a simple one-node-per-shell product, the order-unit dichotomy for that reduced divisor and the cofinal splitting construction, stating that these were not found in the sources it compared and that this is not proof of their absence. It credits the ultraproduct and valuation-domain mechanism for primes of product rings to Finocchiaro, Frisch and Windisch [15] and the valuation-domain prime correspondence to the standard literature [19], and it does not reclaim the interpolation quotient, the maximal ideals or the maximal spectrum. [Sections C](#) and [16](#) state the limits in full; every limitation recorded by any of the six source manuscripts is preserved there.

2 Hahn supports, the surreal interpretation, and the algebra chain

2.1 Foundational setting

The proofs can be carried out in ZFC using only the set-sized field K_Γ . To interpret them inside the surcomplex numbers, take Γ to be a divisible ordered additive subgroup of $\mathbf{No}$ and send

$$\sum_{\gamma} c_{\gamma} t^{\gamma} \mapsto \sum_{\gamma} c_{\gamma} \omega^{-\gamma}. \quad (1)$$

The convention $t^{\gamma} := \omega^{-\gamma}$ is Conway’s monomial map; it is explicitly *not* a surreal exponential. The real and imaginary parts of the image are Conway normal forms, and the standard normal-form arithmetic makes (1) an embedding of fields: the real coefficient subfield $\mathbb{R}((t^{\Gamma}))$ lies in $\mathbf{No}$, and its coefficientwise complexification lies in $\mathbf{No}[i]$ [1, 2]. Conjugation acts on coefficients only. Our explicit examples use displayed subgroups of $\mathbf{No}$, so they do not require an unspecified universal embedding theorem. An element $x \in \mathcal{O}_{\Gamma}$ has a unique expression $x = c + \varepsilon$ with $c \in \mathbb{C}$ and $\varepsilon \in \mathfrak{m}_{\Gamma}$; the residue map $\text{res}(x) = c$ is defined on $\mathcal{O}_{\Gamma}$, not on all of K_{Γ} .

2.2 The support facts actually used

We use the following classical Hahn–Neumann support facts [3][4, Section 3][9]. They are stated explicitly because every infinite construction below depends on them, and they are imported, not proved here.

Lemma 2.1 (Hahn–Neumann support calculus). *Let S, T be well-ordered subsets of an ordered abelian group.*

- (a) *$S \cup T$ and $S + T$ are well ordered. For every η , there are only finitely many pairs $(s, t) \in S \times T$ with $s + t = \eta$.*
- (b) *If $S \subseteq \Gamma_{>0}$, its additive monoid $S^* = \{s_1 + \cdots + s_r : r \geq 0, s_i \in S\}$ is well ordered. For each η , there are only finitely many finite ordered words in S whose sum is η .*
- (c) *Consequently, if u is a Hahn series over a commutative coefficient ring and $v(u) > 0$, the geometric series $\sum_{r \geq 0} u^r$ is strongly summable and is the inverse of $1 - u$.*

Part (b) is a support theorem, not an assertion that $nv(u)$ tends cofinally to infinity. The latter can fail. In $\mathbb{Q}e_1 \oplus \mathbb{Q}e_2$ with $e_2 > ne_1$ for every n , the Hahn identity $\sum_{n \geq 0} t^{ne_1} = (1 - t^{e_1})^{-1}$ is valid but its partial sums do not converge in the intrinsic valuation topology: no tail has valuation greater than e_2 . [Example 4.3](#) turns this observation into a theorem about units.

Lemma 2.2 (Cofinal tails). *Suppose $u_n \in K_\Gamma$ and $v(u_n) \rightarrow +\infty$ cofinally in Γ ; that is, for every $B \in \Gamma$, eventually $v(u_n) > B$. Then (u_n) is strongly summable. Its sum has valuation at least $\min_n v(u_n)$ when at least one term is nonzero.*

Proof. For a fixed B , only finitely many nonzero term supports meet $(-\infty, B]$. A descending sequence in their union, beginning at B , would therefore lie in a finite union of well-ordered supports, which is impossible. Likewise, any exponent occurs in only finitely many term supports. The valuation estimate follows from coefficientwise addition. A nonempty set of term valuations has a minimum because it is a subset of the well-ordered support union. $\square$

We use the usual ZFC equivalence between well-ordering of a linear order and the absence of an infinite strictly descending sequence. All supports and index sets are sets; no proper-class support or proper-class summation is involved.

2.3 Three meanings of an infinite expression

It is essential to keep separate

- (1) a strong Hahn sum, which requires only a well-ordered support union and pointwise finiteness;
- (2) convergence in the intrinsic valuation topology of the fixed field K_Γ , which requires cofinal progress through every threshold of Γ ;
- (3) convergence in the topology inherited from all surreal scales.

These need not agree. [Definition 1.1](#) starts with (1). The *all-point* condition will force (2) for evaluation series ([theorem 3.1](#)), even though (1) alone does not, and on a single disk the two genuinely differ. We make no assertion of convergence in (3), and no density of $\mathbb{C}$ in any fine surreal topology is asserted or used. The repository’s foundational reports emphasize this distinction [\[24, 25\]](#).

2.4 The chain of coefficient algebras

Four algebras of series in Z over K_Γ are kept distinct throughout. Define the *polynomial-coefficient Hahn algebra*

$$\mathcal{A}_\Gamma = \mathbb{C}[X]((t^\Gamma)), \quad (2)$$

whose elements are $A = \sum_{\eta \in S} P_\eta(X)t^\eta$ with S well ordered and every P_η a finite polynomial, with no common bound on $\deg P_\eta$; and the *valuation-restricted algebra*

$$\mathcal{T}_\Gamma = \left\{ \sum_{n \geq 0} a_n Z^n \in K_\Gamma[[Z]] : v(a_n) \rightarrow +\infty \text{ cofinally in } \Gamma \right\}. \quad (3)$$

Taking the coefficient of each X^n gives an injective ring map $\mathcal{A}_\Gamma \hookrightarrow K_\Gamma[[X]]$, and under it $\mathcal{A}_\Gamma$ is exactly the set of series whose coefficient family $(a_n)_n$ is itself strongly summable: finite degree at each Hahn exponent is precisely the pointwise-finiteness half of strong summability. With X and Z identified under that map,

$$K_\Gamma[Z] \subseteq \mathcal{E}_\Gamma \subseteq \mathcal{T}_\Gamma \subseteq \mathcal{A}_\Gamma \subseteq K_\Gamma[[Z]]. \quad (4)$$

The first two inclusions need not be strict; the last two record genuinely different summation requirements. The first two source manuscripts named the ring (2) differently — one as the auxiliary algebra $\mathcal{A}_\Gamma$ in the variable X , the other as the strong disk algebra $\mathcal{H}_\Gamma$ in the variable Z — but it is one ring, and we use the single symbol $\mathcal{A}_\Gamma$. That the inclusions in (4) hold, and that $\mathcal{T}_\Gamma$ and $\mathcal{A}_\Gamma$ are rings, follows from [lemma 2.1](#) together with [proposition 3.5](#) below.

For $A \neq 0$ in $\mathcal{A}_\Gamma$, set $w(A) = \min\{\eta : P_\eta \neq 0\}$; the polynomial $P_{w(A)}$ is its *initial polynomial*. Since $\mathbb{C}[X]$ is a domain,

$$w(AB) = w(A) + w(B), \quad (5)$$

and initial polynomials multiply. Equivalently, writing $\text{in}_0(A) = \overline{t^{-w(A)}A} \in \mathbb{C}[X] \setminus \{0\}$ for the reduction, $\text{in}_0(AB) = \text{in}_0(A)\text{in}_0(B)$. This Gauss multiplicativity is the engine of [section 4](#).

2.5 Notation: radii live in Γ , nodes live in K_Γ

Six manuscripts are merged here, and four of them arrived with letters that collide with this report's. The third arrived with the opposite convention for the two letters that matter most. Its ρ_n was a positive element of the *value group* and its $a_{n,j}$ was a point of the *field*. This report keeps the convention of the first two throughout, in which ρ always names a point of K_Γ and γ always names an element of Γ . Every statement taken from the third, fourth, fifth or sixth manuscript has been rewritten in the convention below; the tables are recorded so that a reader comparing the texts is not misled by a symbol that typesets identically and means something else.

Symbol here	Meaning	In the third manuscript	Note
γ_n	shell radius, an element of Γ ; $0 < \gamma_1 < \gamma_2 < \dots$, cofinal	ρ_n	<i>inverted</i> : its ρ_n is our γ_n
$\rho_{n,j}$	a node, a point of K_Γ with $v(\rho_{n,j}) = -\gamma_n$; written ρ_n when the shell carries one node	$a_{n,j}$	<i>inverted</i> : its $a_{n,j}$ is our $\rho_{n,j}$
$\tau_n = t^{-\gamma_n}$	the shell normalizer, $v(\tau_n) = -\gamma_n$	s_n	
λ_n	the valuation $v(g(\rho_n))$ of (49)	—	its λ_n is a shell <i>parameter</i> , here always written γ_{n+1}
$m_{n,j}, M_n$	node multiplicity and shell total $\sum_j m_{n,j}$	$r_{n,j}, M_n$	r is reserved here for roots and for the jet polynomials of (57)
$m_\gamma(F)$	weighted Gauss valuation (14), $\min_n \{v(a_n) + n\gamma\}$	$w_\beta(F) = \min_k \{v(f_k) - k\beta\}$	<i>sign-inverted</i> : $w_\beta = m_{-\gamma}$ with $\gamma = -\beta$

The fourth manuscript, like the third, writes $a_{n,j}$ for the nodes, and it writes λ_n for the radii. More seriously, two of its letters are *swapped* with two of ours: its C_n is our convex subgroup H_n and its H_n is a correction, our C_n . Several other letters shift. Every statement taken from it has been rewritten as follows, and “shell” is used for its “block”.

Symbol here	Meaning	In the fourth manuscript	Note
H_n	convex subgroup generated by γ_n	C_n	<i>swapped</i> with the next row
$C_{n,d}, \widehat{C}_{n,d}$	sequential correction (32); cardinal correction of section 8.5	—, $H_{n,N}$	<i>swapped</i> : its H_n is our cardinal $\widehat{C}_n$
γ_n	shell radius	λ_n	our λ_n is the valuation (49)
$\rho_{n,j}, \xi_{n,j}, \tau_n$	node, normalized node, normalizer	$a_{n,j}, u_{n,j}, s_n$	our $\xi_{n,j}$ never denotes a boundary coordinate
q_n, M_n	number of nodes and total degree of shell n	r_n, d_n	its M_n is our c_n
c_n	integer with $v \geq -c_n \gamma_n$, as in (25)	M_n	its “scale-moderate” means eventually in V_n
d	damping or suppression exponent	N	
$P_n, \Pi_n, P_D, \widehat{P}_n$	monic shell polynomial, shell factor, canonical product, cofactor P_D/Π_n	W_n, p_n, P, Q_n	our Q_n is the product (29) over already treated nodes
$\Lambda_n, \Lambda_n^\circ, \mathcal{R}_D$	shell algebras, restricted product	$A_n, B_n, \prod'_n(A_n, B_n)$	our $\mathcal{R}_D$ carries the finite factor Λ_0
$R_n, \text{rem}_n, \text{ev}_D$	normalized shell Hermite polynomial, shell class, jet map	$J_n, \mathcal{J}_n, \mathcal{J}_D$	our $J_{n,j}$ are the prescribed jets
$\nu, \varpi_n, \varkappa_n, \Xi_{n,d}$	least coefficient valuation, cofactor class, inverse-shell gain, cardinal remainder	$\nu_n, q_n, \beta_n, R_{n,N}$	our q_n counts nodes; our β_i are reciprocal node valuations
$m_\gamma(F)$	weighted Gauss valuation	$w_r(F)$	<i>sign-inverted</i> : $w_r = m_{-r}$; its $r \leq \lambda_n/2$ is our $\gamma \geq -\gamma_n/2$
ϑ_n	displacement $\sigma_n - \rho_n = t^{\vartheta_n}$	θ_n	our θ_n is a real radius in the second proof of theorem 9.2
$\mathfrak{q}_{\gamma_n}, \kappa_n$	coarsened maximal ideal and residue field	$\mathfrak{p}_n, \kappa_n$	
$\Phi_{\mathcal{U}}, \iota_{\mathcal{U}}, L_{\mathcal{U}}$	ultrafilter character, its restriction to K_Γ , its target field	$\chi_{\mathcal{U}}, \mathcal{J}_{\mathcal{U}}, \prod_{\mathcal{U}} \kappa_n$	its boundary coordinate $\xi_{\mathcal{U}}$ is our $\Phi_{\mathcal{U}}(Z)$
nonprincipal; non-evaluation maximal ideal		free; boundary ideal	not the “free ideals” or disk boundaries of the analytic-geometry report
$\Gamma_\infty = \bigoplus_{j \geq 0} \mathbb{Q}e_j$	H_n has rank $n+1$	$\Gamma_* = \bigoplus_{j \geq 1} \mathbb{Q}e_j$	index shift: its group omits e_0 , and its C_n has rank n

The fifth manuscript’s letters are used only in [section 11](#). Its H_n , a homogeneous block, would have been a third meaning of that letter, after our convex subgroup and the fourth source’s correction; it is renamed.

Symbol here	Meaning	In the fifth manuscript	Note
$F_{(n)}$	homogeneous component of degree n	H_n	H_n is the convex subgroup here
$\mathbf{k}$	trivially valued coefficient field	k	our k is an index
$\Sigma(F), \Sigma_J$	ordinary monomial support, its faces	$E(F), E_J$	our $\mathcal{E}_\Gamma$ is the entire ring
$\Delta/H, \pi_H, \bar{\delta}$	ordered quotient, quotient map, quotient weights $\pi_H(v(z_j))$	Q, π, q	our Q_n , the π of lemma 9.1 , and our q_n are unrelated
$\langle \Sigma, \bar{\delta} \rangle, \text{Adm}_\Sigma(G)$	weight set, admissible weight set	$W_E(q), \mathcal{A}_E(Q)$	our $\mathcal{A}_\Gamma$ is the Hahn algebra
$\mathcal{D}_{\Gamma, \Delta}$	coarsened valuation ring	$\mathcal{O}_H$	the same object
e_*	a new scale above H	τ	our τ_n are normalizers
$\text{Pdir}(F)$	polynomial-direction locus	$\text{Pol}(F)$	avoids the nonabelian-support report's Pol
$\mathcal{Z}_N, Y_m, \mathcal{L}, \varphi_n$	vanishing loci, prescribed algebraic sets, relation lattice, damping sequence	Z_N, V_m, L, β_n	our Z is the variable, V_n the coarsened rings
$\mathcal{D}^{\text{hom}}(F), \omega^x$	homogeneous-block domain, Conway omega-map	$\text{Dom}_{\text{hom}}(F), \Omega(x)$	

The sixth manuscript's letters are used only in [section 12](#). Its nodes, radii, $H_n, V_n, \mathfrak{q}_n$ (our $\mathfrak{q}_{\gamma_n}$), ultrafilters $\mathcal{U}$, residue fields $L_{\mathcal{U}}$, order units and cofinality agree with ours; the following are renamed. The word *spectrum* means prime spectrum in that section, and *free* ultrafilters and fibres are called nonprincipal, as elsewhere here.

Symbol here	Meaning	In the sixth manuscript	Note
$\mathcal{R}_D, \mathcal{R}_D^\Gamma$	sequence ring $\cong \mathcal{E}_\Gamma/(P_D)$	$\mathcal{A}, \mathcal{A}_\Gamma$	our $\mathcal{A}_\Gamma$ is the Hahn algebra
P_D, P_{D_m}	simple and multiple one-node products	P, Q	our P in (85) runs over $j \geq 0$; our Q_n is (29)
$\mathbf{k}$	$\mathbb{R}$ or $\mathbb{C}$	k	our k is an index
$\mathfrak{e}_n$	evaluation prime of $\mathcal{R}_D$	$\mathfrak{p}_n$	the fourth source's $\mathfrak{p}_n$ is our $\mathfrak{q}_{\gamma_n}$
$\mathfrak{n}_\mathcal{U}, \overline{\mathfrak{m}}_\mathcal{U}$	minimal and maximal prime of a fibre	$\mathfrak{n}_\mathcal{U}, \mathfrak{m}_\mathcal{U}$	the inverse image of $\overline{\mathfrak{m}}_\mathcal{U}$ in $\mathcal{E}_\Gamma$ is our $\mathfrak{M}_\mathcal{U}$; our $\mathfrak{m}_\Gamma$ is the maximal ideal of $\mathcal{O}_\Gamma$
$\mathfrak{p}_{\mathcal{U}, \mathcal{C}}, \mathcal{C}, \mathcal{C}'$	fibre prime, convex subgroups	$\mathfrak{p}_{\mathcal{U}, \mathcal{C}}, C, D$	our C_n are corrections, our D a divisor
$\mathcal{C}_{\min}, \mathcal{C}_{\max}, \mathcal{C}_{\text{int}}, \mathcal{C}'_\theta, \mathcal{C}_\theta$	convex extensions, splitting and gap subgroups	$C_{\min}, C_{\max}, C, D_r, C_r$	real parameter $\theta \in (0, 1)$ for its r
$\mathcal{V}_\mathcal{U}, \mathcal{K}_\mathcal{U}, \Gamma_\mathcal{U}, w_\mathcal{U}$	ultraproduct valuation ring, its fraction field, value group, valuation	$W_\mathcal{U}, F_\mathcal{U}, G_\mathcal{U}, w_\mathcal{U}$	our W and F are entire functions
$\text{red}_\mathcal{U}$	quotient map to $\mathcal{V}_\mathcal{U}$	$q_\mathcal{U}$	our q_n counts nodes; our $\Phi_\mathcal{U}$ is $\text{red}_\mathcal{U}$ followed by residues
$\bar{\eta}_n, \zeta_\theta$	gap elements	d_n, d_r	our d is a damping exponent
$m_{-r}(F)$	weighted Gauss valuation	$\mu_r(F)$	<i>sign-inverted</i> , as in the first table
$\Delta_\mathbb{Q} = \bigoplus_{q \in \mathbb{Q}_{\geq 0}} \mathbb{Q}e_q$	rational-indexed enlargement of Γ_∞	$\bigoplus \mathbb{Q}f_q, f_j = e_j$	our f is (83)
$\mathbf{No}[i]$	surcomplex numbers	$\mathbf{No}(i)$	collection notation

Two consequences of the last row of the first table are used repeatedly and are recorded here once. First, $m_\gamma(F)$ is nondecreasing in γ , because each term $v(a_n) + n\gamma$ is. Second, on polynomials and on entire series

$$m_\gamma(F + G) \geq \min\{m_\gamma(F), m_\gamma(G)\}, \quad m_\gamma(FG) \geq m_\gamma(F) + m_\gamma(G),$$

with $m_\gamma(0) = +\infty$; the second is the Cauchy-product estimate, and equality holds by (5), applied to the rescalings $F(t^\gamma X), G(t^\gamma X) \in \mathcal{A}_\Gamma$ of theorem 3.1(2). The first proof of the image theorem never needs the equality; the cardinal second route of section 8.5 does, in the one form $m_\gamma(\widehat{P}_n) = m_\gamma(P_D)$. Only these relations are used, never a maximum-modulus principle.

Finally, we write $F^{[k]}(b) = [Z^k]F(Z + b)$ for the k th Hasse jet coefficient at b , which is $F^{(k)}(b)/k!$ by (7). For $v(b) \geq \gamma$ the same expansion gives the bound

$$v(F^{[k]}(b)) \geq m_\gamma(F) - k\gamma, \tag{6}$$

since every summand $\binom{n}{k}a_n b^{n-k}$ has valuation at least $v(a_n) + (n - k)\gamma = (v(a_n) + n\gamma) - k\gamma$.

3 An exact coefficient criterion and the cofinality dichotomy

For $n \geq 1$, division of $v(a_n)$ by n is legitimate because Γ is divisible and torsion-free. We interpret $(+\infty)/n = +\infty$, so that zero coefficients impose no condition.

Theorem 3.1 (All-point strong summability). *Let $\Gamma \neq 0$ and $F = \sum_{n \geq 0} a_n Z^n \in K_\Gamma[[Z]]$. The following conditions are equivalent.*

- (1) $F \in \mathcal{E}_\Gamma$.
- (2) For every $\gamma \in \Gamma$, the family $(a_n t^{n\gamma})_{n \geq 0}$ is strongly summable; equivalently $F(t^\gamma X) \in \mathcal{A}_\Gamma$.
- (3) For every $h \in \Gamma$, eventually $v(a_n) > nh$.
- (4) For every $\gamma, B \in \Gamma$, eventually $v(a_n) + n\gamma > B$.

In particular, strong entire evaluation is equivalent to the cofinal condition $v(a_n)/n \rightarrow +\infty$, and it is enough to test the monomial points $z = t^\gamma$.

Proof. The implication (1) $\Rightarrow$ (2) is immediate by testing $z = t^\gamma$; the reformulation of (2) inside $\mathcal{A}_\Gamma$ holds because the common Hahn support is the union of the supports of the scaled coefficients, and the coefficient at a fixed Hahn exponent is a polynomial in X exactly when that exponent occurs at only finitely many indices n . The zero series satisfies all four conditions, so assume $F \neq 0$.

Suppose (2) holds. Applying it at $\gamma = 0$ shows that the union of the supports of the a_n is well ordered. Thus

$$m = \min\{v(a_n) : a_n \neq 0\} \in \Gamma$$

exists. Suppose (3) fails. There is $h \in \Gamma$ for which infinitely many positive integers n satisfy $a_n \neq 0$ and $v(a_n) \leq nh$. Choose $q > 0$ with $q > |h|$ and $q > |m|$, where $|\cdot|$ is the ordered group absolute value. By (2), the set of leading exponents

$$\{v(a_n) - 2nq : a_n \neq 0\}$$

is well ordered. Its minimum is attained at some *finite* index k ; write that minimum as μ . Then

$$\mu = v(a_k) - 2kq \geq m - 2kq > -(2k+1)q.$$

Choose a violating index $n > 2k+1$. It gives

$$v(a_n) - 2nq \leq nh - 2nq < -nq < -(2k+1)q < \mu,$$

a contradiction. This proves (2) $\Rightarrow$ (3). Notice that the argument used a minimum attained at a finite index; it did not assume that the multiples of q are cofinal in Γ . See [section A](#).

Suppose (3) holds and fix γ, B . Choose $r > 0$ and set $h = -\gamma + |B| + r$. Eventually

$$v(a_n) + n\gamma > n(|B| + r) > B$$

for $n \geq 1$. Thus (4) holds.

Finally, fix $z \neq 0$ and put $\gamma = v(z)$. Condition (4) implies $v(a_n z^n) = v(a_n) + n\gamma \rightarrow +\infty$. Apply [lemma 2.2](#). At $z = 0$ there is only the constant term. This proves (4) $\Rightarrow$ (1). $\square$

Remark 3.2. Condition (3) is strictly stronger than $v(a_n) \rightarrow +\infty$, that is, than membership in $\mathcal{T}_\Gamma$. Over $\Gamma = \mathbb{Q}$ the coefficients $a_n = t^n$ tend to zero in valuation, but $\sum t^n Z^n$ is not entire: at $Z = t^{-1}$ its terms all have exponent zero. Conversely $a_n = t^{n^2}$ satisfies (3) over $\mathbb{Q}$ and over $\mathbb{R}$, but not over a group containing an element larger than every real number. That last example is exactly the one recorded in the foundations report of this collection; see [section 15](#).

Corollary 3.3 (Cofinality dichotomy). *For $\Gamma \neq 0$, the following are equivalent: $\mathcal{E}_\Gamma \neq K_\Gamma[Z]$; $\mathcal{E}_\Gamma$ contains a nonpolynomial series; Γ has a countable cofinal subset; $\text{cf}(\Gamma) = \aleph_0$. If $\text{cf}(\Gamma) > \aleph_0$, every strongly entire ordinary power series is a polynomial.*

Proof. If infinitely many a_n are nonzero, their valuations divided by their positive indices give, by [theorem 3.1](#), a countable cofinal subset of Γ . Conversely, choose a positive increasing cofinal sequence $(q_n)_{n \geq 1}$. Then

$$1 + \sum_{n \geq 1} t^{nq_n} Z^n$$

satisfies the coefficient criterion and is nonpolynomial. A nonzero ordered group has no finite cofinal subset, because it has no greatest element. $\square$

Example 3.4 (Two different sources of polynomial rigidity). If $\Gamma = 0$, every nonzero coefficient has support $\{0\}$, so strong summability at 1 forces a polynomial. If Γ has uncountable cofinality, there are many infinitesimal coefficients, but countably many coefficient slopes cannot reach all of its scales. The conclusion is again polynomiality, for a different reason. The same countability argument shows that in the uncountable-cofinality case even the larger algebra $\mathcal{T}_\Gamma$ collapses to $K_\Gamma[Z]$; see [\(96\)](#).

Proposition 3.5 (Entire operations). *$\mathcal{E}_\Gamma$ is a K_Γ -subalgebra of $K_\Gamma[[Z]]$, and so are $\mathcal{T}_\Gamma$ and $\mathcal{A}_\Gamma$. Evaluation at each $z \in K_\Gamma$ is a homomorphism on $\mathcal{E}_\Gamma$. Translation $F(Z) \mapsto F(Z+b)$, differentiation, and the primitive with zero constant coefficient preserve $\mathcal{E}_\Gamma$. The translation coefficients are*

$$[Z^k]F(Z+b) = \sum_{n \geq k} \binom{n}{k} a_n b^{n-k} = \frac{F^{(k)}(b)}{k!}. \quad (7)$$

Entire composition $F \circ G$ is also defined and belongs to $\mathcal{E}_\Gamma$. Moreover, at each fixed scale $s \in \Gamma$ the partial sums of $F \in \mathcal{E}_\Gamma$ converge to F in the weighted Gauss valuation m_s of [\(14\)](#).

Proof. The ring assertion for $\mathcal{E}_\Gamma$ follows from [theorem 3.1\(2\)](#): addition and multiplication preserve every rescaled copy of $\mathcal{A}_\Gamma$. Explicitly, at a fixed s write $A_i = v(a_i) + is$ and $B_j = v(b_j) + js$ for two members; both tend cofinally to infinity and have lower bounds A, B . Given β , choose N with $A_i > \beta - B$ and $B_i > \beta - A$ for $i > N$; if $n > 2N$ then each pair $i + j = n$ has $i > N$ or $j > N$, so the convolution coefficient has weighted valuation greater than β . Taking $s = 0$ gives the ring assertion for $\mathcal{T}_\Gamma$; the same tail estimate gives the stated weighted convergence of partial sums. That $\mathcal{A}_\Gamma$ is a ring is [lemma 2.1\(a\)](#).

Evaluation can be performed by choosing $\gamma = v(z)$, writing $z = t^\gamma x$ with $x \in \mathcal{O}_\Gamma$, and applying [lemma 5.1](#).

For translation and a fixed testing scale γ , expand $F(b + t^\gamma X)$ into its finite binomial blocks. Put $\beta = \min\{v(b), \gamma\}$, with $v(0) = +\infty$. Every coefficient in the n th block has valuation at least $v(a_n) + n\beta$, which tends cofinally to infinity, and each block has finitely many X -coefficients. The support-union and local-finiteness proof of [lemma 2.2](#) therefore applies to the blocks with polynomial coefficients. This gives an element of $\mathcal{A}_\Gamma$ at every scale and proves [\(7\)](#). Regrouping the same jointly summable family proves the evaluation identity and the inverse translation by $-b$.

For differentiation, $v(na_n) = v(a_n)$ for positive n , since n is a nonzero complex coefficient of valuation zero. Shifting the index by one preserves the criterion in [theorem 3.1](#): for a fixed h , choose $q > 0$ with $q > |h|$; the bound $v(a_n) > n(h+q)$ for large n implies the required bounds with $n-1$ or

$n + 1$ in place of n . No hidden Archimedean estimate is used. Division of a_n by $n + 1$ also preserves valuation, so termwise integration preserves the criterion as well.

For composition, fix γ and let $G_\gamma = G(t^\gamma X) \in \mathcal{A}_\Gamma$. If $G_\gamma \neq 0$, put $\mu = w(G_\gamma)$. Then $w(a_n G_\gamma^n) = v(a_n) + n\mu \rightarrow +\infty$, so the sum is strongly summable in $\mathcal{A}_\Gamma$. Its coefficients define the same formal composite at every scale, since rescaling commutes with each coefficient's strong sum. The case $G = 0$ is constant. Evaluation and associativity follow from the same support-controlled regrouping. $\square$

4 Restricted series: the sharp order-unit criterion

The trichotomy of [theorem 1.5](#) splits its two nonpolynomial cases by the presence of an order unit. This section proves that the order unit is not an artefact of the proof technique of [section 9](#): it is exactly the condition for a nonconstant element of the restricted algebra $\mathcal{T}_\Gamma$ of [\(3\)](#) to be invertible there. That is what makes case (ii) of the trichotomy sharp rather than merely sufficient, and it is a statement about $\mathcal{T}_\Gamma$, not about $\mathcal{E}_\Gamma$; the ring discipline of [convention 1.2](#) applies.

Theorem 4.1 (Restricted-unit criterion). *Let $f = \sum a_n Z^n \in \mathcal{T}_\Gamma$ be nonconstant. Then f is a unit of $\mathcal{T}_\Gamma$ if and only if $a_0 \neq 0$ and*

$$\delta(f) = \min_{n \geq 1, a_n \neq 0} v(a_n/a_0) \quad (8)$$

is a positive order unit of Γ . Nonzero constant series are always units, and the minimum in (8) exists whenever $a_0 \neq 0$.

Proof. The minimum exists because the coefficient valuations tend cofinally to infinity: below the valuation of any chosen nonzero coefficient there are only finitely many nonzero coefficients.

Suppose $fg = 1$ with $g \in \mathcal{T}_\Gamma$. The constant coefficient forces $a_0 \neq 0$. Divide f by a_0 and multiply g by a_0 , so that both constant coefficients are 1. Then $w(f), w(g) \leq 0$, while [\(5\)](#) gives their sum as zero. Hence both are zero. Their reductions in $\mathbb{C}[X]$ multiply to 1, so both reductions are constant. Every nonconstant coefficient of f and of g consequently has positive valuation; in particular $\delta = \delta(f) > 0$.

Assume for contradiction that δ is not an order unit. The convex subgroup it generates,

$$H_\delta = \{\gamma \in \Gamma : |\gamma| \leq m\delta \text{ for some } m \in \mathbb{N}\}, \quad (9)$$

is then proper. Coarsen the valuation to the ordered quotient Γ/H_δ . Every coefficient of f and g is integral for this coarsening. Their coarsened reductions are *polynomials* over the coarsened residue field: choose $\beta > H_\delta$, meaning $\beta > h$ for every $h \in H_\delta$; since coefficient valuations tend cofinally to infinity in Γ , only finitely many coefficients can have value at most β , and all the remaining ones reduce to zero.

The reduced polynomial of f is nonconstant, because a nonconstant coefficient of valuation $\delta \in H_\delta$ has nonzero coarsened residue. On the other hand the reduced polynomials of f and g multiply to 1 in a polynomial ring over a field, and a nonconstant polynomial cannot be a unit there. This contradiction proves that δ is an order unit.

Conversely, suppose $a_0 \neq 0$ and $\delta(f)$ is a positive order unit. Write $f = a_0(1 + h)$, where h has zero constant coefficient and $w(h) = \delta$. The formal inverse is

$$f^{-1} = a_0^{-1} \sum_{m \geq 0} (-h)^m. \quad (10)$$

Its summands have Gauss valuation $m\delta \rightarrow +\infty$, so the sum exists strongly in $\mathcal{A}_\Gamma$. To prove that it belongs to $\mathcal{T}_\Gamma$, fix a valuation threshold β . All sufficiently large m contribute only coefficients of valuation greater than β ; the remaining finitely many powers of h belong to $\mathcal{T}_\Gamma$, so their coefficients also eventually exceed β . Thus the coefficients of (10) tend cofinally to infinity. The finite geometric identity, followed by the justified strong sum, proves $ff^{-1} = 1$. $\square$

The coarsening argument is what makes this a *necessity* theorem rather than the observation that one particular geometric-series construction fails. It shows that no cancellation in a hypothetical inverse can evade the order-unit obstruction.

Corollary 4.2 (Unit group of the restricted algebra). *The algebra $\mathcal{T}_\Gamma$ has a nonconstant unit if and only if Γ has an order unit; otherwise $\mathcal{T}_\Gamma^\times = K_\Gamma^\times$. The rule “the constant coefficient strictly dominates all the others” is a sufficient unit criterion for every restricted series if and only if Γ is Archimedean.*

Proof. An order unit δ makes $1 - t^\delta Z$ a nonconstant unit by the theorem, and the converse follows by applying the theorem to any nonconstant unit. An ordered group is Archimedean exactly when every positive element has cofinal positive integer multiples; apply that characterization to $1 - t^\delta Z$ for the last assertion. $\square$

Example 4.3 (Zero-free does not mean restricted-invertible). Let $\Gamma = \mathbb{Q}\epsilon \oplus \mathbb{Q}\eta$ with $0 < \epsilon \ll \eta$, the η coordinate being compared first. The polynomial

$$f(Z) = 1 - t^\epsilon Z$$

has residue 1 and no zero in $\mathcal{O}_\Gamma$: its unique zero $t^{-\epsilon}$ has negative valuation. Yet its only formal inverse, $\sum_{n \geq 0} t^{n\epsilon} Z^n$, is not restricted, since $n\epsilon < \eta$ for every n . It does belong to $\mathcal{A}_\Gamma$ and can be strongly evaluated on the integral disk. By contrast $1 - t^\eta Z$ has a restricted inverse, because $n\eta$ is cofinal.

Remark 4.4 (Why a rank-one contraction proof may not be imported). [Example 4.3](#) is the reason the preparation theorem of [section 5](#) is proved by well-founded recursion over a well-ordered monoid and not by a convergent iteration. A rank-one Banach argument containing the phrase “the valuation increases by a positive amount at each step” is not valid here: the standard iteration may improve the error by ϵ repeatedly while remaining forever below η , and its purported convergence is then simply unproved. For a valid valuation-convergence proof the positive amount must be an order unit. The preparation proof below uses positive *supports* and finite-word control instead, and works with no order unit at all. The same care applies to [section 9](#), whose real-valued coarsening is constructed only after an order unit has been assumed.

Remark 4.5 (Two existence questions, genuinely different). Nonpolynomial entire series need only $\text{cf}(\Gamma) = \aleph_0$ ([corollary 3.3](#)); a nonconstant restricted unit needs an order unit ([corollary 4.2](#)). The group Γ_∞ of [section 14](#) has the first and not the second, so the two questions separate. In topological language: the valuation topology of K_Γ has a countable neighbourhood base at zero exactly when $\text{cf}(\Gamma) = \aleph_0$, whereas a nonzero topologically nilpotent element exists exactly when Γ has an order unit, since $v(x^n) = nv(x)$. Neither of these underlying topological facts is claimed as new; the higher-rank framework in which they sit is discussed by Conrad [\[8\]](#).

5 Support-controlled preparation

5.1 The algebra that retains support certificates

Recall $\mathcal{A}_\Gamma = \mathbb{C}[X]((t^\Gamma))$ from (2). The support facts in lemma 2.1 justify multiplication in $\mathcal{A}_\Gamma$ and show that the coefficient map $\mathcal{A}_\Gamma \hookrightarrow K_\Gamma[[X]]$ respects it.

Lemma 5.1 (Units and integral substitution). *A nonzero element of $\mathcal{A}_\Gamma$ is a unit if and only if its initial polynomial is a nonzero constant. Every $A \in \mathcal{A}_\Gamma$ can be evaluated at any $x \in \mathcal{O}_\Gamma$, and this evaluation is a ring homomorphism; if $A \neq 0$ the value has valuation at least $w(A)$. For $x \in \mathcal{O}_\Gamma$, substitution $X \mapsto X + x$ is an automorphism of $\mathcal{A}_\Gamma$.*

Proof. If the initial polynomial is $c \neq 0$ at exponent η , write $A = ct^\eta(1 + u)$ with $w(u) > 0$ and invert $1 + u$ using lemma 2.1(c). Conversely, if $AB = 1$, the product of the two initial polynomials is 1, so both are nonzero constants.

For substitution, put $S = \text{supp}_t A$ and write $x = c + \varepsilon$ with $c \in \mathbb{C}$ and $v(\varepsilon) > 0$, treating $\varepsilon = 0$ separately. For each η , expand the finite polynomial $P_\eta(X + c + \varepsilon)$ in powers of ε . All supports lie in $S + (\text{supp } \varepsilon)^*$, which is well ordered. At a fixed exponent there are only finitely many decompositions from $S + (\text{supp } \varepsilon)^*$, finitely many words in $\text{supp } \varepsilon$ for each such decomposition, and finitely many terms from each P_η . Thus substitution is a legitimate element of $\mathcal{A}_\Gamma$. Finite polynomial identities and strong regrouping prove multiplicativity, and its inverse is substitution by $-x$. Evaluation follows by setting $X = 0$ after this substitution. If $A \neq 0$, every contribution has exponent at least $\min S = w(A)$, because $v(\varepsilon) > 0$; cancellation can only raise the valuation. This gives the claimed bound without any assertion of cofinal tail convergence. $\square$

5.2 Preparation without an Archimedean contraction

Theorem 5.2 (Support-controlled preparation). *Let $A \in \mathcal{A}_\Gamma$ satisfy $w(A) = 0$, and suppose its initial polynomial $P \in \mathbb{C}[X]$ is monic of degree d . There are unique W, U such that*

$$A = WU, \quad W = P + R, \quad R \in \mathfrak{m}_\Gamma[X], \quad \deg R < d, \quad U = 1 + Q, \quad w(Q) > 0.$$

Here W is a monic polynomial of degree d and U is a unit of $\mathcal{A}_\Gamma$. If $S = \text{supp}_t(A - P) \subseteq \Gamma_{>0}$, both corrections are supported in $S^* \setminus \{0\}$.

Proof. Let $M = S^*$, a well-ordered monoid. Write $A = P + \sum_{\eta > 0} A_\eta t^\eta$, with $A_\eta = 0$ off S . Recursively for $\eta \in M \setminus \{0\}$, define polynomials R_η, Q_η by ordinary division by P :

$$H_\eta = A_\eta - \sum_{\substack{\alpha + \beta = \eta \\ \alpha, \beta > 0}} R_\alpha Q_\beta, \tag{11}$$

$$H_\eta = PQ_\eta + R_\eta, \quad \deg R_\eta < d. \tag{12}$$

The sum is finite by lemma 2.1, and every index on its right is smaller than η . Thus this is a well-founded recursion on a set. Set $R = \sum R_\eta t^\eta$ and $Q = \sum Q_\eta t^\eta$. Because the remainders have a uniform degree bound, R is a finite polynomial over $\mathfrak{m}_\Gamma$, and $Q \in \mathcal{A}_\Gamma$. Equations (11)–(12) give $A = (P + R)(1 + Q)$ coefficient by coefficient, and the second factor is a unit by lemma 5.1.

For uniqueness, suppose two pairs differ. The union of their positive supports is well ordered, so there is a least exponent η at which either pair differs. At that exponent, equality of the products gives

$$P(Q_\eta - Q'_\eta) + (R_\eta - R'_\eta) = 0.$$

The second summand has degree less than d . Uniqueness of polynomial division forces both differences to vanish, a contradiction. For $d = 0$, the normalization gives $P = 1$, $W = 1$ and $U = A$. $\square$

This proof is deliberately not a claim that a Banach iteration converges; see [remark 4.4](#). It remains valid when no positive element of Γ has cofinal integer multiples. The uniformly bounded remainder degree is the mechanism that extracts a finite zero scheme from an infinite Hahn series.

Corollary 5.3 (The certificate survives enlargement of the value group). *Let $\Gamma \subseteq \Delta$ be an ordered-group inclusion and regard $A \in \mathcal{A}_\Gamma$ as an element of $\mathcal{A}_\Delta$ under the canonical inclusion. The factorization $A = WU$ of [theorem 5.2](#) computed over Γ is also the factorization computed over Δ : the same W and U , with the same support certificate $S^* \setminus \{0\}$.*

Proof. The recursion (11)–(12) is indexed by S^* , computed inside $\mathbb{C}[X]$, and refers to no element of the value group beyond the exponents already occurring in S . It therefore produces the same data over Δ , and uniqueness over Δ identifies the two factorizations. $\square$

[Corollary 5.3](#) is not a formality: it is exactly what lets the same prepared polynomial be used on both sides of a scalar extension in [theorem 10.7](#), and it is available only because the preparation was proved by a support recursion rather than by an iteration whose convergence would have to be re-established in the larger field.

Example 5.4 (A concrete preparation). Take $P = X^2 - X$, $d = 2$, and

$$A = X^2 - X + t(X^3 + 1) \in \mathcal{A}_\Gamma, \quad (13)$$

so that $S = \{1\}$ and $S^* = \mathbb{N}$. The recursion gives $Q_1 = X + 1$, $Q_2 = -1$, $Q_3 = 3$ and the first three remainders

$$R_1 = X + 1, \quad R_2 = -3X - 1, \quad R_3 = 8X + 2,$$

each of degree less than 2 as required. For instance $X^3 + 1 = (X^2 - X)(X + 1) + (X + 1)$ gives the first step, and $H_2 = -R_1Q_1 = -(X + 1)^2 = (X^2 - X)(-1) + (-3X - 1)$ the second. The recursion is run to perturbation order eight by the accompanying finite check suite ([section B](#)).

6 Exact root counting at every scale

6.1 Initial polynomials on a ball

For $\gamma \in \Gamma$, define the closed valuation ball

$$B_\gamma = \{z \in K_\Gamma : v(z) \geq \gamma\}.$$

A larger ball corresponds to a smaller, typically negative, γ . For $F \in \mathcal{E}_\Gamma \setminus \{0\}$, put

$$m_\gamma(F) = \min_{n: a_n \neq 0} \{v(a_n) + n\gamma\}, \quad (14)$$

$$I_\gamma(F)(X) = \sum_{v(a_n) + n\gamma = m_\gamma(F)} \text{lc}(a_n) X^n. \quad (15)$$

The minimum is attained, and attained at only finitely many indices, and the sum is a finite nonzero polynomial, precisely because the weighted coefficient valuations diverge ([theorem 3.1](#)). Equivalently $I_\gamma(F)$ is the reduction of $t^{-m_\gamma(F)} F(t^\gamma X)$. Write $d_\gamma(F) = \deg I_\gamma(F)$ and $e_\gamma(F) = \text{ord}_{X=0} I_\gamma(F)$.

A zero's multiplicity is the order of the translated formal power series $F(z + Z) \in K_\Gamma[[Z]]$; the translation identity (7) makes this intrinsic.

Corollary 6.1 (Finite zero geometry on the integral disk). *Let $A \in \mathcal{A}_\Gamma$ be nonzero with $w(A) = 0$ and initial polynomial monic of degree d after normalization, and let W be its prepared polynomial. The zeros of A in $\mathcal{O}_\Gamma$ are exactly the d roots of W with multiplicity, all of them integral, and for each $c \in \mathbb{C}$ the number of them with residue c is the multiplicity of c as a root of the initial polynomial. In particular a nonzero element of $\mathcal{A}_\Gamma$ has only finitely many zeros on the integral disk. No complex compactness is used.*

Proof. This is the content of the first two paragraphs of the proof of [theorem 6.2](#) below, applied at $\gamma = 0$. $\square$

Theorem 6.2 (Ball, shell, and residue-direction counts). *For $F \in \mathcal{E}_\Gamma \setminus \{0\}$ and $\gamma \in \Gamma$:*

- (a) B_γ contains exactly $d_\gamma(F)$ zeros of F , counted with multiplicity.
- (b) The open ball $\{z : v(z) > \gamma\}$ contains exactly $e_\gamma(F)$ zeros, and the shell $\{z : v(z) = \gamma\}$ contains $d_\gamma(F) - e_\gamma(F)$ zeros.
- (c) For each $c \in \mathbb{C}$, the zeros for which $v(z) \geq \gamma$ and $\text{res}(t^{-\gamma}z) = c$ have total multiplicity $\text{ord}_{X=c} I_\gamma(F)$.

Proof. Let c_0 be the leading coefficient of $I_\gamma(F)$ and apply [theorem 5.2](#) to

$$A = \frac{F(t^\gamma X)}{c_0 t^{m_\gamma(F)}}.$$

It gives $A = WU$ with $W \in \mathcal{O}_\Gamma[X]$ monic of degree $d_\gamma(F)$, $\text{res}(W) = I_\gamma(F)/c_0$, and U of initial polynomial 1. The field K_Γ is algebraically closed by the standard Hahn-field theorem [4, Corollary 4], so W has exactly its degree many roots in K_Γ , counted with multiplicity.

Every root x of W is integral: if $v(x) < 0$, the leading term x^d has valuation $dv(x)$, strictly smaller than that of every lower-degree term whose coefficient is integral, and such terms cannot sum to zero. On $\mathcal{O}_\Gamma$, evaluation of U has residue 1 and is nonzero. After translation by an integral root, U has nonzero constant term, so its formal local power series is a unit. Hence zeros and multiplicities of A in $\mathcal{O}_\Gamma$ coincide exactly with those of W .

Factor $W = \prod_j (X - x_j)$ and reduce modulo $\mathfrak{m}_\Gamma$. Then $\text{res}(W) = \prod_j (X - \text{res}(x_j))$, which proves the residue-direction count. The direction $c = 0$ is exactly the open ball, and the nonzero directions make up the shell. Rescale back by $Z = t^\gamma X$. $\square$

Corollary 6.3 (Units, faithfulness, and values). *A nonzero entire function without zeros is a nonzero constant. Distinct elements of $\mathcal{E}_\Gamma$ define distinct functions on K_Γ . Every nonconstant element of $\mathcal{E}_\Gamma$ takes every value in K_Γ .*

Proof. If a zero-free F has $a_0 = 0$ it vanishes at zero, a contradiction. If $a_n \neq 0$ for some $n > 0$, choose $\delta > 0$ and

$$\gamma = \frac{v(a_0) - v(a_n)}{n} - \delta.$$

Then $v(a_n) + n\gamma < v(a_0)$, so the constant coefficient is not an active term in (15) and the initial polynomial has positive degree; [theorem 6.2](#) supplies a zero. Therefore F is constant.

For faithfulness, a nonzero F has only finitely many zeros in any B_γ , whereas that ball contains infinitely many elements $t^\gamma c$ with $c \in \mathbb{C}$; apply this to the difference of two series. Finally, for any $b \in K_\Gamma$ a nonconstant $F - b$ cannot be zero-free, so F takes the value b . $\square$

Lemma 6.4 (Division by a linear zero factor). *If $F \in \mathcal{E}_\Gamma$ and $F(r) = 0$, then $F = (Z - r)G$ with $G \in \mathcal{E}_\Gamma$.*

Proof. Set $H(Z) = F(r + Z)$, which is entire by [proposition 3.5](#), and note that $H(0) = 0$. For each γ , write $H(t^\gamma X) = \sum_\eta P_\eta(X)t^\eta \in \mathcal{A}_\Gamma$. Its constant coefficient is zero, so every P_η is divisible by X . Dividing these finite polynomials by X preserves the well-ordered Hahn support and gives an element of $\mathcal{A}_\Gamma$. Consequently

$$(H/Z)(t^\gamma X) = t^{-\gamma} H(t^\gamma X)/X \in \mathcal{A}_\Gamma.$$

By [theorem 3.1\(2\)](#), H/Z is entire. Translating back gives $G(Z) = (H/Z)(Z - r) \in \mathcal{E}_\Gamma$ and $F = (Z - r)G$. $\square$

6.2 Stability of the counts

Corollary 6.5 (Initial-form stability). *If $F, G \in \mathcal{E}_\Gamma$ satisfy $w((F - G)(t^\gamma X)) > m_\gamma(F)$, then $m_\gamma(G) = m_\gamma(F)$ and $I_\gamma(G) = I_\gamma(F)$. All three counts in [theorem 6.2](#) are therefore unchanged.*

Proof. A perturbation supported strictly above the leading exponent changes no coefficient at that exponent; apply [theorem 6.2](#). $\square$

Proposition 6.6 (Locally finite Newton profile). *Let $F \in \mathcal{E}_\Gamma \setminus \{0\}$ and let $[\gamma_0, \gamma_1]$ be an interval of Γ . On that interval, $\gamma \mapsto m_\gamma(F)$ is the minimum of finitely many affine functions $\gamma \mapsto v(a_n) + n\gamma$. The set of active indices changes only at the slopes*

$$\frac{v(a_i) - v(a_j)}{j - i} \in \Gamma \quad (i < j),$$

and between two consecutive such values the three counts of [theorem 6.2](#) are constant.

Proof. By [theorem 3.1\(4\)](#) applied at γ_0 and at γ_1 , only finitely many indices n can attain the minimum (14) for some γ in the interval: all but finitely many satisfy $v(a_n) + n\gamma > m_{\gamma_0}(F) + |m_{\gamma_1}(F) - m_{\gamma_0}(F)|$ throughout it. A minimum of finitely many affine functions is piecewise affine, and two of them exchange dominance exactly at the displayed slope, which lies in Γ by divisibility. [Corollary 6.5](#) gives the constancy. $\square$

This is a valuation statement, a non-Archimedean analogue of a Rouché stability argument; it is not a theorem about Euclidean circles in the fine surcomplex topology. Its purpose here is to retain exact multiplicities in the factorization argument of [section 7](#).

7 Radial divisors, canonical products and factorization

7.1 Which zero sets are admissible?

Definition 7.1. An effective divisor on K_Γ is a function $D : K_\Gamma \rightarrow \mathbb{N}$ with set-sized support. It is *radially finite*, equivalently *ball-finite*, if

$$\sum_{z \in B_\gamma} D(z) < \infty \quad (\gamma \in \Gamma).$$

The sum means a finite total multiplicity, not a Hahn sum. Let $\text{Div}_{\text{rad}}^+(K_\Gamma)$ denote the monoid of these divisors.

The first two source manuscripts used the two displayed adjectives for this one condition; we keep *radially finite* and record *ball-finite* as the synonym. This is a restriction on valuation balls, not on discrete subsets for the fine surreal topology, and the distinction is necessary: fine discreteness alone would impose no useful control on an infinite collection inside a fixed bounded valuation ball. Radial finiteness is exactly what the root count of [theorem 6.2](#) supplies.

Lemma 7.2 (Countability and escape). *An infinite radially finite divisor has countably many points counted with multiplicity. Its nonzero points can be enumerated as $(\rho_j)_{j \geq 1}$, with repetitions for multiplicities, and*

$$v(\rho_j) \longrightarrow -\infty \quad \text{coinitially in } \Gamma.$$

In particular an infinite radially finite divisor forces $\text{cf}(\Gamma) = \aleph_0$; and when $\text{cf}(\Gamma) > \aleph_0$, every radially finite divisor is finite.

Proof. Choose countably many distinct occurrences from the infinite divisor. Their valuations cannot have a common lower bound, for otherwise one ball would contain infinitely many occurrences. They form a countable coinital subset of Γ , whose negatives are a countable cofinal subset. Balls at those scales exhaust K_Γ and each contains only finitely many divisor occurrences, so their union contains the whole divisor and is countable. In any one-to-one enumeration of its occurrences, only finitely many indices have valuation at least any fixed γ , which is the asserted escape. The occurrence at zero, if present, has finite multiplicity and is removed before enumerating the nonzero points. $\square$

7.2 Construction and exact factorization

Theorem 7.3 (Canonical product for a radial divisor). *Let $D \in \text{Div}_{\text{rad}}^+(K_\Gamma)$, let $m = D(0)$, and enumerate its nonzero occurrences as ρ_j . The product*

$$P_D(Z) = Z^m \prod_j \left(1 - \frac{Z}{\rho_j}\right) \tag{16}$$

is a well-defined element of $\mathcal{E}_\Gamma$, independent of the enumeration. Its divisor is exactly D , and the coefficient of Z^m is 1. The product is finite when the nonzero divisor is finite. Its partial products converge to it in the weighted Gauss valuation at every scale. No exponential correction factors are required: the theory is genus zero at every rank.

Proof. Only the infinite case needs a support argument. Put $b_j = \rho_j^{-1}$. Fix $\gamma \in \Gamma$ and substitute $Z = t^\gamma X$. By [lemma 7.2](#), $v(b_j) + \gamma \rightarrow +\infty$. There are only finitely many factors for which

$v(b_j) + \gamma \leq 0$; set them aside. For the rest, the supports of $b_j t^\gamma$ have a well-ordered positive union and are locally finite, by [lemma 2.2](#).

Expand the tail product over finite subsets of indices. Every exponent belongs to the monoid generated by that positive union. For a fixed exponent, [lemma 2.1](#) gives finitely many words in the support, and local finiteness gives finitely many choices of factor labels carrying those exponents. Thus only finitely many product terms contribute, and the coefficient at that exponent is a polynomial in X . The tail product is an element of $\mathcal{A}_\Gamma$ with initial polynomial 1, hence a unit. Multiplying by the finitely many set-aside factors and by $t^{m\gamma} X^m$ gives an element of $\mathcal{A}_\Gamma$; the same estimate shows that the finite partial products approach it in m_γ .

For each fixed ordinary degree, this construction sums the same finite-subset coefficient family multiplied by $t^{k\gamma}$. Consequently the elements obtained for different γ are rescalings of one series in $K_\Gamma[[Z]]$, which belongs to $\mathcal{E}_\Gamma$ by [theorem 3.1\(2\)](#). Strong regrouping also proves independence of the enumeration.

On B_γ , the set-aside factors give precisely the roots in that ball, while the remaining unit is nowhere zero on it. Local translation shows that multiplicities are the displayed factor multiplicities. Since the balls cover K_Γ , the divisor is exactly D . Every factor of the nonzero-root product has constant term 1, so the coefficient of Z^m is 1. $\square$

Theorem 7.4 (Factorization by zeros). *Let $F = \sum a_n Z^n \in \mathcal{E}_\Gamma \setminus \{0\}$ and let $m = \text{ord}_{Z=0} F$. Its zero divisor is radially finite. If ρ_j are all its nonzero zeros, repeated with their multiplicities, then*

$$F(Z) = a_m Z^m \prod_j \left(1 - \frac{Z}{\rho_j}\right). \quad (17)$$

Thus the zero divisor determines F up to a nonzero scalar.

Proof. Radial finiteness is [theorem 6.2](#). Let $G = P_{\text{Div}(F)}$ be the canonical product from [theorem 7.3](#). Since F and G have the same order m at zero, the formal quotient

$$Q = \frac{F/Z^m}{G/Z^m} \in K_\Gamma[[Z]]$$

exists, because G/Z^m has constant coefficient 1.

We claim $Q \in \mathcal{E}_\Gamma$. Fix γ . Preparation of $F(t^\gamma X)$ and $G(t^\gamma X)$ writes them as

$$F(t^\gamma X) = c_F W_F U_F, \quad G(t^\gamma X) = c_G W_G U_G,$$

with $c_F, c_G \in K_\Gamma^\times$, units U_F, U_G of $\mathcal{A}_\Gamma$, and monic W_F, W_G all of whose roots lie in $\mathcal{O}_\Gamma$. Those roots are exactly the normalized roots of F and of G in B_γ , with multiplicity, so $W_F = W_G$. Cancellation in $K_\Gamma[[X]]$, including the common factor X^m , gives

$$Q(t^\gamma X) = \frac{c_F}{c_G} \frac{U_F}{U_G} \in \mathcal{A}_\Gamma.$$

This holds at every scale, so [theorem 3.1\(2\)](#) proves the claim.

The identity $F = GQ$ is now an identity of entire functions. Orders at any translated centre add in the formal power-series ring. Since F and G have identical zero multiplicities, Q has no zeros, so Q is constant by [corollary 6.3](#). Comparing the coefficient of Z^m gives $Q = a_m$. $\square$

The quotient step is important and is the reason the preparation theorem was proved at *every* scale. Equal zero sets alone do not justify dividing two infinite series in the entire category, and no compactness or transfinite convergence assertion replaces that argument. Note also the order of the argument: the quotient is shown strongly evaluable at all scales *before* [theorem 3.1](#) upgrades it to an entire, valuation-convergent series. In particular the proof never assumes that a zero-free restricted series is restricted-invertible — by [example 4.3](#) that is exactly what can fail in higher rank.

Corollary 7.5 (Divisibility, GCDs, and least common multiples). *The map $F \mapsto \text{Div}(F)$ induces an isomorphism of commutative monoids*

$$(\mathcal{E}_\Gamma \setminus \{0\})/K_\Gamma^\times \cong \text{Div}_{\text{rad}}^+(K_\Gamma).$$

For nonzero $F, G \in \mathcal{E}_\Gamma$, G divides F in $\mathcal{E}_\Gamma$ if and only if $\text{Div}(G) \leq \text{Div}(F)$ pointwise. Greatest common divisors and least common multiples are given by the pointwise minimum and maximum of divisors. In particular $\mathcal{E}_\Gamma$ is a GCD domain.

Proof. Surjectivity is [theorem 7.3](#); injectivity modulo constants is [theorem 7.4](#). Divisors add under products by Taylor expansion. If the divisor of G is contained in that of F , their difference is radially finite, so its canonical product supplies the quotient, up to the scalar fixed by [theorem 7.4](#). Pointwise minima and maxima of two radial divisors remain radially finite; their canonical products therefore have the defining universal divisibility properties of a GCD and an LCM. $\square$

Corollary 7.6 (Finite jets and evaluation ideals). *For $x \in K_\Gamma$ and $r \geq 1$, the kernel of the Taylor jet map*

$$\mathcal{E}_\Gamma \longrightarrow K_\Gamma[U]/(U^r), \quad F \longmapsto F(x + U) \bmod U^r$$

is $(Z - x)^r \mathcal{E}_\Gamma$, and the map is surjective. In particular evaluation at x has kernel $(Z - x)\mathcal{E}_\Gamma$ and residue field K_Γ .

Proof. Vanishing of the jet is exactly a zero of order at least r ; apply [corollary 7.5](#). Every jet is realized by a finite polynomial in $Z - x$. $\square$

Corollary 7.7 (Rigidity and value distribution). *Let $F \in \mathcal{E}_\Gamma$ be nonzero.*

- (a) *F has finitely many zeros if and only if F is a polynomial; a zero-free F is a nonzero constant.*
- (b) *If F is nonconstant it is surjective onto K_Γ ; if F is nonpolynomial, every fibre $F^{-1}(b)$ is infinite.*
- (c) *If F is injective as a function on K_Γ , then F is affine of degree one.*

Proof. (a) A polynomial has finitely many zeros, and conversely (17) with a finite product is a polynomial. Zero-freeness is [corollary 6.3](#). (b) Surjectivity is [corollary 6.3](#); if F is nonpolynomial then $F - b$ is nonpolynomial, so by (a) it has infinitely many zeros. (c) An injective F is nonconstant, hence a polynomial by (b), say of degree $d \geq 1$ with leading coefficient a_d . For each b , injectivity makes the divisor of $F - b$ a single point r_b with multiplicity d , so $F(Z) - b = a_d(Z - r_b)^d$ by [theorem 7.4](#). Comparing two values $b \neq b'$ gives $a_d[(Z - r_b)^d - (Z - r_{b'})^d] = b' - b$, a nonzero constant. For $d \geq 2$ the left side has a nonzero coefficient of Z^{d-1} unless $r_b = r_{b'}$, and $r_b = r_{b'}$ makes it zero; either way we contradict $b \neq b'$. Hence $d = 1$. $\square$

Part (b) and part (c) are familiar non-Archimedean value distribution, stated here for completeness at arbitrary rank and not offered as independent priority claims.

Corollary 7.8 (Conjugation and real-coefficient functions). *A nonzero entire series has all coefficients in $\mathbb{R}((t^\Gamma))$ if and only if its zero divisor is invariant under coefficientwise conjugation and its first nonzero coefficient lies in $\mathbb{R}((t^\Gamma))$.*

Proof. Conjugating coefficients conjugates each value and each zero with its multiplicity. Conversely, under the stated conditions the conjugate series has the same divisor and the same first nonzero coefficient, so uniqueness in [theorem 7.4](#) makes the two series equal. $\square$

7.3 What “factorization” does not imply

Proposition 7.9. *If $\mathcal{E}_\Gamma$ contains a nonpolynomial function, it is not Noetherian and is not atomic. Its irreducible elements are precisely the scalar multiples of linear polynomials.*

Proof. By [theorem 7.4](#), a nonpolynomial entire function has infinitely many zeros with multiplicity. Let $\rho_1, \rho_2, \dots$ be an escaping sequence of nonzero roots and put $T_n(Z) = \prod_{j \geq n} (1 - Z/\rho_j)$. Then $T_n = (1 - Z/\rho_n)T_{n+1}$, so

$$(T_1) \subsetneq (T_2) \subsetneq (T_3) \subsetneq \dots,$$

strictness following either by evaluating multiplicities at ρ_n or from [corollary 7.5](#). Thus the ring is not Noetherian.

A nonunit has a zero. If its divisor has at least two occurrences, separating one occurrence from the remaining divisor gives a product of two nonunits. A singleton divisor gives a scalar multiple of one linear factor and is irreducible. Consequently a function with an infinite divisor cannot be a *finite* product of irreducibles, which is the failure of atomicity. $\square$

The canonical product theorem is therefore not a unique-factorization-domain theorem. Its product is infinite in a precisely controlled Hahn sense; an algebraic UFD requires finite irreducible factorizations.

8 The higher-rank obstruction and the exact image of the jet map

8.1 A sequence of genuinely new Archimedean scales

For positive $\alpha, \beta \in \Gamma$, write

$$\beta \gg_A \alpha \iff \beta > k\alpha \text{ for every integer } k \geq 1.$$

This is Archimedean domination: a genuinely new Archimedean class, not merely a large finite ratio.

Lemma 8.1. *Suppose $\text{cf}(\Gamma) = \aleph_0$ and Γ has no order unit. There is a positive cofinal sequence $(\gamma_n)_{n \geq 1}$ with $\gamma_{n+1} \gg_A \gamma_n$ for every n .*

Proof. Choose a positive cofinal sequence (p_n) . After selecting γ_n , the set $\{k\gamma_n : k \geq 1\}$ is not cofinal, since γ_n is not an order unit; hence it has a strict upper bound $b_n \in \Gamma$. Choose $\gamma_{n+1} > \max\{b_n, p_{n+1}, \gamma_n\}$ and start with $\gamma_1 \geq p_1 > 0$. The resulting sequence has the required domination and is cofinal because it dominates the p_n . $\square$

8.2 Shells, normalized nodes, and the coarsened valuation rings

The rest of this section identifies, exactly, the image of the full jet map of a radially finite divisor. The objects it needs are finite algebras attached to one valuation shell at a time, together with one coarsened valuation ring per shell. They are introduced here. Throughout this subsection D is a radially finite effective divisor in the sense of [definition 7.1](#), and D is infinite unless the contrary is said; Γ is arbitrary, and in particular the radii below are *not* assumed to satisfy the Archimedean domination of [lemma 8.1](#).

Lemma 8.2 (Shell decomposition). *Let D be an infinite radially finite divisor. There is a unique decomposition*

$$D = D_0 + \sum_{n \geq 1} D_n, \quad D_n = \sum_{j=1}^{q_n} m_{n,j} [\rho_{n,j}], \quad v(\rho_{n,j}) = -\gamma_n,$$

in which D_0 is a finite divisor supported in $\mathcal{O}_\Gamma$, the q_n and $m_{n,j}$ are positive integers, the points $\rho_{n,1}, \dots, \rho_{n,q_n}$ are distinct, and

$$0 < \gamma_1 < \gamma_2 < \dots, \quad (\gamma_n) \text{ cofinal in } \Gamma.$$

Each shell total $M_n = \sum_{j \leq q_n} m_{n,j}$ is finite. In particular an infinite radially finite divisor forces $\text{cf}(\Gamma) = \aleph_0$.

Proof. Radial finiteness at $\gamma = 0$ makes D_0 , the restriction of D to $\mathcal{O}_\Gamma$, a finite divisor; it contains the occurrence at 0, whose valuation is $+\infty$. The remaining support has negative valuations, and their negatives form the set of positive radii. Only finitely many radii lie below any fixed bound, by radial finiteness, so every nonempty set of radii has a least member: choose one member and minimize over the finitely many members at most it. Listing the radii by repeatedly taking the next least member exhausts them, because each has only finitely many predecessors. The list is infinite and cofinal, since otherwise one ball would carry the whole infinite divisor. Each shell lies in a ball and therefore has finite total multiplicity. Uniqueness is the uniqueness of the valuation of a point. Cofinality of (γ_n) is the last assertion; it also follows from [lemma 7.2](#), which this lemma refines by grouping the enumeration there into shells. $\square$

Fix the normalizing monomials and the normalized shell data

$$\tau_n = t^{-\gamma_n}, \quad \xi_{n,j} = \rho_{n,j}/\tau_n, \quad P_n(T) = \prod_{j=1}^{q_n} (T - \xi_{n,j})^{m_{n,j}}. \quad (18)$$

Each $\xi_{n,j}$ is a unit of $\mathcal{O}_\Gamma$, so P_n is monic of degree M_n with coefficients in $\mathcal{O}_\Gamma$ and $P_n(0)$ a unit. Distinct normalized nodes need not have distinct residues in $\mathbb{C}$, and their differences may have arbitrarily large positive valuation; nothing below assumes otherwise.

A prescribed jet at $\rho_{n,j}$ is a polynomial $J_{n,j}(U) = \sum_{\ell < m_{n,j}} b_{n,j,\ell} U^\ell$ over K_Γ , the intended meaning being $F(\rho_{n,j} + U) \equiv J_{n,j}(U) \pmod{U^{m_{n,j}}}$. Finite Hermite interpolation over the field K_Γ supplies a unique *normalized shell Hermite polynomial* R_n with $\deg R_n < M_n$ and

$$R_n(\xi_{n,j} + U) \equiv J_{n,j}(\tau_n U) \pmod{U^{m_{n,j}}} \quad (1 \leq j \leq q_n). \quad (19)$$

It exists for arbitrary data, with no integrality assumption: the question answered below is whether its coefficients lie in a particular subring. If $F \in \mathcal{E}_\Gamma$ realizes the jets then $R_n = \text{rem}(F(\tau_n T), P_n)$, since $F(\tau_n(\xi_{n,j} + U)) = F(\rho_{n,j} + \tau_n U)$.

Definition 8.3 (Killing one Archimedean scale). For $\gamma > 0$ in Γ put

$$H_\gamma = \{\eta \in \Gamma : |\eta| \leq m\gamma \text{ for some } m \in \mathbb{N}\}, \quad (20)$$

$$V_\gamma = \{x \in K_\Gamma : v(x) \geq -m\gamma \text{ for some } m \in \mathbb{N}\}. \quad (21)$$

Here $\mathbb{N}$ contains 0, and $v(0) = +\infty$ puts 0 in V_γ . Write $H_n = H_{\gamma_n}$ and $V_n = V_{\gamma_n}$ for the radii of a fixed divisor.

H_γ is the convex subgroup of Γ generated by γ , and it is divisible because dividing by a positive integer does not increase absolute value. Both families increase with n , and $\bigcup_n H_n = \Gamma$, $\bigcup_n V_n = K_\Gamma$, because (γ_n) is cofinal. Membership in V_n is a one-sided bound; arbitrarily small elements are admitted. The unit condition is two-sided:

$$x \in V_n^\times \iff x \neq 0 \text{ and } |v(x)| \leq m\gamma_n \text{ for some } m \in \mathbb{N}. \quad (22)$$

Lemma 8.4 (Coarsened residues). V_γ is the valuation ring of the coarsening of v by H_γ , with value group Γ/H_γ . Its maximal ideal and residue field are

$$\mathfrak{q}_\gamma = \{x : v(x) > m\gamma \text{ for every } m \in \mathbb{N}\}, \quad \kappa_\gamma = V_\gamma/\mathfrak{q}_\gamma \cong \mathbb{C}((t^{H_\gamma})),$$

the residue map being restriction of the Hahn support to H_γ .

Proof. An element of V_γ has nonnegative class modulo H_γ , so all of its support exponents lie in nonnegative cosets. Restricting the support to the zero coset is additive, is surjective onto the full Hahn field on H_γ , and is multiplicative because two nonnegative cosets sum to zero only when both are zero. Its kernel consists of the series all of whose exponents exceed H_γ , which is the displayed $\mathfrak{q}_\gamma$. This also gives the valuation-ring and unit descriptions. $\square$

This is the same computation as [lemma 10.3](#), performed inside Γ by a convex subgroup rather than across an ordered-group extension; the two coarsenings are analogous, not the same object, and no statement is transferred between them. Both produce a residue field of the form $\mathbb{C}((t^H))$ rather than $\mathbb{C}$, and in both cases that is the point.

Define the two finite algebras of shell n :

$$\Lambda_n = K_\Gamma[T]/(P_n), \quad \Lambda_n^\circ = V_n[T]/(P_n) \subseteq \Lambda_n. \quad (23)$$

The inclusion is injective because division by a monic polynomial gives a unique representative of degree below M_n over either coefficient ring. A class in Λ_n therefore lies in Λ_n° exactly when every coefficient of its degree-bounded representative lies in V_n . Over K_Γ , finite Chinese remainders identify Λ_n with the product of the local jet rings of the shell, the identification with the original jets including the rescaling in [\(19\)](#). Over V_n the analogous decomposition may fail, because two normalized nodes can differ by a nonunit of V_n . That failure is the source of the close-node obstruction of [theorem 8.31](#), not a technical inconvenience to be discarded.

For the finite part set $\Lambda_0 = \prod_{x \in \text{supp } D_0} K_\Gamma[U]/(U^{D(x)})$, omitting the factor when D_0 is empty; the same convention is used in every formula below that names Λ_0 .

Definition 8.5 (The shellwise restricted product).

$$\mathcal{R}_D = \Lambda_0 \times \left\{ (r_n) \in \prod_{n \geq 1} \Lambda_n : r_n \in \Lambda_n^\circ \text{ for all sufficiently large } n \right\}, \quad (24)$$

equivalently the union over N of $\Lambda_0 \times \prod_{n < N} \Lambda_n \times \prod_{n \geq N} \Lambda_n^\circ$.

This is a ring under componentwise operations, and a K_Γ -algebra because a fixed scalar lies in V_n for all large n . The phrase “restricted product” here asserts *no* locally compact topology, no adelic structure and no Banach norm; it names the algebraic union just displayed and nothing else.

8.3 The exact image of the infinite jet map

Every finite projection of the jet map of a radially finite divisor is surjective, by ordinary finite Hermite interpolation over the field K_Γ . Neither a finite truncation nor a finite root calculation can therefore detect a global failure of surjectivity; see [section B](#). What the finite projections cannot see is a condition on the *tail* of the family of shells, and the following theorem states it.

Theorem 8.6 (Shellwise Hermite interpolation; the exact image). *Let D be an infinite radially finite divisor, with the shell data of [lemma 8.2](#), and prescribe arbitrary finite jets $J_{n,j}$ at all of its nodes together with an arbitrary element of Λ_0 at its finite part. Let R_n be the normalized shell Hermite polynomials [\(19\)](#). There is an $F \in \mathcal{E}_\Gamma$ realizing every prescribed jet if and only if*

$$R_n \in V_n[T] \quad \text{for all sufficiently large } n, \quad (25)$$

that is, if and only if for all sufficiently large n there is an integer $c_n \geq 0$ with

$$v([T^j]R_n) \geq -c_n \gamma_n \quad (0 \leq j < M_n).$$

No condition is imposed on D_0 or on finitely many exceptional shells, and no uniform bound on q_n , on $m_{n,j}$, on M_n or on c_n is required. Consequently the image of the full jet map

$$\text{ev}_D : \mathcal{E}_\Gamma \longrightarrow \Lambda_0 \times \prod_{n \geq 1} \Lambda_n$$

is exactly $\mathcal{R}_D$.

The condition does not depend on the choice of normalizing monomial. Replacing τ_n by any τ'_n of the same valuation replaces T by uT for a unit u of $\mathcal{O}_\Gamma$, which multiplies each coefficient by a power of a valuation-zero unit and leaves membership in $V_n[T]$ unchanged.

Theorem 8.7 (The entire quotient is the restricted product). *For a radially finite divisor D with canonical product P_D of [theorem 7.3](#), the shellwise jet map induces an isomorphism of K_Γ -algebras*

$$\mathcal{E}_\Gamma / (P_D) \cong \mathcal{R}_D,$$

so that $0 \rightarrow P_D \mathcal{E}_\Gamma \rightarrow \mathcal{E}_\Gamma \xrightarrow{\text{ev}_D} \mathcal{R}_D \rightarrow 0$ is exact. Every finite projection of ev_D is surjective; the full map into $\Lambda_0 \times \prod_n \Lambda_n$ need not be.

Proof. The image is [theorem 8.6](#). The kernel is $P_D \mathcal{E}_\Gamma$ by [corollaries 7.5](#) and [7.6](#): vanishing of all prescribed jets is exactly divisibility of the divisor. That ev_D is an algebra homomorphism is [proposition 3.5](#) together with the multiplicativity of the truncated jet map. Surjectivity of a finite projection is finite Hermite interpolation. $\square$

The kernel statement invoked in that proof is the report’s own [corollary 7.5](#), which is downstream of support-controlled preparation ([theorem 5.2](#)) and of algebraic closedness of K_Γ ([theorem 6.2](#)). A second and genuinely different proof of the same statement, restricted to radially finite canonical products, is recorded next. Both are kept. The second uses neither preparation nor algebraic closedness, which is what makes [proposition 8.42](#) available; the first proves far more, namely that *every* nonzero entire function is such a product ([theorem 7.4](#)), and that stronger statement does use algebraic closedness.

Proposition 8.8 (Principal vanishing ideal, second proof). *Let D be radially finite with canonical product P_D of [theorem 7.3](#), and let $F \in \mathcal{E}_\Gamma$. If every jet prescribed by D vanishes on F , then $F \in P_D \mathcal{E}_\Gamma$. The converse is the jet homomorphism. Let P denote the subproduct of P_D over the nodes of negative valuation, listed with multiplicity, and let $0 < \beta_1 \leq \beta_2 \leq \dots$ be the valuations $v(1/x)$ of the reciprocals of those nodes. The proof uses only the coefficient bound*

$$v([Z^k]P) \geq \beta_1 + \dots + \beta_k, \quad (26)$$

which also re-proves the entireness half of [theorem 7.3](#) for P without preparation, since for $k \geq 2$ the right-hand side divided by k is at least $\beta_{\lfloor k/2 \rfloor}/2$, and that tends cofinally to $+\infty$.

Proof. Divide first by $Z^{D(0)}$ and by the finitely many factors at nonzero nodes of nonnegative valuation, using [lemma 6.4](#); the quotient is entire and retains the required zeros at every remaining node. We may therefore replace P_D by P , which has $P(0) = 1$ and only negative-valuation nodes; note $\beta_i \rightarrow +\infty$ cofinally by [lemma 7.2](#).

Expanding P over finite subsets of the reciprocals $1/x$ gives (26): every degree- k term is a product of k of the listed reciprocals with distinct labels, so its valuation is at least the sum of the k smallest β_i , and the valuation of a Hahn sum is at least the minimum of its terms. Among $\beta_1, \dots, \beta_k$ more than half are at least $\beta_{\lfloor k/2 \rfloor}$ and the rest are positive, so the average is at least $\beta_{\lfloor k/2 \rfloor}/2$; [theorem 3.1](#) then gives entireness of P directly.

Since $P(0) = 1$ there is a unique formal quotient $G = F/P \in K_\Gamma[[Z]]$; we show $G \in \mathcal{E}_\Gamma$. Let $P_{\leq N}$ be the finite product over the first N shells and $F_N = F/P_{\leq N}$, entire by finitely many applications of [lemma 6.4](#). Every coefficient of the formal inverse $P_{\leq N}^{-1}$ is integral, because each factor $(1 - Z/x)^{-1}$ expands with $v(1/x) > 0$; so if $A \geq 0$ bounds $-v$ on the coefficients of F , then every coefficient of every F_N has valuation at least $-A$, *independently of N* . Write T_N for the formal inverse of the tail $P_{>N}$. By (26) applied to that tail, $v([Z^k]P_{>N}) \geq k\gamma_{N+1}$, and the reciprocal recursion gives the same bound for $[Z^k]T_N$. In $K_\Gamma[[Z]]$ we have $G = F_N T_N$.

Fix a target $\gamma \in \Gamma$ and choose $\theta > \max\{\gamma, 0\}$; choose N with $\gamma_{N+1} > 4\theta + 2A$. As F_N is entire, for all large j its coefficient has valuation at least $4j\theta$. Split the convolution for $[Z^k]G$ at $j = k/2$: a summand with $j \geq k/2$ has valuation at least $4j\theta \geq 2k\theta$, and a summand with $j < k/2$ has valuation at least

$$-A + (k - j)\gamma_{N+1} > -A + k\gamma_{N+1}/2 \geq k(\gamma_{N+1}/2 - A) > 2k\theta.$$

Hence $v([Z^k]G) > k\gamma$ for all large k ; cancellation only improves the bound. As γ was arbitrary, [theorem 3.1](#) gives $G \in \mathcal{E}_\Gamma$. $\square$

The theorem identifies the missing object rather than a necessary inequality inside it. It also explains why solving each finite Chinese remainder problem and passing to an unspecified limit is not enough: the relevant restricted product is in general a proper subring of the full product. The element $g \bmod f$ of [theorem 8.29](#) maps to an everywhere nonzero sequence, hence to a unit of the full product ring, but it is not a unit of $\mathcal{E}_\Gamma/(f)$: an inverse there would give representatives satisfying a Bézout identity. Thus the elementary implication “invertible at each node implies invertible in the global quotient” is exactly where unrestricted infinite Chinese remainder reasoning fails, and [theorem 8.26](#) below says precisely how much more is needed.

8.4 Proof of the exact image theorem

Two finite-algebra facts and one assembly lemma carry the proof.

Lemma 8.9 (Integral polynomial operations). *Let $P \in \mathcal{O}_\Gamma[T]$ be monic of degree M .*

- (a) *If every coefficient of $G \in K_\Gamma[T]$ has valuation at least B , then so does every coefficient of $\text{rem}(G, P)$. In particular rem maps $\mathcal{O}_\Gamma[T]$ into $\mathcal{O}_\Gamma[T]$.*
- (b) *Suppose in addition $P = \prod_j (T - \xi_j)^{m_j}$ with $v(\xi_j) = 0$, and let $Q(T) = \prod_h (T - \eta_h)^{m'_h}$ be a finite product with $v(\eta_h) > 0$, where $\eta_h = 0$ is permitted. Then T and Q are units of $\mathcal{O}_\Gamma[T]/(P)$, and for every integer $d \geq 0$ the inverse of $T^d Q$ has a representative in $\mathcal{O}_\Gamma[T]$ of degree below M .*

Proof. (a) Monic division subtracts only integral multiples of P , so the bound B survives every step.

(b) Consider multiplication on the free $\mathcal{O}_\Gamma$ -module $\mathcal{O}_\Gamma[T]/(P)$ with basis $1, T, \dots, T^{M-1}$; its matrices have entries in $\mathcal{O}_\Gamma$. After extending scalars to K_Γ , the determinant of multiplication by $L \in K_\Gamma[T]$ is

$$\det(m_L) = \prod_j L(\xi_j)^{m_j} = \text{Res}(P, L), \quad (27)$$

because in $K_\Gamma[U]/(U^{m_j})$ multiplication by $L(\xi_j + U)$ is triangular with constant diagonal $L(\xi_j)$, and finite Chinese remainders over K_Γ decompose Λ_n accordingly. For $L = T$ each ξ_j is a unit of $\mathcal{O}_\Gamma$. For $L = Q$ each $\xi_j - \eta_h$ has valuation zero, since $v(\xi_j) = 0 < v(\eta_h)$. The determinants are therefore units of $\mathcal{O}_\Gamma$, and the adjugate supplies inverses over $\mathcal{O}_\Gamma$. Multiplying the two inverses gives the assertion for $T^d Q$. $\square$

No lower bound on $v(\xi_j - \xi_k)$ for two nodes of the *current* shell was used; they may be separated only at an arbitrarily higher scale. Only the already-treated nodes, of strictly smaller radius, enter Q , and they are uniformly separated from the current shell after normalization. This is the one place where the same finite algebra that creates the close-node obstruction also makes the construction possible.

Lemma 8.10 (Assembly). *Let $C_n \in K_\Gamma[Z]$ be polynomials with $Z^n \mid C_n$, and suppose that for every $\gamma \in \Gamma$ the weighted Gauss valuations $m_\gamma(C_n)$ tend cofinally to $+\infty$. Then for any polynomial S the series $F = S + \sum_n C_n$ is a well-defined element of $\mathcal{E}_\Gamma$, every coefficient of F being a finite sum, and evaluation and each fixed finite jet may be interchanged with the sum over n .*

Proof. Only C_n with $n \leq k$ contributes to the coefficient of Z^k , so each coefficient is a finite sum. Fix γ and B . All but finitely many C_n have $m_\gamma(C_n) > B$, and the finitely many exceptions have finite degree; beyond those degrees every weighted coefficient of F exceeds B . This is [theorem 3.1\(4\)](#), so $F \in \mathcal{E}_\Gamma$. For evaluation at b and for a fixed jet order, choose $\gamma \leq v(b)$; estimate [\(6\)](#) makes the family of jet coefficients cofinally summable, and the full family of evaluated monomials is strongly summable because below a fixed exponent only finitely many polynomials contribute, each with finitely many terms. [Lemma 2.2](#) and regrouping give the interchange. $\square$

Proof of necessity in [theorem 8.6](#). Let $F = \sum_k a_k Z^k \in \mathcal{E}_\Gamma$ be nonzero and fix n . Put $E_{n,k}(T) = \text{rem}(T^k, P_n)$, whose coefficients lie in $\mathcal{O}_\Gamma$ by [lemma 8.9\(a\)](#). By [theorem 3.1](#) the valuations $v(a_k \tau_n^k) = v(a_k) - k\gamma_n$ tend cofinally to $+\infty$, so

$$\text{rem}_n(F) = \sum_{k \geq 0} a_k \tau_n^k E_{n,k}(T) \quad (28)$$

is a polynomial of degree below M_n whose coefficients are strong Hahn sums ([lemma 2.2](#)). At each normalized node its finite jets agree with those of $F(\tau_n T)$: the finite remainder identity holds term

by term, and jet evaluation at the integral nodes preserves the cofinal estimates by (6). Hence $\text{rem}_n(F)$ is the normalized shell Hermite polynomial R_n of the jets of F .

By strong summability at $Z = 1$ there is a fixed $A \geq 0$ with $v(a_k) \geq -A$ for every k . Every coefficient of (28) has valuation at least

$$m_{-\gamma_n}(F) = \min_k \{v(a_k) - k\gamma_n\},$$

a minimum attained at a finite index k_n , which may depend on n . For all n with $\gamma_n > A$ this gives

$$m_{-\gamma_n}(F) \geq -A - k_n\gamma_n \geq -(k_n + 1)\gamma_n,$$

so every coefficient of R_n lies in V_n . The zero series is immediate, and D_0 imposes nothing. $\square$

The attained minimum is load-bearing. An infimum of infinitely many estimates lying strictly above a cut need not itself lie above that cut in a non-Archimedean ordered group; here there is an actual minimizing index, not an appeal to an unproved compactness principle. This is the same point as in lemma 8.22 and is recorded again in section A.

For the converse, suppose a polynomial S already fits every node of radius smaller than γ_n , together with D_0 . Put

$$Q_n(T) = \prod_{a \text{ already treated}} (T - a/\tau_n)^{D(a)}, \quad (29)$$

$$\Theta_n(T) = R_n(T) - \text{rem}(S(\tau_n T), P_n). \quad (30)$$

Every already-treated node has strictly smaller radius, so each a/τ_n has positive valuation or is zero, and Q_n satisfies the hypothesis of lemma 8.9(b). For an integer $d \geq 0$ set

$$\Psi_{n,d}(T) = \text{rem}\left(\Theta_n(T)(T^d Q_n(T))^{-1}, P_n\right), \quad (31)$$

$$C_{n,d}(Z) = (Z/\tau_n)^d Q_n(Z/\tau_n) \Psi_{n,d}(Z/\tau_n), \quad (32)$$

the inverse in (31) being taken in Λ_n with the integral representative supplied by lemma 8.9(b). Then every earlier prescribed jet of $C_{n,d}$ vanishes, the current-shell remainder of $C_{n,d}$ is exactly Θ_n , so $S + C_{n,d}$ fits shell n as well, and Z^d divides $C_{n,d}$. The parameter d changes none of these identities; it is free, and is used only for size.

Lemma 8.11 (Uniform damping estimate). *Suppose every coefficient of Θ_n has valuation at least B . Then the same holds for $\Psi_{n,d}$, independently of d , and*

$$m_{-\gamma_n/2}(C_{n,d}) \geq B + d\gamma_n/2. \quad (33)$$

If $\Theta_n \in V_n[T]$, so that $B = -c\gamma_n$ may be taken with an integer $c \geq 0$, then every $d \geq \max\{n, 2c + 2\}$ gives

$$m_{-\gamma_n/2}(C_{n,d}) \geq \gamma_n. \quad (34)$$

Proof. Multiplication by an integral inverse and integral remainder reduction preserve the lower bound B , by lemma 8.9. Under $T = Z/\tau_n = t^{\gamma_n} Z$ the monomial T has weighted value $\gamma_n - \gamma_n/2 = \gamma_n/2 > 0$ at $\gamma = -\gamma_n/2$; the substituted Q_n has weighted value at least 0 because its coefficients are integral; and the substituted $\Psi_{n,d}$ has weighted value at least B . Adding these, by the product estimate of section 2.5, gives (33). For the last assertion, $-c\gamma_n + d\gamma_n/2 \geq \gamma_n$ holds as soon as $d \geq 2c + 2$. $\square$

The estimate never uses $d\gamma_n \rightarrow +\infty$ in Γ for fixed n ; that assertion is false when γ_n is not an order unit, and it is exactly the false step that [lemma 8.1](#) forbids. The finite damping needed at stage n lives inside the convex subgroup H_n generated by γ_n . It is the stages, not the powers at one stage, that traverse the cofinal hierarchy.

Proof of sufficiency in [theorem 8.6](#). Suppose $R_n \in V_n[T]$ for every $n > N$. Fit D_0 and the first N shells by a finite Hermite polynomial S_N over K_Γ . Since S_N has finitely many coefficients and $\bigcup_n V_n = K_\Gamma$, there is an $L > N$ with all coefficients of S_N in V_L .

Treat the finitely many shells $N < n \leq L$ by (29)–(32) with $d = 0$. Throughout these steps the coefficients stay in V_L : for such n we have $R_n \in V_n[T] \subseteq V_L[T]$, and both τ_n and τ_n^{-1} are units of V_L because $\gamma_n \in H_L$, so normalizing, taking integral remainders and multiplying by the integral inverse of [lemma 8.9\(b\)](#) all preserve V_L -coefficients. This produces a polynomial S_L fitting D_0 and the first L shells, with coefficients in V_L . Choose $A \geq 0$ in H_L — a sufficiently large finite multiple of γ_L will do — so that every coefficient of S_L has valuation at least $-A$. This one bound is maintained for the rest of the construction.

Let $n > L$ and suppose S_{n-1} fits everything of radius below γ_n and has all coefficient valuations at least $-A$. Since $A \in H_L \subseteq H_n$, all coefficients of S_{n-1} lie in V_n , and τ_n is a unit of V_n ; hence Θ_n of (30) lies in $V_n[T]$. Choose $d_n \geq n$ by [lemma 8.11](#) so that $C_n = C_{n,d_n}$ satisfies $m_{-\gamma_n/2}(C_n) \geq \gamma_n$, and set $S_n = S_{n-1} + C_n$. Then every coefficient $c_{n,k}$ of C_n obeys

$$v(c_{n,k}) \geq \gamma_n + k\gamma_n/2 \geq \gamma_n > 0, \quad (35)$$

so the bound $-A$ is preserved and the induction continues. A zero residual is handled by $C_n = 0$.

Put $F = S_L + \sum_{n>L} C_n$. Since Z^n divides C_n , each coefficient is a finite sum. For fixed γ we eventually have $\gamma \geq -\gamma_n/2$, and m_γ is nondecreasing, so

$$m_\gamma(C_n) \geq m_{-\gamma_n/2}(C_n) \geq \gamma_n \longrightarrow +\infty \quad \text{cofinally.}$$

[Lemma 8.10](#) makes F entire and lets every fixed finite jet pass through the sum. At any prescribed node the required jet is attained at a finite stage, and every later correction has zero jet there. Hence F realizes all the prescribed data. $\square$

Remark 8.12 (Why the finite initialization is not a formality). An exceptional early jet may carry a coefficient at a scale larger than many later radii. “Absorb finitely many exceptions into a polynomial” does not by itself justify the later damping estimates, which need one permanent lower bound. The intermediate finite run over V_L , followed by the single fixed bound $-A \in H_L$, is what prevents such a coefficient from propagating into an uncontrolled obstruction. This is the only initialization issue, and no uniform bound on the later jet orders is required.

Corollary 8.13 (Exact one-node interpolation). *Suppose each shell carries one node ρ_n , of multiplicity m_n , and prescribe jets $\sum_{\ell < m_n} b_{n,\ell}(Z - \rho_n)^\ell$. These are realized by an entire function if and only if, for all sufficiently large n ,*

$$b_{n,\ell} \in V_n \quad \text{for every } \ell < m_n.$$

For values alone ($m_n = 1$ for all n) this reads: the prescription $F(\rho_n) = b_n$ is realizable if and only if $b_n \in V_n$ eventually, that is, if and only if for all large n there is an integer $c_n \geq 0$ with $v(b_n) \geq -c_n\gamma_n$.

Proof. The normalized Hermite polynomial is $R_n(T) = \sum_{\ell < m_n} b_{n,\ell} \tau_n^\ell (T - \xi_n)^\ell$. Both τ_n and ξ_n are units of V_n , and the change between powers of T and powers of $T - \xi_n$ is given by mutually inverse triangular matrices over V_n . Hence $R_n \in V_n[T]$ if and only if every $b_{n,\ell}$ lies in V_n . Apply [theorem 8.6](#). $\square$

Corollary 8.14 (Coarsely separated shells). *Suppose that on every sufficiently large shell the differences $\xi_{n,j} - \xi_{n,k}$, $j \neq k$, are units of V_n . Then arbitrary prescribed jets are realizable exactly when all of their coefficients, in the original coordinate, lie in V_n eventually.*

Proof. Under that hypothesis the ideals $(T - \xi_{n,j})^{m_{n,j}}$ are already pairwise comaximal in $V_n[T]$: two linear ideals are comaximal because the difference of their generators is a unit, and their powers are comaximal by expanding a sufficiently high power of an identity $u + w = 1$. The finite Chinese remainder isomorphism therefore holds over V_n , and rescaling jets by powers of the V_n -unit τ_n does not change membership. Apply [theorem 8.6](#). $\square$

The hypothesis is *coarsened* separation, not separation in v . Normalized differences may have arbitrarily large positive fine valuation, provided that valuation still lies in H_n . Without it the conclusion is false, by [theorem 8.31](#).

Corollary 8.15 (Order units and the disappearance of the restriction). *If Γ has an order unit h , then $H_\gamma = \Gamma$ and $V_\gamma = K_\Gamma$ for every $\gamma \geq h$, so $V_n = K_\Gamma$ and $\Lambda_n^\circ = \Lambda_n$ for all large n . Condition (25) is then vacuous, $\mathcal{R}_D$ is the full product, and every radially finite Hermite prescription is realized.*

Proof. An order unit makes every element of Γ bounded in absolute value by a finite multiple of h , hence of any $\gamma \geq h$; a cofinal (γ_n) eventually exceeds h . Apply [theorems 8.6](#) and [8.7](#). $\square$

This recovers [theorem 9.2](#) and is discussed again in [section 9](#), where the independent Gauss-norm proof is retained. At the other extreme, if $\text{cf}(\Gamma) > \aleph_0$ then [lemma 8.2](#) leaves no infinite radially finite divisor at all and every interpolation problem here is finite. The new content lies in the intervening groups.

Corollary 8.16 (Without an order unit the restriction never disappears). *Conversely, if Γ has no order unit, then every H_n is a proper subgroup of Γ . In particular, for a divisor with one simple node ρ_n on each shell and empty finite part, the image of the value map $F \mapsto (F(\rho_n))_n$ is a proper subring of $K_\Gamma^{\mathbb{N}_{>0}}$.*

Proof. If $H_n = \Gamma$, every element of Γ is bounded in absolute value by a finite multiple of γ_n , so γ_n is an order unit. If every H_n is proper, choose $\eta_n > 0$ outside H_n . It exceeds every finite multiple of γ_n , so $t^{-\eta_n} \notin V_n$ for every n , and [corollary 8.13](#) shows that the values $F(\rho_n) = t^{-\eta_n}$ are realized by no entire F . $\square$

This converse is the fourth source's; [corollary 8.23\(a\)](#) is its instance $\eta_n = \gamma_{n+1}$ for the escaping sequence of [lemma 8.1](#).

8.5 A second proof of sufficiency: cardinal corrections

The fourth source proved [theorem 8.6](#) independently. Its necessity argument is the one above, an attained Gauss minimum. Its sufficiency construction is different, and is recorded here as a second route. The first route corrects shell by shell with *polynomials* that vanish at every earlier node,

and needs the finite initialization over V_L . The second builds, for each shell separately, an *entire* correction that vanishes to the prescribed orders on every other shell: the canonical product with that shell removed, times a finite polynomial. One integral estimate for the inverse of that cofactor on its own shell replaces the sequential bookkeeping, and the construction comes with an explicit truncation bound. Neither route is offered as a correction of the other, and like the first, the second uses neither preparation nor algebraic closedness.

Throughout this subsection D is an infinite radially finite divisor with the shell data of [lemma 8.2](#), and $D_0 = 0$ until the last step. Put

$$\Pi_n(Z) = \frac{P_n(Z/\tau_n)}{P_n(0)} = \prod_{j=1}^{q_n} \left(1 - \frac{Z}{\rho_{n,j}}\right)^{m_{n,j}}, \quad P_D = \prod_{n \geq 1} \Pi_n, \quad \hat{P}_n = \prod_{h \neq n} \Pi_h. \quad (36)$$

The cofactor $\hat{P}_n$ is the canonical product of the radially finite divisor $D - D_n$, entire by [theorem 7.3](#); it is obtained by removing finitely many factors, not by an entire division theorem. For a class $r \in \Lambda_n$ write $\nu(r)$ for the least valuation of the coefficients of its representative of degree below M_n , with $\nu(0) = +\infty$. Monic division never lowers that minimum ([lemma 8.9\(a\)](#)), so $\nu(rs) \geq \nu(r) + \nu(s)$. For $F \in \mathcal{E}_\Gamma$ let $\text{rem}_n(F) \in \Lambda_n$ be the shell class ([28](#)). It is the n th component of ev_D , hence a K_Γ -algebra homomorphism, and the necessity proof above shows

$$\nu(\text{rem}_n(F)) \geq m_{-\gamma_n}(F). \quad (37)$$

Finally put

$$\varpi_n = \text{rem}_n(\hat{P}_n) \in \Lambda_n, \quad \varkappa_n = \sum_{h < n} M_h(\gamma_n - \gamma_h) \geq 0. \quad (38)$$

Lemma 8.17 (Summing entire corrections). *Let $F_n \in \mathcal{E}_\Gamma$ and suppose that $m_\gamma(F_n) \rightarrow +\infty$ cofinally for every $\gamma \in \Gamma$. Then the coefficientwise sum $F = \sum_n F_n$ is defined and belongs to $\mathcal{E}_\Gamma$, $m_\gamma(F - \sum_{n \leq N} F_n) \rightarrow +\infty$ for every γ , and $\text{rem}_m(F) = \sum_n \text{rem}_m(F_n)$ for every shell m , the right side being a coefficientwise strong sum.*

Proof. At $\gamma = 0$ the hypothesis makes the valuations in each coefficient family $([Z^k]F_n)_n$ tend cofinally to $+\infty$, so [lemma 2.2](#) defines its sum. Fix γ and B . All sufficiently late F_n have every weighted coefficient above B , hence so does the sum of those late terms, and the finitely many earlier F_n are entire, so they have this property in all sufficiently large degrees. Thus the weighted coefficients of F tend cofinally to $+\infty$, which is [theorem 3.1\(4\)](#); the same argument applied to the late terms alone gives the tail assertion. For the last assertion, (37) makes the coefficient families of $(\text{rem}_m(F_n))_n$ cofinally small, and $\nu(\text{rem}_m(F) - \sum_{n \leq N} \text{rem}_m(F_n)) \geq m_{-\gamma_m}(F - \sum_{n \leq N} F_n) \rightarrow +\infty$; a class whose ν exceeds every element of Γ is zero. $\square$

Compare [lemma 8.10](#): there the corrections are polynomials divisible by Z^n ; here they are entire functions, and only their weighted Gauss valuations are controlled.

Lemma 8.18 (Integral inverse of the cofactor). *The class ϖ_n is a unit of Λ_n , and*

$$\nu(\varpi_n^{-1}) \geq \varkappa_n. \quad (39)$$

Proof. Write $A_n^{\mathcal{O}_\Gamma} = \mathcal{O}_\Gamma[T]/(P_n)$, a finite free $\mathcal{O}_\Gamma$ -algebra inside Λ_n in which T is a unit, by [lemma 8.9\(b\)](#). Every element of $1 + \mathfrak{m}_\Gamma A_n^{\mathcal{O}_\Gamma}$ is a unit of $A_n^{\mathcal{O}_\Gamma}$: the determinant of its multiplication matrix in the basis $1, T, \dots, T^{M_n-1}$ is 1 modulo $\mathfrak{m}_\Gamma$, hence a unit of $\mathcal{O}_\Gamma$, and the adjugate gives an integral inverse. No geometric series is used.

Split $\widehat{P}_n = P_{<n}P_{>n}$ into the finite product over earlier shells and the canonical product of the later ones. For a node $\rho = \rho_{h,j}$ of an earlier shell, the class of its factor is

$$1 - \frac{\tau_n}{\rho}T = -\frac{\tau_n}{\rho}T \left(1 - \frac{\rho}{\tau_n}T^{-1}\right), \quad v(\rho/\tau_n) = \gamma_n - \gamma_h > 0,$$

so the last factor lies in $1 + \mathfrak{m}_\Gamma A_n^{\mathcal{O}_\Gamma}$. For a node ρ of a later shell h the class $1 - (\tau_n/\rho)T$ itself lies in $1 + \mathfrak{m}_\Gamma A_n^{\mathcal{O}_\Gamma}$, since $v(\tau_n/\rho) = \gamma_h - \gamma_n > 0$; so do the classes u_N of the finite partial products of $P_{>n}$. Those partial products converge to $P_{>n}$ in every m_γ ([theorem 7.3](#)), so by (37) some u_N satisfies $\nu(\text{rem}_n(P_{>n}) - u_N) > 0$, and then $\text{rem}_n(P_{>n})$ lies in $1 + \mathfrak{m}_\Gamma A_n^{\mathcal{O}_\Gamma}$ as well. Hence, with

$$c_n = \prod_{h < n} \prod_{j=1}^{q_h} (-\tau_n/\rho_{h,j})^{m_{h,j}}, \quad v(c_n) = -\varkappa_n,$$

the class ϖ_n/c_n is a power of the integral unit T times an element of $1 + \mathfrak{m}_\Gamma A_n^{\mathcal{O}_\Gamma}$. Its inverse is integral, and multiplying by c_n^{-1} gives (39). $\square$

The argument never inverts a difference of two nodes of the *same* shell. Those may be arbitrarily close, and whatever they cost is already measured by R_n . Only the strict radial separation of distinct shells enters, exactly as in [lemma 8.9\(b\)](#).

Lemma 8.19 (Cardinal corrections). *Let $r \in \Lambda_n$ with representative R of degree below M_n , let $d \geq 0$ be an integer, and put*

$$\Xi_{n,d}(T) = \text{rem}(T^{-d}\varpi_n^{-1}R, P_n), \quad (40)$$

$$\widehat{C}_{n,d}(Z) = \widehat{P}_n(Z) (Z/\tau_n)^d \Xi_{n,d}(Z/\tau_n), \quad (41)$$

the inverses being taken in Λ_n . Then $\widehat{C}_{n,d} \in \mathcal{E}_\Gamma$, $\text{rem}_n(\widehat{C}_{n,d}) = r$, every prescribed jet of D at a node outside shell n vanishes on $\widehat{C}_{n,d}$, and for every $\gamma \geq -\gamma_n/2$

$$m_\gamma(\widehat{C}_{n,d}) \geq m_\gamma(P_D) + \nu(r) + \varkappa_n + d(\gamma_n + \gamma). \quad (42)$$

Proof. It is an entire function times a polynomial. Its class on shell n is $\varpi_n \cdot T^d \cdot \Xi_{n,d} = r$, and on any other shell h it vanishes to the prescribed orders because $\widehat{P}_n$ contains the factor Π_h . By [lemma 8.18](#) and integral multiplication, $\nu(\Xi_{n,d}) \geq \nu(r) + \varkappa_n$. Under $T = Z/\tau_n = t^{\gamma_n}Z$ the coefficient of Z^k acquires the factor $t^{k\gamma_n}$, so its weighted value at γ exceeds the valuation of the original coefficient by $k(\gamma_n + \gamma) \geq 0$. Hence $m_\gamma(\Xi_{n,d}(Z/\tau_n)) \geq \nu(\Xi_{n,d})$, and $m_\gamma((Z/\tau_n)^d) = d(\gamma_n + \gamma)$. Next, $m_\gamma(\Pi_n) = 0$ for $\gamma > -\gamma_n$: the constant term is 1, and the coefficient of Z^k , $k \geq 1$, is a sum of products of k reciprocals of nodes of valuation γ_n , so its weighted value is at least $k(\gamma_n + \gamma) > 0$. The equality case of Gauss multiplicativity ([section 2.5](#)) applied to $P_D = \Pi_n \widehat{P}_n$ gives $m_\gamma(\widehat{P}_n) = m_\gamma(P_D)$. The product estimate now gives (42). $\square$

Second proof of sufficiency in theorem 8.6, for $D_0 = 0$. Suppose $R_n \in V_n[T]$ for all $n > N_0$. For such n with $R_n \neq 0$ choose an integer $c_n \geq 0$ with $\nu(R_n) \geq -c_n\gamma_n$ and put

$$d_n = 2c_n + 2; \quad (43)$$

for $n \leq N_0$ put $d_n = 0$. Let $\widehat{C}_n = \widehat{C}_{n,d_n}$ be the cardinal correction of the class of R_n , and $\widehat{C}_n = 0$ if $R_n = 0$. For $n > N_0$ and $\gamma \geq -\gamma_n/2$, [lemma 8.19](#) and $\varkappa_n \geq 0$ give

$$m_\gamma(\widehat{C}_n) \geq m_\gamma(P_D) - c_n\gamma_n + (2c_n + 2)\frac{\gamma_n}{2} = m_\gamma(P_D) + \gamma_n.$$

For fixed γ the condition $\gamma \geq -\gamma_n/2$ holds for all large n , and $m_\gamma(P_D) + \gamma_n \rightarrow +\infty$ cofinally. By [lemma 8.17](#), $F = \sum_n \hat{C}_n$ lies in $\mathcal{E}_\Gamma$ and $\text{rem}_m(F) = \sum_n \text{rem}_m(\hat{C}_n) = R_m$ for every m , since exactly one summand has a nonzero class on shell m . By the identification in the necessity proof, F realizes every prescribed jet. $\square$

Corollary 8.20 (An explicit truncation certificate). *In that construction, if $L \geq N_0$ and $\gamma \geq -\gamma_{L+1}/2$, then*

$$m_\gamma\left(F - \sum_{n \leq L} \hat{C}_n\right) \geq m_\gamma(P_D) + \gamma_{L+1}.$$

Arbitrarily large shell degrees M_n and integers c_n do not weaken it.

Proof. Every remaining summand satisfies the bound of the preceding proof with $\gamma_n \geq \gamma_{L+1}$, and a strong sum has weighted valuation at least the least weighted valuation of its terms. $\square$

Completion of the second route: a nonzero finite part. Let $P_0 = Z^{D(0)} \prod_x (Z - x)^{D(x)}$, the product over the nonzero points of $\text{supp } D_0$, and let F_0 be a polynomial with the prescribed jets on D_0 . A solution has the form $F = F_0 + P_0 G$, where G must carry the shell data $(R_n - \text{rem}_n(F_0)) \text{rem}_n(P_0)^{-1}$. Each class $\text{rem}_n(P_0)$ is a unit of Λ_n° : it is $\tau_n^{\deg P_0}$ times $T^{\deg P_0}$ times the classes of the factors $1 - (x/\tau_n)T^{-1}$, where $v(x/\tau_n) = v(x) + \gamma_n > 0$ because $v(x) \geq 0$, and τ_n is a unit of V_n . Since $\text{rem}_n(F_0) \in \Lambda_n^\circ$ for large n by the necessity half, the new data lie in $\mathcal{R}_D$ exactly when the original data do. Apply the case $D_0 = 0$ to $D - D_0$ to obtain G . The kernel half of [theorem 8.7](#) is [proposition 8.8](#), which the fourth source proved in the same way, by coefficient division without preparation. $\square$

Remark 8.21 (One simple node per shell). If shell n carries one simple node ρ_n and the data are values b_n , then $\Lambda_n = K_\Gamma$, $\text{rem}_n(G) = G(\rho_n)$, and the cardinal correction is

$$\hat{C}_{n,d}(Z) = b_n \frac{\hat{P}_n(Z)}{\hat{P}_n(\rho_n)} \left(\frac{Z}{\rho_n}\right)^d.$$

For instance, in the group Γ_∞ of [section 14](#), with $\rho_n = t^{-e_n}$, the admissible values $b_n = t^{-n!e_n}$ of [corollary 8.23\(b\)](#) are interpolated explicitly by

$$F(Z) = \sum_{n \geq 1} t^{-n!e_n} \frac{\hat{P}_n(Z)}{\hat{P}_n(t^{-e_n})} (t^{e_n} Z)^{2n!+2},$$

whereas the values $t^{-e_{n+1}}$ are admissible at no shell. For a single shell, d_n is an ordinary finite exponent: it absorbs a negative coefficient valuation bounded by a finite multiple of γ_n , but not one below every such multiple. That is why coarsening by H_n is both necessary and sufficient, rather than a convenient growth condition.

8.6 A universal bound on values at escaping monomials

The next lemma was the original core obstruction, and its necessity half is still what the negative results use. It is short, but it uses both halves of [definition 1.1](#): a common lower bound on the coefficient supports, and legitimate evaluation at the large argument. [Theorem 8.6](#) now supplies its converse, so the bound can be stated as an exact criterion.

Lemma 8.22 (Archimedean growth barrier, exact form). *Let $(\gamma_n)_{n \geq 1}$ be positive and strictly increasing with cofinal range, so that in particular $\gamma_n \rightarrow +\infty$ cofinally.*

- (a) (Necessity.) *For every $F \in \mathcal{E}_\Gamma$ there is an n_0 such that $F(t^{-\gamma_n}) \in V_n$ for all $n \geq n_0$; that is, for each such n some integer $c_n \geq 0$ satisfies $v(F(t^{-\gamma_n})) \geq -c_n\gamma_n$. In particular, if $\lambda_n > 0$ satisfies $\lambda_n \gg_A \gamma_n$ for every sufficiently large n , then eventually*

$$v(F(t^{-\gamma_n})) > -\lambda_n. \quad (44)$$

The value $+\infty$ is allowed when F vanishes at the argument.

- (b) (Sufficiency.) *Conversely, assume (γ_n) is strictly increasing, and let $b_n \in K_\Gamma$ satisfy $b_n \in V_n$ for all sufficiently large n . Then there is an $F \in \mathcal{E}_\Gamma$ with $F(t^{-\gamma_n}) = b_n$ for every $n \geq 1$, with no condition at all on the finitely many exceptional indices.*

Hence for a strictly increasing positive cofinal (γ_n) the prescription $F(t^{-\gamma_n}) = b_n$ is solvable in $\mathcal{E}_\Gamma$ if and only if $b_n \in V_n$ eventually.

Proof. (a) The assertion is immediate for $F = 0$. Write $F = \sum_{k \geq 0} a_k Z^k$ and let $m = \min_{a_k \neq 0} v(a_k)$, which exists by strong summability at $Z = 1$. The terms' valuations at $Z = t^{-\gamma_n}$ tend cofinally to infinity by [theorem 3.1](#), so their nonempty set has a minimum, attained at a finite index k_n , and the valuation of their Hahn sum is at least that minimum. For large n we have $\gamma_n > |m|$ and therefore

$$v(F(t^{-\gamma_n})) \geq m - k_n\gamma_n > -(k_n + 1)\gamma_n,$$

which is the first assertion with $c_n = k_n + 1$. If moreover $\lambda_n \gg_A \gamma_n$, then $-(k_n + 1)\gamma_n > -\lambda_n$, which is (44).

(b) The points $t^{-\gamma_n}$ are distinct and their divisor is radially finite with one simple node per shell, so [corollary 8.13](#) applies and supplies the interpolant. $\square$

Part (a) is the form used in [theorem 8.29](#) and in [corollary 8.23](#); it is elementary and independent of the constructive half of [theorem 8.6](#). Part (b) is not elementary, and the dependency is recorded in [section A](#). The monotonicity hypothesis is stated because the proof of (a) needs γ_n to exceed a fixed bound *eventually*, which a merely cofinal set of values does not supply; every application in this report — [lemma 8.1](#) and [lemma 8.2](#) — produces a strictly increasing sequence. No conclusion has been weakened by making the hypothesis explicit.

The eventual condition on γ_n cannot be replaced by cofinality of its range. In the reverse-lexicographic group $\bigoplus_{j \geq 0} \mathbb{Q}e_j$, set $\gamma_{2n} = e_{n+2}$, $\gamma_{2n+1} = e_0$ and $\lambda_{2n} = e_{n+3}$, $\lambda_{2n+1} = e_1$ for $n \geq 0$. The range of (γ_n) is cofinal and $\lambda_n \gg_A \gamma_n$ always, but the constant entire function $H = t^{-e_2}$ violates the conclusion at every odd index. The strictly increasing sequences used here avoid this failure.

Corollary 8.23 (Explicit failure of value interpolation, and its exact boundary). *Assume $\text{cf}(\Gamma) = \aleph_0$ and Γ has no order unit, and take the sequence (γ_n) of [lemma 8.1](#).*

- (a) (The negative half.) *The prescription*

$$F(t^{-\gamma_n}) = t^{-\gamma_{n+1}} \quad (n \geq 1)$$

has no solution $F \in \mathcal{E}_\Gamma$, although each finite subcollection of these prescriptions has a polynomial solution.

- (b) (The positive half.) *For every sequence (d_n) of nonnegative integers the prescription $F(t^{-\gamma_n}) = t^{-d_n\gamma_n}$ does have an entire solution. No rate restriction on (d_n) is imposed. More generally one may prescribe, at each node, finitely many derivatives whose number grows arbitrarily with n , provided each shell's finitely many coefficients satisfy the same scale condition.*

Together, (a) and (b) are the two sides of the criterion of [lemma 8.22](#): what an entire function may not do is jump repeatedly to a strictly larger Archimedean class, while arbitrarily high finite powers of the current scale are always available.

Proof. (a) The prescribed values have valuation $-\gamma_{n+1}$, and $\gamma_{n+1} \gg_A \gamma_n$, so they lie in no V_n ; this contradicts [lemma 8.22\(a\)](#). The nodes are distinct, so ordinary finite Lagrange interpolation over the field K_Γ realizes any finite subcollection. (b) Each $t^{-d_n \gamma_n}$ lies in V_n by definition, so [lemma 8.22\(b\)](#) applies; the jet version is [corollary 8.13](#). $\square$

This failure is not caused by nodes accumulating in a bounded ball: their divisor is radially finite and is therefore the divisor of an entire function by [theorem 7.3](#). Realizability of zeros is strictly weaker than realizability of values, and [theorem 8.6](#) measures the difference exactly.

8.7 Two functions with no common zero and no Bézout identity

The quotient description now decides the Bézout question for a canonical product against an arbitrary entire second factor. Two finite unit computations are needed first.

Lemma 8.24 (Units of a shell algebra). *For $R \in V_n[T]$ of degree below M_n the following are equivalent: its class is a unit of Λ_n° ; $\text{Res}(P_n, R)$ is a unit of V_n ; every value $R(\xi_{n,j})$ is a unit of V_n .*

Proof. Multiplication by R on the finite free V_n -module Λ_n° is invertible if and only if its determinant is a unit, by determinants in one direction and the adjugate in the other. Formula (27) gives $\text{Res}(P_n, R) = \prod_j R(\xi_{n,j})^{m_{n,j}}$. All the $\xi_{n,j}$ and all the coefficients lie in V_n , so every factor lies in that local ring, and a finite product in a local ring is a unit exactly when each factor is. The case of a vanishing factor is included. $\square$

The statement keeps repeated roots and does not require the normalized nodes to have distinct residues. The determinant appears here as an exact finite-algebra unit test, not as an approximate stability threshold.

Lemma 8.25 (Units of the restricted product). *An element $(r_0, r_1, r_2, \dots)$ of $\mathcal{R}_D$ is a unit if and only if r_0 is a unit of Λ_0 , every r_n is a unit of Λ_n , and $r_n \in (\Lambda_n^\circ)^\times$ for all sufficiently large n .*

Proof. A ring inverse must be the componentwise inverse. If both the element and its inverse lie in $\mathcal{R}_D$, then both components lie in Λ_n° for all large n , making those components units there. Conversely those conditions put the componentwise inverse in $\mathcal{R}_D$. $\square$

Theorem 8.26 (Sharp Bézout criterion). *Let D be radially finite with canonical product P_D , and let $F \in \mathcal{E}_\Gamma$. There are $A, B \in \mathcal{E}_\Gamma$ with*

$$AP_D + BF = 1 \tag{45}$$

if and only if both of the following hold:

- (1) $F(x) \neq 0$ at every node x of D ;
- (2) for all sufficiently large occupied shells n there is a finite integer $c_n \geq 0$ with

$$v(F(\rho_{n,j})) \leq c_n \gamma_n \quad (1 \leq j \leq q_n). \tag{46}$$

For a finite divisor the second condition is vacuous, and for a Γ with an order unit it is automatic. Moreover the criterion depends only on the support of D and on the values of F there: it is unchanged by any reassignment of positive finite multiplicities, even unbounded ones.

Proof. For a finite divisor, finite Hermite interpolation supplies the inverse jets when all values are nonzero, and [corollary 7.5](#) converts them into (45); necessity is evaluation. Assume D infinite. An identity (45) exists exactly when $F \bmod P_D$ is a unit of $\mathcal{E}_\Gamma/(P_D)$, so [theorem 8.7](#) and [lemma 8.25](#) apply. The unit condition in Λ_0 and in each Λ_n is nonvanishing of the constant terms of all local jets, which is (1).

For the tail condition, necessity in [theorem 8.6](#) puts the normalized shell remainder $\text{rem}_n(F)$ in $V_n[T]$ for all large n , and its values at the normalized nodes are $F(\rho_{n,j})$. By [lemma 8.24](#) its inverse lies in Λ_n° exactly when those values are units of V_n . The lower halves of the unit condition (22) are automatic from membership in V_n ; what remains is exactly the upper bound (46), and a maximum over the finitely many nodes of a shell produces one c_n per shell.

Conversely, under (1) and (2) the local inverse jets of F have shell representatives in $V_n[T]$ for all large n , so [theorem 8.6](#) yields a $B \in \mathcal{E}_\Gamma$ carrying them. Then $1 - BF$ vanishes on D to the required orders, and [corollary 7.5](#) supplies A .

Finally, both conditions refer only to $\text{supp } D$ and to the values of F there, and any assignment of positive finite multiplicities to a radially finite set of nodes is again radially finite. $\square$

The simplification to a pointwise test is special to the *inverse jets of an already entire function*. Arbitrary Hermite data require the full coefficient criterion of [theorem 8.6](#); replacing it by a pointwise condition would lose [theorem 8.31](#).

The fourth source proved [theorem 8.26](#) independently, as a consequence of its own proof of [theorem 8.7](#), and extended it to finitely many generators and to membership of arbitrary elements.

Theorem 8.27 (Finitely many generators). *Let D be an infinite radially finite divisor with $D_0 = 0$ and canonical product P_D , and let $G_1, \dots, G_s \in \mathcal{E}_\Gamma$, $s \geq 1$. Then $(P_D, G_1, \dots, G_s) = \mathcal{E}_\Gamma$ if and only if*

- (1) *at every node of D at least one G_ℓ is nonzero, and*
- (2) *for every node $\rho_{n,j}$ of every sufficiently large shell, at least one G_ℓ satisfies $G_\ell(\rho_{n,j}) \neq 0$ and $v(G_\ell(\rho_{n,j})) \leq c\gamma_n$ for some integer $c \geq 0$; equivalently, $v(G_\ell(\rho_{n,j})) \in H_n$.*

On a sufficiently large shell, the condition in (2) is equivalent to

$$\gcd(\overline{P_n}, \overline{\text{rem}_n(G_1)}, \dots, \overline{\text{rem}_n(G_s)}) = 1 \quad \text{in } \kappa_n[T],$$

the bars denoting reduction modulo $\mathfrak{q}_{\gamma_n}$; this holds even when distinct normalized nodes have the same reduction.

Proof. By [theorem 8.7](#) the ideal is the unit ideal exactly when the images $g_\ell = \text{ev}_D(G_\ell)$ generate the unit ideal of $\mathcal{R}_D$, that is, when there are $a_{\ell,n} \in \Lambda_n$ with $\sum_\ell a_{\ell,n} g_{\ell,n} = 1$ for every n and $a_{\ell,n} \in \Lambda_n^\circ$ for all large n . In Λ_n , which is a finite product of local jet rings over K_Γ , the $g_{\ell,n}$ generate the unit ideal exactly when at each node one of them has a nonzero constant term; this is (1).

For large n every $g_{\ell,n}$ lies in Λ_n° , by the necessity half of [theorem 8.6](#). The reduction of Λ_n° modulo $\mathfrak{q}_{\gamma_n}$ is $\kappa_n[T]/(\overline{P_n})$, and $\overline{P_n}$ splits at the reductions of the $\xi_{n,j}$. The reduced classes generate the unit ideal exactly when they have no common root among those reductions, which is the gcd

condition. The value of $g_{\ell,n}$ at $\xi_{n,j}$ is $G_{\ell}(\rho_{n,j}) \in V_n$, whose coarsened residue is nonzero exactly when its valuation lies in H_n ; so the gcd condition is (2) at shell n , repeated residues included. A residue identity lifts coefficientwise to $\sum_{\ell} a_{\ell} g_{\ell,n} = 1 + e$ with $e \in \mathfrak{q}_{\gamma_n} \Lambda_n^{\circ}$. The multiplication determinant of $1 + e$ is 1 modulo $\mathfrak{q}_{\gamma_n}$, so $1 + e$ is a unit of Λ_n° and the identity can be divided by it. Hence the $g_{\ell,n}$ generate the unit ideal of Λ_n° exactly when (2) holds at shell n .

If (1) and (2) hold, choose the $a_{\ell,n}$ in Λ_n for the finitely many exceptional shells and in Λ_n° afterwards, and realize each tuple $(a_{\ell,n})_n$ by some $B_{\ell} \in \mathcal{E}_{\Gamma}$ (theorem 8.6). Then $1 - \sum_{\ell} B_{\ell} G_{\ell}$ has zero image, so it lies in $P_D \mathcal{E}_{\Gamma}$. Conversely, a global identity reduces to identities in every Λ_n and, for large n , in Λ_n° . $\square$

For $s = 1$ this is theorem 8.26 in the case $D_0 = 0$. As there, finite nilpotent directions do not enter: a unit of a local jet ring is detected by its constant term, so only values matter, even for unbounded multiplicities.

Corollary 8.28 (Membership, one simple node per shell). *Suppose D has one simple node ρ_n on each shell and $D_0 = 0$, and let $F, G \in \mathcal{E}_{\Gamma}$. Then $F \in (P_D, G)$ if and only if $F(\rho_n) = 0$ whenever $G(\rho_n) = 0$, and the ratios $F(\rho_n)/G(\rho_n)$ at the remaining nodes lie in V_n for all sufficiently large n .*

Proof. If $F = AP_D + BG$, then $F(\rho_n) = B(\rho_n)G(\rho_n)$, and $B(\rho_n) \in V_n$ for all large n by the necessity half of corollary 8.13. Conversely, prescribe $B(\rho_n)$ to be the ratio where $G(\rho_n) \neq 0$ and 0 elsewhere; corollary 8.13 realizes B , and $F - BG$ vanishes at every node, so it lies in $P_D \mathcal{E}_{\Gamma}$ by theorem 8.7. $\square$

Theorem 8.29 (An explicit pointwise Bézout failure). *Assume the hypotheses of lemma 8.1 and choose its sequence (γ_n) . Define*

$$\rho_n = t^{-\gamma_n}, \quad \sigma_n = t^{-\gamma_n} + t^{\gamma_{n+1}}, \quad (47)$$

$$f(Z) = \prod_{n \geq 1} (1 - Z/\rho_n), \quad g(Z) = \prod_{n \geq 1} (1 - Z/\sigma_n). \quad (48)$$

Then $f, g \in \mathcal{E}_{\Gamma}$, both have constant coefficient 1, their roots are simple and disjoint, and

$$\gcd(f, g) = 1, \quad (f, g) \neq \mathcal{E}_{\Gamma}.$$

In particular there are no $A, B \in \mathcal{E}_{\Gamma}$ with $Af + Bg = 1$. The failure is exactly a failure of condition (2) of theorem 8.26: condition (1) holds, while

$$\lambda_n := v(g(\rho_n)) = \gamma_{n+1} + (2 - n)\gamma_n + \sum_{j < n} \gamma_j \quad (49)$$

exceeds every finite multiple of γ_n .

Proof. We have $v(\rho_n) = v(\sigma_n) = -\gamma_n$, so both root sequences escape every valuation ball and theorem 7.3 defines the two entire functions. Different indices have different valuations; at the same index $\sigma_n - \rho_n = t^{\gamma_{n+1}} \neq 0$. Their zero sets are consequently disjoint and simple, and corollary 7.5 gives their GCD.

We compute $v(g(\rho_n))$ exactly. For $j < n$, ρ_n/σ_j has negative valuation, so

$$v(1 - \rho_n/\sigma_j) = \gamma_j - \gamma_n.$$

For $j = n$,

$$1 - \rho_n/\sigma_n = \frac{\sigma_n - \rho_n}{\sigma_n}, \quad v(1 - \rho_n/\sigma_n) = \gamma_{n+1} + \gamma_n.$$

For $j > n$ the ratio has positive valuation, and the infinite tail product has residue 1 and valuation zero by the positive-support product argument used in [theorem 7.3](#). This gives (49). The sum there is finite. Since $0 < \gamma_j < \gamma_n$ for $j < n$, the part of (49) other than γ_{n+1} is bounded in absolute value by a finite multiple of γ_n . Hence $\lambda_n \gg_A \gamma_n$.

Suppose $Af + Bg = 1$. Evaluate at ρ_n . The values are well defined and evaluation is multiplicative, so

$$B(\rho_n) = \frac{1}{g(\rho_n)}, \quad v(B(\rho_n)) = -\lambda_n.$$

For all sufficiently large n this contradicts [lemma 8.22\(a\)](#). Thus the ideal is proper.

This argument is self-contained and uses only part (a) of [lemma 8.22](#). Alternatively, and equivalently, $D = \sum_n [\rho_n]$ has one simple node per shell, $f = P_D$ exactly, since $D(0) = 0$ in (16); condition (1) of [theorem 8.26](#) holds because g has no zero in common with f , and (49) with $\lambda_n \gg_A \gamma_n$ is exactly the failure of (46). The criterion therefore both recovers this example and says what would have had to be true instead. $\square$

The cancellation-sensitive step is (49). No unproved assertion that nearby zeros force a small value is used: the contribution of each earlier root is retained, and the infinite tail is proved to be a unit. The earlier-root terms can lower the valuation by finite multiples of γ_n , but cannot offset the new Archimedean scale γ_{n+1} .

The fourth source turned this single instance into an exact threshold.

Theorem 8.30 (The exact threshold for paired zeros). *Let $0 < \gamma_1 < \gamma_2 < \dots$ be cofinal in Γ , with no domination assumed, let $\vartheta_n > 0$ in Γ , and put*

$$\rho_n = t^{-\gamma_n}, \quad \sigma_n = \rho_n + t^{\vartheta_n}, \quad f = \prod_{n \geq 1} (1 - Z/\rho_n), \quad g_\vartheta = \prod_{n \geq 1} (1 - Z/\sigma_n).$$

Then $f, g_\vartheta \in \mathcal{E}_\Gamma$ have simple and disjoint zero sets,

$$v(g_\vartheta(\rho_n)) = \vartheta_n + (2 - n)\gamma_n + \sum_{h < n} \gamma_h, \tag{50}$$

and $(f, g_\vartheta) = \mathcal{E}_\Gamma$ if and only if $\vartheta_n \in H_n$ for all sufficiently large n .

Proof. Both root sequences have valuation $-\gamma_n$ at index n , distinct indices give distinct valuations, and $\sigma_n - \rho_n \neq 0$; [theorem 7.3](#) gives the zero assertions. The computation of (49) applies verbatim with γ_{n+1} replaced by ϑ_n : the factor of index $h < n$ has valuation $\gamma_h - \gamma_n$, the factor of index n has valuation $\vartheta_n + \gamma_n$, and the tail is a unit of valuation zero. The terms of (50) other than ϑ_n lie in H_n , so $v(g_\vartheta(\rho_n)) \in H_n$ exactly when $\vartheta_n \in H_n$. Apply [theorem 8.26](#) to $D = \sum_n [\rho_n]$, for which $f = P_D$ and condition (1) holds. $\square$

Under the hypotheses of [lemma 8.1](#) the displacement $\vartheta_n = \gamma_{n+1}$ lies in no H_n , and this is [theorem 8.29](#). At the other extreme a displacement $\vartheta_n = n!\gamma_n$ causes no obstruction. The dividing line is not the size of an ordinary numerical coefficient but whether the displacement leaves the convex subgroup of the current radius, and infinitely many such shells suffice to make the ideal proper.

Theorem 8.31 (A close pair with individually harmless values). *Keep ρ_n, σ_n of (47) and regard $\{\rho_n, \sigma_n\}$ as one shell of two simple nodes, with $\tau_n = t^{-\gamma_n}$, normalized nodes 1 and $1 + \varepsilon_n$, and*

$$\varepsilon_n = t^{\gamma_{n+1} + \gamma_n}.$$

The prescription $F(\rho_n) = 0$, $F(\sigma_n) = c_n$ is realized by some $F \in \mathcal{E}_\Gamma$ if and only if

$$c_n/\varepsilon_n \in V_n \quad \text{for all sufficiently large } n. \quad (51)$$

In particular there is no $F \in \mathcal{E}_\Gamma$ with $F(\rho_n) = 0$ and $F(\sigma_n) = 1$ for all n , even though every prescribed value lies in every V_n and every finite subproblem has a polynomial solution. By contrast $c_n = \varepsilon_n$ and $c_n = t^{\gamma_{n+1}}$ are both realizable.

Proof. The unique shell interpolation polynomial of degree below two with $R_n(1) = 0$ and $R_n(1 + \varepsilon_n) = c_n$ is $R_n(T) = (c_n/\varepsilon_n)(T - 1)$, and both of its coefficients are $\pm c_n/\varepsilon_n$. Theorem 8.6 gives (51). For $c_n = 1$ both nonzero coefficients have valuation $-\gamma_{n+1} - \gamma_n$, which lies strictly below $-m\gamma_n$ for every $m \in \mathbb{N}$ because $\gamma_{n+1} \gg_A \gamma_n$; the criterion fails at every shell. Finite subproblems are ordinary Lagrange interpolation over K_Γ . For $c_n = \varepsilon_n$ one gets $R_n = T - 1$, and for $c_n = t^{\gamma_{n+1}}$ one gets $R_n = t^{-\gamma_n}(T - 1)$; both lie in $V_n[T]$. $\square$

This obstruction is genuinely joint. Neither the list of node radii nor the separate valuations of the prescribed values detects it: the divided difference carries an additional negative Archimedean scale. It is also the exact reason why corollary 8.14 needs its hypothesis, and why applying corollary 8.13 node by node on a multi-node shell is wrong.

8.8 Maximal ideals that are not evaluations

Corollary 8.32 (A finitely generated ideal invisible to point tests). *In the situation of theorem 8.29 there is a proper two-generated ideal contained in no evaluation maximal ideal $(Z - x)\mathcal{E}_\Gamma$. Every maximal ideal containing it is therefore not the kernel of evaluation at any $x \in K_\Gamma$.*

Proof. The ideal (f, g) is proper but f and g do not vanish simultaneously. By corollary 7.6, containment in $(Z - x)\mathcal{E}_\Gamma$ would mean that both vanish at x . A maximal ideal containing a proper ideal exists by Zorn's lemma; none of these maximal ideals is an evaluation ideal. $\square$

This is a failure of the assertion that *every finitely generated proper ideal has a common zero*. We do not call it a metric corona counterexample, because no norm lower bound and no bounded-function algebra has been specified. Nor do we claim that non-evaluation maximal ideals are absent in the rank-one entire ring; the statement concerns a *two-generated* ideal already lacking a common zero.

Corollary 8.32 produces its maximal ideals by Zorn's lemma and names none of them. It is kept because it is weaker and cheaper: it needs only corollary 7.6 and theorem 8.29, and in particular none of the image theorem. The next theorem constructs such maximal ideals explicitly, with their residue fields, at the cost of using theorem 8.6 and a nonprincipal ultrafilter.

For the rest of this subsection let D have one simple node ρ_n per occupied shell and empty finite part, so that

$$P_D = \prod_{n \geq 1} (1 - Z/\rho_n), \quad \mathcal{R}_D = \{(b_n) \in K_\Gamma^{\mathbb{N}_{>0}} : b_n \in V_n \text{ eventually}\}$$

by theorem 8.7 and corollary 8.13. Adjoining a finite part changes none of the constructions below. Fix a nonprincipal ultrafilter $\mathcal{U}$ on the positive integers — this is an ordinary ZFC choice, not an algorithmically computed object — and set

$$L_{\mathcal{U}} = \left(\prod_{n \geq 1} \kappa_n \right) / \mathcal{U}, \quad \kappa_n = \mathbb{C}((t^{H_n})), \quad (52)$$

two sequences being identified when they agree on a member of $\mathcal{U}$. This quotient is a field: a nonzero class has nonzero components on a member of $\mathcal{U}$, and the reciprocals there give its inverse.

Theorem 8.33 (Explicit non-evaluation maximal ideals). *The formula*

$$\Phi_{\mathcal{U}}(F) = [\text{res}_n(F(\rho_n))]_{\mathcal{U}}, \quad (53)$$

where $\text{res}_n : V_n \rightarrow \kappa_n$ is the coarsened residue map of lemma 8.4 and the residue is taken at all sufficiently large n , defines a surjective ring homomorphism $\mathcal{E}_{\Gamma} \rightarrow L_{\mathcal{U}}$. Arbitrary values at the finitely many remaining indices do not affect the class. Its kernel is the maximal ideal

$$\mathfrak{M}_{\mathcal{U}} = \{F \in \mathcal{E}_{\Gamma} : \{n : v(F(\rho_n)) > m\gamma_n \text{ for every } m \in \mathbb{N}\} \in \mathcal{U}\}, \quad (54)$$

which contains P_D and is not the kernel of evaluation at any point of K_{Γ} . Distinct nonprincipal ultrafilters give distinct maximal ideals, and $\mathcal{E}_{\Gamma}/\mathfrak{M}_{\mathcal{U}} \cong L_{\mathcal{U}}$.

Proof. Necessity in theorem 8.6, in the form lemma 8.22(a), gives the eventual membership $F(\rho_n) \in V_n$ that (53) needs. Addition and multiplication are respected on a common cofinite set of indices, so $\Phi_{\mathcal{U}}$ is a ring homomorphism, and changing finitely many components does not change the class at a nonprincipal ultrafilter. For surjectivity, given $c_n \in \kappa_n$ use the canonical inclusion $\kappa_n = \mathbb{C}((t^{H_n})) \subseteq K_{\Gamma}$ to lift to $b_n \in V_n$ and apply corollary 8.13. Lemma 8.4 identifies the kernel with (54), and since $L_{\mathcal{U}}$ is a field that kernel is maximal. It contains P_D , which vanishes at every node.

Suppose the kernel were evaluation at a point x . Then $P_D(x) = 0$, so $x = \rho_r$ for some r by theorem 7.3. The bounded prescription $b_r = 1$ and $b_n = 0$ for $n \neq r$ has an entire interpolant F_r by corollary 8.13; its residue class is zero at a nonprincipal ultrafilter, so $F_r \in \mathfrak{M}_{\mathcal{U}}$, while $F_r(\rho_r) = 1$. This excludes every evaluation point.

For distinct nonprincipal ultrafilters $\mathcal{U}$ and $\mathcal{W}$, choose $S \in \mathcal{U}$ with $S \notin \mathcal{W}$ and interpolate the values 1 on S and 0 off S ; the image is 1 under $\Phi_{\mathcal{U}}$ and 0 under $\Phi_{\mathcal{W}}$, so the kernels differ. The residue-field identification is the first isomorphism theorem. $\square$

Remark 8.34 (How the coefficient field acts). The copy of K_{Γ} inside $L_{\mathcal{U}}$ is not a constant-coordinate embedding into each κ_n . A scalar $c \in K_{\Gamma}$ maps to the class of its eventual support restrictions $[c|_{H_n}]_{\mathcal{U}}$. For $c \neq 0$ the leading exponent of c lies in H_n for all large n , so those residues are nonzero; the map is an embedding and supplies the K_{Γ} -algebra structure on the target. It remains valid when $\text{supp } c$ is contained in no single H_n .

Corollary 8.35 (Finite-rank residue fields in an infinite-rank example). *Take $\Gamma_{\infty} = \bigoplus_{j \geq 0} \mathbb{Q}e_j$ of section 14.1 with $\gamma_n = e_n$ for $n \geq 1$. Then*

$$H_n = \bigoplus_{j=0}^n \mathbb{Q}e_j, \quad \mathcal{E}_{\Gamma}/\mathfrak{M}_{\mathcal{U}} \cong \left(\prod_{n \geq 1} \mathbb{C}((t^{\bigoplus_{j=0}^n \mathbb{Q}e_j})) \right) / \mathcal{U}.$$

Every nonprincipal $\mathfrak{M}_{\mathcal{U}}$ contains both of the functions f, g of (48), although they have no common zero.

Proof. In the largest-index order, every element supported in $e_0, \dots, e_n$ is bounded in absolute value by a finite multiple of e_n , while an element with a nonzero coordinate of index above n is not; this is H_n . Note that the index 0 is included: H_n has rank $n+1$, not n . Theorem 8.33 gives the residue field. Now $f(\rho_n) = 0$ for every n , so $f \in \mathfrak{M}_{\mathcal{U}}$; and (49) with $\lambda_n \gg_A \gamma_n$ puts $g(\rho_n)$ in $\mathfrak{q}_{\gamma_n}$ for every n , so $g \in \mathfrak{M}_{\mathcal{U}}$ as well. $\square$

Each of these residue fields is a full Hahn field on a convex subgroup of Γ , exactly the shape produced by the coarsening of [lemma 10.3](#) in an entirely different situation. The coincidence of shape is worth noticing and is not an identification: one coarsening happens inside Γ at a shell radius, the other across an ordered-group extension.

This construction gives a specified family of maximal ideals together with a field presentation of each. By itself it does *not* classify all maximal ideals of $\mathcal{E}_\Gamma$, and it does not assert that every non-evaluation maximal ideal arises this way. The fourth source's [theorem 8.36](#) below shows that nothing else occurs *above* P_D ; the maximal ideals of $\mathcal{E}_\Gamma$ that contain no such product remain undescribed. The ultrafilter mechanism itself is standard algebra; what is new is the analytic surjection onto $\mathcal{R}_D$ that feeds it. No model-theoretic transfer principle is used.

8.9 All maximal ideals above a one-node-per-shell product

This subsection is the fourth source's. It keeps the one-node-per-shell hypothesis of [section 8.8](#) and the empty finite part, but allows each node ρ_n an arbitrary multiplicity $m_n \geq 1$, so that $P_D = \prod_n (1 - Z/\rho_n)^{m_n}$. In the coordinate $U = (Z - \rho_n)/\rho_n = (T - \xi_n)/\xi_n$, where ξ_n is a unit of V_n , triangular changes of basis over V_n , as in the proof of [corollary 8.13](#), identify Λ_n with $K_\Gamma[U]/(U^{m_n})$ and Λ_n° with $V_n[U]/(U^{m_n})$. By [theorem 8.7](#),

$$\mathcal{E}_\Gamma/(P_D) \cong \mathcal{R}_D = \left\{ (r_n) \in \prod_n K_\Gamma[U]/(U^{m_n}) : r_n \in V_n[U]/(U^{m_n}) \text{ eventually} \right\}, \quad (55)$$

and each $V_n[U]/(U^{m_n})$ is a local ring with maximal ideal $(\mathfrak{q}_{\gamma_n}, U)$ and residue field κ_n . Formula [\(53\)](#) still defines Φ_U , since $F(\rho_n) \in V_n$ eventually by [corollary 8.13](#).

Theorem 8.36 (Every maximal ideal above P_D). *The maximal ideals of $\mathcal{E}_\Gamma$ containing P_D are exactly*

- (a) *the evaluation ideals $(Z - \rho_n)\mathcal{E}_\Gamma$, $n \geq 1$, with residue field K_Γ ; and*
- (b) *the kernels $\mathfrak{M}_U$ of Φ_U , one for each nonprincipal ultrafilter $\mathcal{U}$ on $\mathbb{N}_{>0}$, with $\mathcal{E}_\Gamma/\mathfrak{M}_U \cong L_U = (\prod_n \mathbb{C}((t^{H_n}))) / \mathcal{U}$.*

Distinct ultrafilters give distinct ideals, and inserting arbitrary elements of κ_n at finitely many coordinates of [\(53\)](#) does not change Φ_U .

Proof. For simple nodes, (b) is [theorem 8.33](#), and its proof works verbatim with multiplicities: [corollary 8.13](#) realizes arbitrary values in V_n with zero higher jets, which gives surjectivity, and the kernel and distinctness arguments use values only. The evaluation ideals contain P_D and are maximal by [corollary 7.6](#).

For exhaustiveness it suffices to find the maximal ideals M of $\mathcal{R}_D$. For $S \subseteq \mathbb{N}_{>0}$ let $e_S \in \mathcal{R}_D$ be the idempotent equal to 1 on S and 0 elsewhere; its coordinates lie in every $V_n[U]/(U^{m_n})$. In the field $\mathcal{R}_D/M$ each e_S maps to 0 or 1, and $e_S + e_{\mathbb{N}_{>0} \setminus S} = 1$, $e_{S \cap T} = e_S e_T$; hence $\mathcal{U}_M = \{S : e_S \notin M\}$ is an ultrafilter. If $\mathcal{U}_M$ is principal at n , then $1 - e_{\{n\}} \in M$, the quotient factors through $K_\Gamma[U]/(U^{m_n})$, whose unique maximal ideal is (U) , and M corresponds to $(Z - \rho_n)\mathcal{E}_\Gamma$.

Suppose $\mathcal{U}_M$ is nonprincipal. If $re_S = 0$ with $S \in \mathcal{U}_M$, the image of e_S is 1 and so the image of r is 0; thus M contains every tuple vanishing on a member of $\mathcal{U}_M$. Modulo those tuples, $\mathcal{R}_D$ becomes the ultraproduct $(\prod_n V_n[U]/(U^{m_n})) / \mathcal{U}_M$, because changing finitely many coordinates makes any element of $\mathcal{R}_D$ integral without changing its class. An ultraproduct of local rings is local: a tuple

is a unit exactly when its coordinates are units on a member of $\mathcal{U}_M$, the nonunits are the tuples whose constant coefficients lie in $\mathfrak{q}_{\gamma_n}$ on a member of $\mathcal{U}_M$, and they form an ideal with quotient $(\prod_n \kappa_n)/\mathcal{U}_M$. So M is the kernel of $\Phi_{\mathcal{U}_M}$. The idempotents recover $\mathcal{U}_M$ from M , which gives distinctness again. $\square$

Write $\beta\mathbb{N}_{>0}$ for the set of all ultrafilters on $\mathbb{N}_{>0}$ with the basic clopen sets $\widehat{S} = \{\mathcal{U} : S \in \mathcal{U}\}$: the Stone-Čech space of the discrete set $\mathbb{N}_{>0}$, whose principal ultrafilters form a copy of $\mathbb{N}_{>0}$.

Theorem 8.37 (The maximal spectrum of the quotient). *With the Zariski topology on the left, the bijection of theorem 8.36 is a homeomorphism*

$$\text{Max}(\mathcal{E}_\Gamma/(P_D)) \cong \beta\mathbb{N}_{>0},$$

sending $(Z - \rho_n)\mathcal{E}_\Gamma$ to the principal ultrafilter at n . The evaluation ideals form a discrete dense subset.

Proof. Work in (55). For $r = (r_n)$ let c_n be the constant coefficient of r_n and put $Z_r = \{n : c_n = 0\} \subseteq T_r = \{n : c_n \in \mathfrak{q}_{\gamma_n}\}$. At the principal ultrafilter at n , r lies outside the maximal ideal exactly when $n \notin Z_r$; at a nonprincipal $\mathcal{U}$, exactly when $\mathbb{N}_{>0} \setminus T_r \in \mathcal{U}$, finitely many nonintegral coordinates being irrelevant. So the basic open set $D(r)$ corresponds to

$$\widehat{\mathbb{N}_{>0} \setminus T_r} \cup \{\text{principal ultrafilter at } n : n \in T_r \setminus Z_r\},$$

which is open in $\beta\mathbb{N}_{>0}$ because principal points are open. Conversely $D(e_S)$ corresponds exactly to $\widehat{S}$. Every nonempty $\widehat{S}$ contains the principal ultrafilter at some $n \in S$, which gives density. $\square$

The ultrafilter topology is familiar. The new information is the varying residue fields, which a topological description of the spectrum does not see.

Corollary 8.38 (The whole maximal-ideal locus of a paired example). *In theorem 8.30 let $S = \{n : \vartheta_n \notin H_n\}$. The maximal ideals of $\mathcal{E}_\Gamma$ containing both f and g_ϑ are exactly the $\mathfrak{M}_\mathcal{U}$ with $\mathcal{U}$ nonprincipal and $S \in \mathcal{U}$. For finite S there are none and $(f, g_\vartheta) = \mathcal{E}_\Gamma$; for infinite S they exist. In the situation of theorem 8.29, $S = \mathbb{N}_{>0}$, and the locus is all of $\beta\mathbb{N}_{>0} \setminus \mathbb{N}_{>0}$.*

Proof. No evaluation ideal contains both functions, since they have no common zero. By (50), $v(g_\vartheta(\rho_n))$ is ϑ_n plus an element of H_n . Since $\vartheta_n > 0$, its class modulo H_n is positive exactly when $n \in S$, so $\Phi_\mathcal{U}(g_\vartheta) = 0$ exactly when $S \in \mathcal{U}$. Apply theorem 8.36. A nonprincipal ultrafilter contains no finite set, and the ultrafilter lemma extends an infinite S to a nonprincipal ultrafilter. $\square$

In particular every maximal ideal that corollary 8.32 obtains from Zorn's lemma is one of the explicit ideals of theorem 8.33, and corollary 8.35 lists all of them. The description still uses ultrafilters and choice; it computes no particular nonprincipal ultrafilter.

Corollary 8.39 (Algebraic type of the residue fields). *Every field $L_\mathcal{U}$ is algebraically closed. Over the real Hahn field (remark 8.43) the corresponding field $(\prod_n \mathbb{R}((t^{H_n})))/\mathcal{U}$ is real closed for its ultraproduct order.*

Proof. Each H_n is divisible, so $\mathbb{C}((t^{H_n}))$ is algebraically closed by the Hahn-field theorem [4, Corollary 4]; $\mathbb{R}((t^{H_n}))$ is ordered and its complexification is that field, so it is real closed. A monic polynomial of fixed degree over an ultraproduct has monic coordinates of that degree on a member of the ultrafilter, and roots can be chosen coordinatewise; the same argument with square roots of positive elements and roots of odd-degree polynomials gives real closedness. $\square$

Proposition 8.40 (A transcendental boundary coordinate). *Let $\mathcal{U}$ be nonprincipal and let $\iota_{\mathcal{U}} : K_{\Gamma} \rightarrow L_{\mathcal{U}}$ be the restriction of $\Phi_{\mathcal{U}}$, injective by [remark 8.34](#). Then $\Phi_{\mathcal{U}}(Z)$ is transcendental over $\iota_{\mathcal{U}}(K_{\Gamma})$. Over the real Hahn field with the monomial nodes $\rho_n = t^{-\gamma_n}$, $\Phi_{\mathcal{U}}(Z)$ exceeds every element of $\iota_{\mathcal{U}}(\mathbb{R}((t^{\Gamma})))$ in the ultraproduct order.*

Proof. Let $h = \sum_{j \leq e} c_j Z^j \in K_{\Gamma}[Z]$ with $c_e \neq 0$. Once γ_n exceeds the finitely many differences $v(c_j) - v(c_e)$, the term $c_e \rho_n^e$ is the unique one of least valuation, so $v(h(\rho_n)) = v(c_e) - e\gamma_n$, which lies in H_n for large n . Its coarsened residue is then nonzero, so $\Phi_{\mathcal{U}}(h) \neq 0$. Since $\Phi_{\mathcal{U}}$ restricted to $K_{\Gamma}[Z]$ is the K_{Γ} -algebra map sending Z to $\Phi_{\mathcal{U}}(Z)$, no nonzero polynomial over $\iota_{\mathcal{U}}(K_{\Gamma})$ vanishes there. In the real monomial case, for fixed c and large n the element $\rho_n - c$ has leading exponent $-\gamma_n \in H_n$ with coefficient 1, so its coarsened residue is positive, and the ultraproduct inequality gives $\Phi_{\mathcal{U}}(Z) > \iota_{\mathcal{U}}(c)$. $\square$

The transcendence has a classical analogue in Henriksen’s Theorem 5 for the ring of complex entire functions [\[13\]](#), and it is not claimed as a new phenomenon. What is specific here is that the coefficient embedding and the residue fields are computed through the varying coarsenings.

Remark 8.41 (Not an illicit infinite substitution). The equation $\Phi_{\mathcal{U}}(P_D) = 0$ is consistent with the transcendence of $\Phi_{\mathcal{U}}(Z)$. The character is a ring homomorphism and respects finite algebra; it has not been shown to preserve strong Hahn sums, and $\Phi_{\mathcal{U}}(P_D)$ must not be computed by substituting $\Phi_{\mathcal{U}}(Z)$ termwise into the infinite product or its power series. These maximal ideals form an algebraic spectrum. No additional zero reachable by strong evaluation in some field extension is claimed, and the characters are not asserted to be evaluation maps on any larger surreal domain.

What this subsection does not do. The classification is stated for one node per shell. With several nodes on a shell the finite residue algebra $\kappa_n[T]/(\overline{P_n})$ can have several maximal ideals, and a description of the spectrum must track that varying residue-direction structure together with the ultrafilter; this is not done. Prime ideals below these maximal ideals are not classified, especially for unbounded multiplicities. Maximal ideals of $\mathcal{E}_{\Gamma}$ that contain no product of this form are not described. The unit-ideal criteria of [section 8.7](#) are not a corona theorem: no bounded function algebra and no norm lower bound is involved. See [question 16.4](#).

Re-scoped by the sixth source. The fourth source’s non-claim about prime ideals is kept above as it was written. For one *simple* node per shell with empty finite part, the prime ideals below these maximal ideals are now classified in [section 12](#) ([theorem 12.9](#)); the classification transfers to uniformly bounded multiplicities as a statement about prime spaces and dimension, and for unbounded multiplicities [theorem 12.40](#) shows only that primes not containing the reduced product exist. Shells with several nodes, unbounded multiplicities, and maximal or prime ideals of $\mathcal{E}_{\Gamma}$ containing no product of this form remain unclassified.

8.10 Real Hahn coefficients

Proposition 8.42 (Real form). *Theorems 8.6, 8.7 and 8.26 and the interpolation corollaries corollaries 8.13 to 8.15, together with theorem 8.31, remain true with K_{Γ} replaced by the real Hahn field $\mathbb{R}((t^{\Gamma}))$ and $\mathcal{E}_{\Gamma}$ by its ring of strongly entire series, provided all interpolation nodes and all prescribed data lie in that field.*

Proof. Every algebraic inverse used above concerns polynomials that are split by hypothesis at the given nodes, not arbitrary polynomials; finite Hermite interpolation uses only field arithmetic; and

the coefficient, support and coarsening arguments are unchanged. For the kernel statement use the second proof recorded in [proposition 8.8](#), which does not invoke [theorem 5.2](#) or [theorem 6.2](#). $\square$

No algebraic closedness of the coefficient field is used anywhere in this subsection's list. This is a genuine difference from the factorization theory of [section 7](#): [theorems 6.2](#) and [7.4](#) do use algebraic closedness essentially, as recorded in [section 16.3](#), and [proposition 8.42](#) makes no claim about them. Under the embedding (1), the close pair of [theorem 8.31](#) in Γ_∞ consists of the real surreal numbers $\rho_n = \omega^{\omega^n}$ and $\sigma_n = \omega^{\omega^n} + \omega^{-\omega^{n+1}}$ of (82). So: no entire power series in that fixed real Hahn workspace takes the value 0 at every ρ_n and 1 at every σ_n , while changing the second values to $\omega^{-\omega^{n+1}}$ makes the problem solvable. These remain statements about a fixed set-sized workspace embedded in **No** and **No**[i]; nothing here asserts a nonpolynomial series summable at every point of the whole proper class, and the distinction is the one already drawn in [section 2.3](#).

Remark 8.43 (The fourth source over the real field). The fourth source proved all of its results for the real and the complex Hahn field simultaneously. Accordingly the second route of [section 8.5](#), [corollary 8.16](#), [theorems 8.27](#) and [8.30](#) and [corollary 8.28](#), [theorem 8.33](#) and all of [section 8.9](#) hold over $\mathbb{R}((t^\Gamma))$, with nodes and data in that field and residue fields $\mathbb{R}((t^{H_n}))$ and their ultraproducts, which are real closed by [corollary 8.39](#). Their proofs use only the ingredients listed in the proof of [proposition 8.42](#), Gauss multiplicativity over $\mathbb{R}[X]$, and, for the type of the residue fields alone, algebraic closedness of the complexified Hahn fields $\mathbb{C}((t^{H_n}))$. The fourth source also proves, without root counts, that distinct strongly entire series define distinct functions, so this part of [corollary 6.3](#) holds over $\mathbb{R}((t^\Gamma))$ as well: for $F \neq 0$ and a scale γ , choose a real $c \neq 0$ at which the initial polynomial $I_\gamma(F)$ of (15) does not vanish; then $F(t^\gamma c)$ has valuation $m_\gamma(F)$ and is nonzero.

9 The positive case: an order unit restores interpolation

9.1 A rank-one coarsening with the same entire functions

It would be incorrect to infer that the preceding obstruction occurs for every higher-rank group. A group may have several convex levels and still have a greatest Archimedean class. In that case a single cofinal scale permits ordinary non-Archimedean analytic methods.

Lemma 9.1 (Real-valued coarsening). *Suppose $h > 0$ is an order unit of Γ . There is an order-preserving homomorphism $\pi : \Gamma \rightarrow \mathbb{R}$ with $\pi(h) = 1$, whose kernel is the convex subgroup*

$$H = \{\gamma \in \Gamma : n|\gamma| < h \text{ for all } n \geq 1\}.$$

The formula

$$|a|_\pi = \exp(-\pi(v(a))) \quad (a \neq 0), \quad |0|_\pi = 0$$

defines a nontrivial real-valued non-Archimedean absolute value on K_Γ . Its topology agrees with the intrinsic valuation topology, and K_Γ is complete for it. Furthermore

$$\mathcal{E}_\Gamma = \left\{ \sum_{n \geq 0} a_n Z^n : |a_n|_\pi R^n \longrightarrow 0 \text{ for every real } R > 0 \right\}. \quad (56)$$

Proof. Being an order unit bounds every γ above and below by integer multiples of h . Define

$$\pi(\gamma) = \sup\{q \in \mathbb{Q} : qh \leq \gamma\} \in \mathbb{R}.$$

Rational upper and lower approximations show $\pi(\alpha + \beta) = \pi(\alpha) + \pi(\beta)$ and $\pi(h) = 1$; monotonicity is immediate. The kernel consists exactly of the elements infinitesimal relative to h , giving the displayed H . In particular $\pi(\gamma) > n$ implies $\gamma > nh$, and $\gamma > (n+1)h$ implies $\pi(\gamma) > n$. The balls specified by integer multiples of h are cofinal among all valuation neighbourhoods of zero, and these inequalities show that the real-valued and ordered-group topologies agree.

For completeness, take a Cauchy sequence for $|\cdot|_\pi$ and choose a subsequence (x_{j_n}) with $v(x_{j_{n+1}} - x_{j_n}) > nh$. The differences have valuations tending cofinally to infinity, so their Hahn sum exists by lemma 2.2; adding it to x_{j_1} gives a limit of the subsequence, and the Cauchy property gives the same limit for the whole sequence. Algebraic closedness of K_Γ was already supplied by the Hahn-field theorem.

By theorem 3.1, the left side of (56) is characterized by $v(a_n)/n \rightarrow +\infty$ in Γ . This is equivalent to $\pi(v(a_n))/n \rightarrow +\infty$ in $\mathbb{R}$: integer multiples of h test the forward implication, and for the reverse one chooses a real bound strictly exceeding $\pi(\gamma)$ for a prescribed $\gamma \in \Gamma$. The usual elementary coefficient test identifies this last condition with the right side of (56): explicitly, it implies $\pi(v(a_n)) - n \log R \rightarrow +\infty$ for every R ; conversely, failure of the ratio condition gives infinitely many ratios at most some C , and then $R = e^{C+1}$ violates the coefficient test. $\square$

This is a *coarsening*, not a valuation-preserving embedding of Γ in $\mathbb{R}$. The kernel H may be nonzero, and residue-direction data is not identified across the two valuations: the coarsening changes residue information even while preserving the topology and the entire ring. It is the cofinality of the topology and the all-radius coefficient test that permit its use here.

9.2 A complete Hermite interpolation argument

This theorem is the order-unit special case of theorem 8.6: an order unit makes every coarsened ring V_n equal to K_Γ on all large shells, so the shellwise condition (25) is vacuous. That is the first proof below, and it is short. The second proof, which is the one the source manuscripts wrote, is kept in full: it is a genuinely different argument, by a real-valued coarsening and Gauss norms rather than by valuation estimates inside Γ , it records the damping mechanism that fails in section 8.7 in its own idiom, and it is self-contained — in particular it does not depend on theorem 8.6, so the two routes to case (ii) of theorem 1.5 remain independent. Neither proof is offered as a correction of the other.

Theorem 9.2 (Arbitrary radial Hermite interpolation). *Assume Γ has an order unit. Let (x_n) be distinct points of a radially finite set, and assign a positive integer m_n to each point so that the divisor $\sum_n m_n[x_n]$ is radially finite. For arbitrary jets $J_n \in K_\Gamma[U]/(U^{m_n})$ there exists $F \in \mathcal{E}_\Gamma$ with*

$$F(x_n + U) \equiv J_n(U) \pmod{U^{m_n}} \quad \text{for every } n.$$

First proof, from the exact image theorem. If the divisor is finite this is ordinary Hermite interpolation over the field K_Γ by a polynomial. Otherwise lemma 8.2 supplies its shell decomposition and corollary 8.15 says that (25) holds automatically, because $V_n = K_\Gamma$ for all n with $\gamma_n \geq h$. Apply theorem 8.6. Note the direction of the dependence: this uses the constructive half of theorem 8.6, whose damping estimate (33) is carried out inside the convex subgroup H_n and never assumes that the multiples of one γ_n are cofinal — the order unit is used only to make the shellwise condition empty, not to run the construction. $\square$

Second proof, by a real-valued coarsening and Gauss norms. Use $|\cdot| = |\cdot|_\pi$ from lemma 9.1. A radially finite infinite set can be enumerated with $|x_n| \rightarrow \infty$: a ball of any fixed real absolute-value

radius is contained in some original valuation ball, and conversely an original ball has bounded $|\cdot|_\pi$ radius. Handle the finitely many nodes with $|x_n| \leq 1$, including zero, first by a finite Hermite polynomial, and include their factors in every later vanishing product. Reindex the remaining nodes by $n \geq 1$, so that $|x_n| > 1$ and $|x_n| \rightarrow \infty$.

Suppose a polynomial S_{n-1} has the required jets at earlier nodes and put

$$Q_{n-1}(Z) = \prod_{j < n} (Z - x_j)^{m_j},$$

including the finitely many initially handled nodes. It is a unit in the local jet ring at x_n , so there is a unique jet

$$B_n(U) = \frac{J_n(U) - S_{n-1}(x_n + U)}{Q_{n-1}(x_n + U)} \pmod{U^{m_n}}.$$

For a positive integer N define the polynomial of degree less than m_n

$$r_{n,N}(U) = B_n(U)(1 + U/x_n)^{-N} \pmod{U^{m_n}} \quad (57)$$

and the polynomial correction

$$p_{n,N}(Z) = Q_{n-1}(Z)(Z/x_n)^N r_{n,N}(Z - x_n). \quad (58)$$

It vanishes to the required orders at all earlier nodes, and by (57) its jet at x_n is the required residual. The parameter N can therefore be used purely to control its size.

Set $\theta_n = |x_n|^{1/2}$, so $1 < \theta_n < |x_n|$ and $\theta_n \rightarrow \infty$. (The radius is written θ_n , not R_n , because R_n denotes the normalized shell Hermite polynomial of (19) throughout this report.) For a polynomial $P = \sum p_k Z^k$ write $\|P\|_R = \max_k |p_k| R^k$. If $B_n(U) = \sum_{j < m_n} b_j U^j$, the coefficient of U^ℓ in (57) has absolute value at most $\max_{j \leq \ell} |b_j| |x_n|^{j-\ell}$. This bound is independent of N , because the binomial coefficients $\binom{-N}{k}$ are integers and so have absolute value at most 1. Since $\theta_n < |x_n|$, substitution $U = Z - x_n$ gives

$$\|r_{n,N}(Z - x_n)\|_{\theta_n} \leq \max_{j < m_n} |b_j| |x_n|^j =: C'_n,$$

and consequently

$$\|p_{n,N}\|_{\theta_n} \leq \|Q_{n-1}\|_{\theta_n} C'_n \left(\frac{\theta_n}{|x_n|} \right)^N. \quad (59)$$

The right side tends to zero as the ordinary integer N tends to infinity. Choose N_n so that this norm is at most 2^{-n} and put $S_n = S_{n-1} + p_{n,N_n}$.

For every fixed real $R > 0$, eventually $R \leq \theta_n$, so $\|p_{n,N_n}\|_R \leq 2^{-n}$. Hence (S_n) is Cauchy in the Gauss norm at every radius. The space of series satisfying $|a_k| R^k \rightarrow 0$ is complete for that norm: coefficientwise limits exist by completeness of K_Γ , and uniform approximation by a finite polynomial bounds the tail of the limit. Thus the limits for all radii are one common coefficient series F satisfying (56), so $F \in \mathcal{E}_\Gamma$.

Finally, every specified finite jet is preserved. For a fixed node x and order m , choose $R > |x|$. Coefficients of the translated polynomial up to degree $m - 1$ are continuous for $\|\cdot\|_R$, by the finite binomial formula and its convergent tail. All sufficiently late S_n have the prescribed jet, so their limit does too. $\square$

Theorem 9.3 (Bézout in the order-unit case). *If Γ has an order unit then $\mathcal{E}_\Gamma$ is a Bézout domain. In particular, if $f, g \in \mathcal{E}_\Gamma$ have no common zero there are $A, B \in \mathcal{E}_\Gamma$ with $Af + Bg = 1$.*

This is the order-unit special case of [theorem 8.26](#): with an order unit, condition (2) of that criterion is automatic, so only the common-zero condition (1) remains. Both proofs are kept, for the reason given before [theorem 9.2](#).

First proof, from the sharp criterion. Let f, g be nonzero with no common zero and let $D = \text{Div}(f)$, which is radially finite by [theorem 6.2](#) and has $f = a_m P_D$ for a scalar $a_m \in K_\Gamma^\times$ by [theorem 7.4](#). Condition (1) of [theorem 8.26](#) is the absence of a common zero. For condition (2), an order unit h makes $H_{\gamma_n} = \Gamma$ for every $\gamma_n \geq h$, so every nonzero $g(\rho_{n,j})$ has $|v(g(\rho_{n,j}))| \leq c_n \gamma_n$ for some integer $c_n \geq 0$; a maximum over the finitely many nodes of a shell gives one c_n . The criterion supplies A', B with $A' P_D + Bg = 1$, hence $(A'/a_m)f + Bg = 1$. The reduction of an arbitrary pair to this case, and the induction on the number of generators, are as in the second proof below. $\square$

Second proof, from the order-unit interpolation theorem. First suppose f, g are nonzero with no common zero. At each zero x of f let m_x be its multiplicity. Since $g(x) \neq 0$, the formal inverse of $g(x + U)$ exists modulo U^{m_x} . By [theorem 9.2](#) choose $B \in \mathcal{E}_\Gamma$ whose jet is that inverse at every such x . Then $1 - Bg$ has a zero of order at least m_x at each zero of f , so it is divisible by f in $\mathcal{E}_\Gamma$ by [corollary 7.5](#); write $1 - Bg = Af$.

For arbitrary nonzero f, g , divide both by their GCD d , which exists by [corollary 7.5](#). The quotients have no common zero, so their ideal is the unit ideal; multiplying the resulting identity by d shows $(f, g) = (d)$. Cases involving zero are immediate. Induct on the finite number of generators to obtain the Bézout property. $\square$

For a radially finite divisor D the full jet map is therefore surjective:

$$\mathcal{E}_\Gamma / (P_D) \cong \prod_{x \in \text{supp } D} K_\Gamma[U] / (U^{D(x)}).$$

The kernel is again given by divisibility, and surjectivity is [theorem 9.2](#). Equivalently, in the language of [section 8.3](#), an order unit collapses the restricted product $\mathcal{R}_D$ onto the full product, since $\Lambda_n^\circ = \Lambda_n$ for all large n ([corollary 8.15](#)). This exact unrestricted Chinese remainder statement is the boundary case of [theorem 8.7](#), and it is exactly what fails, in a now completely measured way, when Γ has countable cofinality and no order unit.

10 Change of workspace: the universal scalar-extension domain

The adjective “entire” must name its field of arguments. A formal series with coefficients in one Hahn field may be entire there and fail even strong summability at a single point of a larger field. This section gives the exact boundary, and then the exact description of everything inside it.

Throughout, $\Gamma \subseteq \Delta$ are nonzero divisible ordered abelian groups with the canonical inclusion $K_\Gamma \subseteq K_\Delta$, both full Hahn fields over the *same* coefficient field $\mathbb{C}$.

10.1 The sharp criterion for remaining entire

Theorem 10.1 (Sharp change-of-workspace criterion). *With the above notation,*

$$\mathcal{E}_\Delta \cap K_\Gamma[[Z]] = \begin{cases} \mathcal{E}_\Gamma, & \Gamma \text{ is cofinal in } \Delta, \\ K_\Gamma[Z], & \Gamma \text{ is not cofinal in } \Delta. \end{cases} \quad (60)$$

In the second case a single monomial argument of K_Δ makes every nonpolynomial series with coefficients in K_Γ fail strong summability.

Proof. If Γ is cofinal in Δ , a sequence of elements of $\Gamma \cup \{+\infty\}$ tends cofinally to infinity in Γ if and only if it does so in Δ . Apply [theorem 3.1](#) to the coefficient slopes.

If Γ is not cofinal, choose $\delta \in \Delta$ strictly larger than every element of Γ . For a nonpolynomial $F = \sum a_n Z^n \in K_\Gamma[[Z]]$ consider evaluation at $Z = t^{-\delta}$. For any two nonzero coefficients with $m > n$,

$$(v(a_m) - m\delta) - (v(a_n) - n\delta) = v(a_m) - v(a_n) - (m - n)\delta < 0,$$

since $v(a_m) - v(a_n) \in \Gamma$ and $(m - n)\delta \geq \delta$. The leading exponents of the nonzero terms therefore form an infinite strictly descending sequence, so their support union is not well ordered. No nonpolynomial series over K_Γ can be strongly entire over K_Δ . Polynomials are entire over either field, proving (60). $\square$

10.2 The convex-hull valuation ring

Definition 10.2. The convex hull of Γ in Δ is

$$H = \text{conv}_\Delta(\Gamma) = \{s \in \Delta : |s| \leq \gamma \text{ for some } \gamma \in \Gamma_{>0}\}, \quad (61)$$

a divisible convex subgroup of Δ . Coarsening v_Δ by H gives a valuation with value group Δ/H .

Lemma 10.3. The valuation ring of that coarsening is

$$\mathcal{D}_{\Gamma,\Delta} = \{0\} \cup \{z \neq 0 : v_\Delta(z) \geq \gamma \text{ for some } \gamma \in \Gamma\}. \quad (62)$$

Its maximal ideal consists of zero together with the elements satisfying $v_\Delta(z) > \gamma$ for every $\gamma \in \Gamma$. Its residue field is canonically $\mathbb{C}((t^H))$. Moreover $\mathcal{D}_{\Gamma,\Delta} = K_\Delta$ exactly when Γ is cofinal in Δ .

Proof. A value s has nonnegative class modulo H exactly when it is bounded below by some member of H , and every member of H is itself bounded below by a member of Γ ; this proves (62). The class is strictly positive exactly when $s > H$, equivalently $s > \Gamma$.

For an element of this valuation ring, discard all support exponents outside H . No negative coset can occur, and the discarded positive cosets form the coarsened maximal ideal. The remaining Hahn series has support in H . This is a surjective residue homomorphism onto $\mathbb{C}((t^H))$ with the stated kernel. Finally the ring is the whole field exactly when $H = \Delta$, which by (61) is equivalent to cofinality of Γ in Δ . $\square$

The residue field here is generally *not* $\mathbb{C}$. It records all scales in the convex hull H , whereas the original fine valuation's residue field records only the exponent zero.

10.3 Exact strong evaluation, not merely a sufficient domain

Theorem 10.4 (Universal scalar-extension domain). *Let $f = \sum a_n Z^n \in \mathcal{E}_\Gamma$ be nonpolynomial. In K_Δ the family $(a_n z^n)_n$ is strongly summable if and only if $z \in \mathcal{D}_{\Gamma,\Delta}$. The domain is therefore independent of the choice of nonpolynomial f , and the values on it again lie in $\mathcal{D}_{\Gamma,\Delta}$.*

Proof. Suppose $z \neq 0$ lies in the stated ring. Choose $\gamma \in \Gamma$ with $v_\Delta(z) \geq \gamma$ and put $u = t^{-\gamma}z \in \mathcal{O}_\Delta$. Since f is entire over K_Γ , [theorem 3.1\(2\)](#) gives $f(t^\gamma X) \in \mathcal{A}_\Gamma \subseteq \mathcal{A}_\Delta$, and [lemma 5.1](#) gives strong summability of its evaluation family at the integral point u . That family is the original family $(a_n z^n)_n$. With $\mu = m_\gamma(f) \in \Gamma$, the same lemma gives $v_\Delta(f(z)) \geq \mu$ unless the value is zero, so the value lies in $\mathcal{D}_{\Gamma, \Delta}$. At $z = 0$ the assertion is immediate.

Conversely suppose $s = v_\Delta(z) < \gamma$ for every $\gamma \in \Gamma$. List the infinitely many nonzero coefficients at increasing indices $n_1 < n_2 < \dots$ and write $\alpha_j = v(a_{n_j}) \in \Gamma$. Since $s < 0$,

$$\begin{aligned} (\alpha_{j+1} + n_{j+1}s) - (\alpha_j + n_j s) &= \alpha_{j+1} - \alpha_j + (n_{j+1} - n_j)s \\ &\leq \alpha_{j+1} - \alpha_j + s < 0, \end{aligned}$$

the final inequality because s is below the element $\alpha_j - \alpha_{j+1}$ of Γ . These are actual leading exponents of the nonzero summands $a_{n_j} z^{n_j}$, regardless of the higher terms of z . Their strict descent prevents the support union from being well ordered, so the family is not strongly summable. $\square$

Corollary 10.5 (Cofinal extensions are exactly the entire extensions). *For nonpolynomial $f \in \mathcal{E}_\Gamma$, the same coefficient series belongs to $\mathcal{E}_\Delta$ if and only if Γ is cofinal in Δ . Polynomials always survive.*

Proof. Combine [theorem 10.4](#) with [lemma 10.3](#); polynomial evaluations are finite sums. $\square$

Remark 10.6 (The negative and the positive statement are compatible). [Theorem 10.1](#) and [theorem 10.4](#) must be read together, and [remark 1.8](#) already stated how. The first says that in the noncofinal case a nonpolynomial f is *not entire over K_Δ* , so that $\mathcal{E}_\Delta \cap K_\Gamma[[Z]]$ collapses to $K_\Gamma[Z]$; the second says that the very same series remains strongly summable on the proper subring $\mathcal{D}_{\Gamma, \Delta}$, with values in $\mathcal{D}_{\Gamma, \Delta}$, and [theorem 10.7](#) below says it keeps every one of its zeros there. There is no disagreement about whether the function survives: it survives on $\mathcal{D}_{\Gamma, \Delta}$ and nowhere else, and “not entire over K_Δ ” is precisely the statement $\mathcal{D}_{\Gamma, \Delta} \neq K_\Delta$. Indeed the noncofinal half of [theorem 10.1](#) is the special case $z = t^{-\delta}$ of the necessity half of [theorem 10.4](#), and its independent descending-exponent proof is retained because it exhibits the single witnessing point explicitly.

A cofinal extension need not preserve rank, and a noncofinal one need not introduce any algebraic element: the obstruction lies entirely in the ordered hierarchy of scales. Nor does a strongly defined value in the larger field have to be a limit there. If Δ contains a positive value above all of Γ , the original cofinal coefficient valuations in Γ no longer tend to infinity in Δ .

10.4 No new zeros, even for an infinite divisor

Theorem 10.7 (Zero conservation under arbitrary enlargement). *Let $0 \neq f \in \mathcal{E}_\Gamma$. Every zero of its strongly defined extension on $\mathcal{D}_{\Gamma, \Delta}$ belongs to K_Γ , with unchanged multiplicity. This remains true when f has infinitely many zeros.*

Proof. Suppose $z \in \mathcal{D}_{\Gamma, \Delta}$ and $f(z) = 0$. Choose $\gamma \in \Gamma$ with $v_\Delta(z) \geq \gamma$. Preparation of $f(t^\gamma X)$ over K_Γ ([theorem 5.2](#)) gives

$$f(t^\gamma X) = c t^\mu W_\gamma(X) U_\gamma(X),$$

with W_γ monic over $\mathcal{O}_\Gamma$ and U_γ a unit of $\mathcal{A}_\Gamma$ with initial polynomial 1. By [corollary 5.3](#) this is also the preparation over Δ , and by [lemma 5.1](#) the unit factor is nonzero at every point of $\mathcal{O}_\Delta$. Therefore $W_\gamma(t^{-\gamma}z) = 0$. The polynomial W_γ already splits into linear factors over the algebraically closed field K_Γ , so $t^{-\gamma}z$, and hence z , lies in K_Γ .

The same prepared polynomial has the same linear-factor multiplicities in both fields, and the evaluated unit is nonzero in both, so local orders of vanishing are unchanged. The argument uses only the finitely many zeros in one containing ball and applies separately to each zero; it does not require the global divisor to be finite. $\square$

This is the one statement of the section that imports algebraic closedness. [Section A](#) records that dependency.

10.5 No alternative entire continuation

Proposition 10.8 (Identity on constants). *Two formal series over K_Δ lying in $\mathcal{A}_\Delta$ and agreeing at infinitely many distinct points of $\mathcal{O}_\Delta$ are equal as formal series. In particular agreement at all ordinary complex constants suffices.*

Proof. Their difference A lies in $\mathcal{A}_\Delta$. If it were nonzero, multiplication by a nonzero scalar would normalize its Gauss value to zero and its initial polynomial to monic. By [theorem 5.2](#), the normalized series is WU with W a nonzero finite-degree polynomial and U a unit. Evaluation on $\mathcal{O}_\Delta$ is a ring homomorphism by [lemma 5.1](#), so U never vanishes there. Thus A has at most $\deg W$ distinct integral zeros, a contradiction. This finite upper bound uses no splitting of W and hence no algebraic-closedness hypothesis. $\square$

Theorem 10.9 (Obstruction to any entire-series continuation). *Suppose Γ is not cofinal in Δ and $f \in \mathcal{E}_\Gamma$ is nonpolynomial. There is no $g \in \mathcal{E}_\Delta$ agreeing with f at all points of K_Γ . In fact there is no such g agreeing with f merely at all points of $\mathbb{C}$.*

Proof. The coefficient series of f belongs to $\mathcal{A}_\Delta$, since its strong support certificate at scale zero over Γ remains valid under an ordered embedding; the same is true of g . Agreement on $\mathbb{C}$ would force the two coefficient series to be identical by [proposition 10.8](#). But [corollary 10.5](#) says that this coefficient series is not entire over K_Δ , a contradiction. $\square$

This rules out an alternative entire *power-series* continuation. It does not rule out arbitrary functions extending the pointwise values, or continuations in a different analytic category, and it asserts no density of $\mathbb{C}$ in any fine surreal topology.

10.6 The boundary is intrinsic to the function

Corollary 10.10 (Intrinsic description of the extension boundary). *Let $f \in \mathcal{E}_\Gamma$ be nonpolynomial. In any larger ordered group Δ the following three subsets generate the same convex subgroup:*

$$\Gamma, \quad \{v(a_n)/n : n \geq 1, a_n \neq 0\}, \quad \{-v(r) : r \neq 0, f(r) = 0\}.$$

Consequently the universal domain of [theorem 10.4](#) can be recovered either from the coefficient slopes alone or from the zero divisor alone.

Proof. All the displayed values lie in Γ . The slopes are cofinal in Γ by [theorem 3.1](#). There are infinitely many zeros by [corollary 7.7\(a\)](#), and their negated valuations are cofinal by [lemma 7.2](#). A cofinal subset of an ordered group generates its full convex hull in any containing ordered group. $\square$

This makes the extension obstruction intrinsic. It is not an artefact of having embedded the coefficients in an unnecessarily large workspace whose extra scales the function never detects.

10.7 Consequences for the Bézout counterexample and for $\mathbf{No}[i]$

Corollary 10.11 (No repair of the counterexample by a Hahn extension). *Let f, g be the pair of [theorem 8.29](#). If Γ is cofinal in Δ then $f, g \in \mathcal{E}_\Delta$ and still no $A, B \in \mathcal{E}_\Delta$ satisfy $Af + Bg = 1$. If Γ is not cofinal in Δ then neither f nor g is entire on K_Δ , although both remain strongly defined on $\mathcal{D}_{\Gamma, \Delta}$ with all their zeros ([theorems 10.4](#) and [10.7](#)).*

Proof. In the cofinal case (γ_n) remains cofinal in Δ and $\gamma_{n+1} \gg_A \gamma_n$ is unchanged, so the evaluation formula [\(49\)](#) is unchanged. The proof of [lemma 8.22](#) now applies to any candidate $B \in \mathcal{E}_\Delta$, even with coefficients outside K_Γ , giving the same contradiction. In the noncofinal case apply [theorem 10.1](#) to the two nonpolynomial series, and [theorems 10.4](#) and [10.7](#) for the positive half. $\square$

Corollary 10.12 (Whole-class rigidity on the surcomplex numbers). *Let $F(Z) = \sum_{n \geq 0} a_n Z^n$ be an ordinary power series with coefficients in $\mathbf{No}[i]$. If the family $(a_n z^n)_n$ is strongly normal-form-summable for every $z \in \mathbf{No}[i]$, then only finitely many a_n are nonzero.*

Proof. The union of the normal-form exponent supports of the real and imaginary parts of the countably many coefficients is a *set*. Its divisible additive span together with 1, with signs chosen to match $t^\gamma = \omega^{-\gamma}$, is a nonzero set-sized ordered subgroup $\Gamma \subseteq \mathbf{No}$, and all a_n lie in K_Γ under [\(1\)](#); the hypothesis makes F entire over K_Γ . Every set of surreal numbers has a surreal strict upper bound: the set-sized cut $\delta = \{\Gamma \cup \{0\} \mid \emptyset\}$ supplies a positive $\delta \in \mathbf{No}$ larger than every member of Γ . Enlarge Γ to its divisible span with δ . If F were nonpolynomial, the descending-leading-exponent argument of [theorem 10.1](#) — equivalently [theorem 10.4](#) — would prohibit strong evaluation at $z = \omega^\delta = t^{-\delta}$, contradicting the hypothesis. $\square$

Remark 10.13 (This rigidity is a re-derivation, not a new theorem). [Corollary 10.12](#) is presented as a consequence of the extension theorem, not as an independent contribution. The same conclusion is already in this collection twice over: the *Analysis* report proves all-scale polynomial rigidity directly, and the *Foundations* report records the exact boundary of the phenomenon with a pair of series over $\mathbb{C}((t^{\mathbb{Q}}))$ differing only in the sign of an exponent. The first two source manuscripts merged here credit the repository’s pre-existing all-scale theme rather than claiming the corollary, and the fifth does the same for the whole-family half of its several-variable version, [proposition 11.32](#). The step that makes the reduction legitimate is the localization principle recorded in the foundations report — every *set* of surcomplex numbers lies in one divisible set-sized workspace — and the reduction consumes exactly that principle plus the cut δ . What is proposed as new here is only the exact cofinal/noncofinal criterion for a specified pair of set-sized workspaces ([theorems 10.1](#) and [10.4](#)) and the persistence of the Bézout obstruction under every cofinal extension ([corollary 10.11](#)). See [section 15](#).

11 Several ordinary variables: exact extension domains and exceptional directions

This section is the fifth source’s contribution, together with the finite-variable criteria that earlier versions of this report kept in an appendix, now [section 11.2](#). It answers the scalar-extension clause of [question 16.5](#), the clause beginning “In particular”, and nothing else of that question: preparation, interpolation, divisor and ideal theory in several variables remain open ([section 11.8](#)). Three warnings govern the whole section.

- (1) *Weaker hypotheses.* The fifth source assumes neither divisibility of the value group nor anything about the coefficient field beyond its trivial valuation. Nothing in this section uses divisibility, algebraic closedness, preparation or root counts, except where a statement says otherwise. Conversely, nothing proved elsewhere in the report under its standing hypotheses is transferred here. In particular zero conservation ([theorem 10.7](#)), which needs preparation and algebraic closedness, and the inclusion $f(\mathcal{D}_{\Gamma,\Delta}) \subseteq \mathcal{D}_{\Gamma,\Delta}$ of [theorem 10.4](#) are not asserted in this setting.
- (2) *Two summation conventions.* “Entire” means summability of the *whole* monomial family, as in [definition 1.1](#). [Section 11.5](#) studies a second, explicitly different convention, homogeneous-block evaluation. It is never called entire, and a statement about one convention is not a statement about the other.
- (3) *Notation.* The fifth source’s letters are translated by the last table of [section 2.5](#). In particular its homogeneous block H_n is written $F_{(n)}$ here, because H_n is the convex subgroup of [definition 8.3](#).

11.1 Setting

In this section only, $\mathbf{k}$ is an arbitrary field with the trivial valuation, Γ is a nonzero set-sized ordered abelian group, not assumed divisible, of finite rank, or to have an order unit, and $K_\Gamma = \mathbf{k}((t^\Gamma))$; for $\mathbf{k} = \mathbb{C}$ and divisible Γ this is the field of the rest of the report. The number $d \geq 1$ of ordinary variables is finite; for $\alpha \in \mathbb{N}^d$ write $|\alpha| = \alpha_1 + \cdots + \alpha_d$ and $X^\alpha = \prod_j X_j^{\alpha_j}$. For $F = \sum_\alpha a_\alpha X^\alpha \in K_\Gamma[[X_1, \dots, X_d]]$ the *ordinary monomial support* is $\Sigma(F) = \{\alpha : a_\alpha \neq 0\}$; each nonzero a_α also has its own Hahn exponent support in Γ , and the two kinds of support play different roles.

Definition 11.1 (Whole-family entireness). $\mathcal{E}_{\Gamma,d}$ is the set of $F \in K_\Gamma[[X_1, \dots, X_d]]$ for which the *whole* family $(a_\alpha x^\alpha)_{\alpha \in \mathbb{N}^d}$ is strongly summable for every $x \in K_\Gamma^d$. Summing first along one coordinate and then another, or first within each total degree, is not the definition. For $x_j = 0$ the convention $x_j^0 = 1$ applies, so only the face $\alpha_j = 0$ of the support survives.

Fix an ordered-group inclusion $\Gamma \subseteq \Delta$ and the canonical inclusion $K_\Gamma \subseteq K_\Delta = \mathbf{k}((t^\Delta))$. The convex hull $H = \text{conv}_\Delta(\Gamma)$ is defined by [\(61\)](#); it is a convex subgroup, with no divisibility needed, and Γ is cofinal in it. Write $\pi_H : \Delta \rightarrow \Delta/H$ for the quotient map of ordered groups and

$$\mathcal{D}_{\Gamma,\Delta} = \{0\} \cup \{z \in K_\Delta^\times : \pi_H(v(z)) \geq 0\},$$

the coarsened valuation ring of [\(62\)](#); it equals K_Δ exactly when Γ is cofinal in Δ . No splitting of $H \subseteq \Delta$ is assumed. A convenient model of a new scale is $\Gamma = \mathbb{Q}$ and $\Delta = \mathbb{R}e_* \oplus \mathbb{Q}$, ordered lexicographically with the e_* coordinate dominant: then $H = \mathbb{Q}$, $\Delta/H \cong \mathbb{R}$, and t^{-e_*} is larger in magnitude than every old monomial $t^{-\gamma}$.

The support calculus of [lemma 2.1](#) is stated for arbitrary ordered abelian groups and is used unchanged. Two further elementary facts are used, for families indexed by arbitrary sets.

Lemma 11.2 (Tail certificate and regrouping). *Let $(f_i)_{i \in I}$ be a family of Hahn series over an ordered abelian group G .*

- (a) *If for each $b \in G$ only finitely many nonzero f_i satisfy $v(f_i) \leq b$, the family is strongly summable.*
- (b) *A strongly summable family may be partitioned into arbitrary subfamilies, summed within each part and then summed over the parts, with the original sum as result. The product of two strongly summable families is jointly strongly summable on the product index set, and its sum is the product of the two sums.*

Proof. (a) Only the finitely many terms of valuation at most b can contain the exponent b , which gives finite incidence. A descending sequence $b_1 > b_2 > \dots$ in the support union would lie in the finitely many supports of terms of valuation at most b_1 , and a finite union of well-ordered sets is well ordered by lemma 2.1(a). (b) Regrouped supports lie in the original well-ordered union, and at each exponent only finitely many original indices, hence finitely many parts, contribute. Product supports lie in the sumset of the two support unions, which is well ordered with finitely many decompositions of each exponent by lemma 2.1(a), and each decomposition has finitely many contributing indices. $\square$

The converse of (b) is false, and example 11.22 exhibits the failure that matters here.

11.2 Finite-variable coefficient and unit criteria

The two numbered statements of this subsection were Appendix B of earlier versions of this report, proved there under the report's standing hypotheses, with the coefficient criterion in ratio form; their labels are kept. The fifth source reproved the coefficient criterion in a division-free form under the weaker hypotheses of this section, and that form is printed here as the statement. The proof of the earlier appendix was the same descending-leading-exponent argument and is not printed twice.

Theorem 11.3 (Finite-variable coefficient criterion). *For $F = \sum_{\alpha} a_{\alpha} X^{\alpha} \in K_{\Gamma}[[X_1, \dots, X_d]]$ the following are equivalent:*

- (i) $F \in \mathcal{E}_{\Gamma, d}$;
- (ii) *the family is strongly summable at every diagonal monomial point $(t^{\gamma}, \dots, t^{\gamma})$, $\gamma \in \Gamma$;*
- (iii) *for all $\gamma, b \in \Gamma$ the set*

$$\{\alpha \in \Sigma(F) : v(a_{\alpha}) + |\alpha|\gamma \leq b\} \text{ is finite.} \quad (63)$$

If Γ is divisible, they are also equivalent to $v(a_{\alpha})/|\alpha| \rightarrow +\infty$ cofinally as $|\alpha| \rightarrow \infty$ through nonzero coefficients of positive degree. Nonpolynomial members of $\mathcal{E}_{\Gamma, d}$ exist exactly when $\text{cf}(\Gamma) = \aleph_0$.

Proof. (i) $\Rightarrow$ (ii) is immediate. Assume (ii), fix γ , and put $b_{\alpha} = v(a_{\alpha}) + |\alpha|\gamma$. Strong summability at the diagonal point gives a common lower bound $L \in \Gamma$ for these leading exponents. Suppose infinitely many $\alpha \in \Sigma(F)$ satisfy $b_{\alpha} \leq b$. Only finitely many multi-indices have bounded total degree, so there are such $\alpha_1, \alpha_2, \dots$ with strictly increasing degrees $n_r = |\alpha_r|$. Choose $\delta > 0$ in Γ with $\delta > b - L$. Then

$$(b_{\alpha_{r+1}} - n_{r+1}\delta) - (b_{\alpha_r} - n_r\delta) \leq b - L - \delta < 0,$$

and these are actual leading exponents at the diagonal point of valuation $\gamma - \delta$. Their strict descent contradicts (ii); so (iii) holds.

For (iii) $\Rightarrow$ (i), take a point with nonzero coordinates and let $\gamma = \min_j v(x_j)$. Then $v(a_{\alpha} x^{\alpha}) \geq v(a_{\alpha}) + |\alpha|\gamma$, and (63) with lemma 11.2(a) gives strong summability. Zero coordinates discard terms and cause no further difficulty.

If Γ is divisible, the ratio condition implies (63) by testing a ratio bound above $-\gamma$ and a positive element dominating $|b|$; conversely, for $c \in \Gamma$, (63) with $\gamma = -c$ and $b = 0$ gives $v(a_{\alpha}) > |\alpha|c$ for all but finitely many α . Finally, a nonpolynomial F has infinitely many nonzero coefficients, and (63) at $\gamma = 0$ makes their valuations a countable cofinal subset of Γ ; a nonzero ordered group has no finite cofinal subset. The converse is proposition 11.4. $\square$

Proposition 11.4 (Every ordinary support occurs). *Suppose $\text{cf}(\Gamma) = \aleph_0$. There is a sequence $(\varphi_n)_{n \geq 0}$ in Γ such that for every $\Sigma \subseteq \mathbb{N}^d$ and every choice of $c_\alpha \in \mathbf{k}^\times$, $\alpha \in \Sigma$, the series*

$$F_\Sigma(X) = \sum_{\alpha \in \Sigma} c_\alpha t^{\varphi_{|\alpha|}} X^\alpha$$

is entire and has ordinary support exactly Σ .

Proof. Choose a nondecreasing positive cofinal sequence $u_n \in \Gamma$ and put $\varphi_0 = 0$, $\varphi_n = n^2 u_n$ for $n \geq 1$. For fixed γ, b we eventually have $u_n > |\gamma| + |b|$, and then for $n \geq 2$

$$\varphi_n + n\gamma \geq n^2 u_n - n|\gamma| > n(n-1)u_n > b.$$

Nonzero elements of $\mathbf{k}$ have valuation zero, whatever their size in any other sense, so (63) holds. $\square$

Over $\mathbb{Q}$ the choice $\varphi_n = n^2$ works. The arbitrary-rank construction matters because $n^2\eta$ need not be cofinal when η is not an order unit. For $\Gamma = 0$ strong summability at $(1, \dots, 1)$ already forces a finite ordinary support, and every extension question below is trivial.

For the unit criterion put

$$\mathcal{T}_{\Gamma,d} = \left\{ \sum_{\alpha} a_{\alpha} X^{\alpha} : v(a_{\alpha}) \rightarrow +\infty \text{ as } |\alpha| \rightarrow \infty \right\}.$$

The following theorem is stated, as in the earlier appendix, under the report's standing hypotheses, Γ divisible and $\mathbf{k} = \mathbb{C}$; the fifth source does not treat it.

Theorem 11.5. *A nonconstant $F \in \mathcal{T}_{\Gamma,d}$ is a unit of that algebra if and only if $a_0 \neq 0$ and $\min_{\alpha \neq 0, a_{\alpha} \neq 0} v(a_{\alpha}/a_0)$ is a positive order unit of Γ .*

Proof. Regrouping into $\mathbb{C}[X_1, \dots, X_d]((t^{\Gamma}))$ gives multiplicativity of the Gauss valuation. If $FG = 1$, normalize both constant coefficients to 1; their Gauss values are nonpositive and sum to zero, so their initial polynomials multiply to 1 and are constant, and all nonconstant coefficient values are positive. If their minimum δ is not an order unit, coarsen by the proper convex subgroup (9) it generates: restrictedness reduces both series to finite polynomials over the coarsened residue field, and at least one nonconstant coefficient of F survives, contradicting invertibility of its reduction. Conversely an order-unit gap makes the geometric inverse restricted, with precisely the tail estimates of theorem 4.1; finite-variable convolution has finitely many terms at each multi-index, so those estimates are unchanged. $\square$

Finite-dimensionality is used in the finiteness of bounded-degree index sets. These statements assert nothing for infinitely many variables, and they do not extend the one-variable divisor classification to higher-dimensional zero sets. The single-variable uniform extension domain of theorem 10.4 is *false* in several variables, since a series independent of one coordinate imposes no constraint in that coordinate; its exact replacement is theorem 11.8.

11.3 The exact whole-family extension domain

Lemma 11.6 (Uniform unit twisting). *Let $(b_i)_{i \in I}$ be a family in K_{Δ} , let $\alpha(i) \in \mathbb{N}^d$, and let $u_1, \dots, u_d \in 1 + \mathfrak{m}_{\Delta}$, where $\mathfrak{m}_{\Delta} = \{x \in K_{\Delta} : v(x) > 0\}$. Then $(b_i)_i$ is strongly summable if and only if $(b_i \prod_j u_j^{\alpha(i)_j})_i$ is. Multiplying each term by a nonzero element of $\mathbf{k}$ also preserves and reflects strong summability.*

Proof. Let B be the support union of the first family and put $S = \bigcup_j \text{supp}(u_j - 1) \subseteq \Delta_{>0}$, a finite union of well-ordered sets. Every finite product $\prod_j u_j^{\alpha(i)_j}$ has support in the one monoid S^* , regardless of i , so the twisted support union lies in the well-ordered set $B + S^*$ (lemma 2.1). A fixed exponent has finitely many decompositions $b + s$ with $b \in B$, $s \in S^*$, and each b lies in finitely many of the original supports; this gives finite incidence. For the converse apply the same argument with $u_j^{-1} \in 1 + \mathfrak{m}_\Delta$. Nonzero scalars from $\mathbf{k}$ do not change supports. $\square$

Every nonzero $z_j \in K_\Delta$ has a unique leading-term decomposition $z_j = c_j t^{\delta_j} u_j$ with $c_j \in \mathbf{k}^\times$, $\delta_j = v(z_j)$ and $u_j \in 1 + \mathfrak{m}_\Delta$. Consequently $(a_\alpha z^\alpha)_\alpha$ is strongly summable if and only if $(a_\alpha t^{\langle \alpha, \bar{\delta} \rangle})_\alpha$ is, for arbitrary coefficient families. This removes the tails of the arguments, not those of the coefficients.

Lemma 11.7 (The finite lattice estimate). *Let $\delta_1, \dots, \delta_d \in \Delta$, put $\bar{\delta}_j = \pi_H(\delta_j)$, and let $\mathcal{L} = \{m \in \mathbb{Z}^d : \sum_j m_j \bar{\delta}_j = 0\}$. There are a positive integer $c_\mathcal{L}$ and $\eta \in \Gamma_{>0}$ with*

$$\left| \sum_j m_j \delta_j \right| \leq c_\mathcal{L} \|m\|_1 \eta \quad (m \in \mathcal{L}), \quad \|m\|_1 = \sum_j |m_j|. \quad (64)$$

Proof. A subgroup of $\mathbb{Z}^d$ is free of finite rank, by induction on d through the projection to the last coordinate. Choose a basis $\ell_1, \dots, \ell_r$ of $\mathcal{L}$. Each $h_s = \sum_j (\ell_s)_j \delta_j$ lies in H , so one $\eta \in \Gamma_{>0}$ bounds all $|h_s|$. The matrix with columns ℓ_s has a rational left inverse, and clearing its denominators shows that the integer coordinates of $m = \sum_s k_s \ell_s$ satisfy $\sum_s |k_s| \leq c_\mathcal{L} \|m\|_1$ for a positive integer $c_\mathcal{L}$ independent of m . Only this numerical estimate uses rational linear algebra; no division in Γ or Δ occurs. Hence $|\sum_j m_j \delta_j| = |\sum_s k_s h_s| \leq c_\mathcal{L} \|m\|_1 \eta$. If $\mathcal{L} = 0$, take $c_\mathcal{L} = 1$ and any η . $\square$

Theorem 11.8 (Exact multivariate extension domain). *Let $F = \sum_\alpha a_\alpha X^\alpha \in \mathcal{E}_{\Gamma, d}$ and $z \in (K_\Delta^\times)^d$. Put $\bar{\delta}_j = \pi_H(v(z_j))$ and*

$$\langle \Sigma(F), \bar{\delta} \rangle = \{ \langle \alpha, \bar{\delta} \rangle : \alpha \in \Sigma(F) \} \subseteq \Delta/H.$$

Then

$$(a_\alpha z^\alpha)_\alpha \text{ is strongly summable} \iff \langle \Sigma(F), \bar{\delta} \rangle \text{ is well ordered.} \quad (65)$$

Here $\langle \Sigma(F), \bar{\delta} \rangle$ is a set of distinct weights, not an indexed family: infinitely many multi-indices may have the same weight. If some coordinates of z vanish, restrict to the face of $\Sigma(F)$ on which those coordinates have exponent zero, delete them, and apply the same criterion; if no variable remains, evaluation is the constant coefficient.

Proof. Write $\delta_j = v(z_j)$. By lemma 11.6 we may replace z_j by t^{δ_j} , so the α th term becomes $b_\alpha = a_\alpha t^{\langle \alpha, \bar{\delta} \rangle}$, whose exponents all lie in $\langle \alpha, \bar{\delta} \rangle + \Gamma$ and map under π_H to $\langle \alpha, \bar{\delta} \rangle$.

Necessity. A strongly summable family has a well-ordered support union, and its image under the order-preserving map π_H is well ordered, because a strictly descending sequence of images lifts to a strictly descending sequence of chosen preimages. That image is $\langle \Sigma(F), \bar{\delta} \rangle$, since every term with $\alpha \in \Sigma(F)$ is nonzero.

Sufficiency within one fibre. Assume the weight set is well ordered, fix a weight w in it and some $\alpha^0 \in \Sigma(F)$ of weight w . For every α of the same weight, $m = \alpha - \alpha^0$ lies in the lattice $\mathcal{L}$ of lemma 11.7. After multiplying the whole fibre by $t^{-\langle \alpha^0, \bar{\delta} \rangle}$, its terms have all their exponents in H and valuations at least

$$v(a_\alpha) + \langle \alpha - \alpha^0, \bar{\delta} \rangle \geq (v(a_\alpha) - c_\mathcal{L} |\alpha| \eta) - c_\mathcal{L} |\alpha^0| \eta. \quad (66)$$

Condition (63) with $\gamma = -c_{\mathcal{L}}\eta$ says that the bracket exceeds any given bound in Γ except at finitely many indices, and since Γ is cofinal in H the same holds for bounds in H : for $b \in H$ choose $B \in \Gamma$ with $B \geq b + c_{\mathcal{L}}|\alpha^0|\eta$. Lemma 11.2(a), applied in the ordered group H , makes the fibre strongly summable.

Assembly of the fibres. Different fibres lie in different cosets of H . A nonempty subset of the full support union has a least coset because the weight set is well ordered, and within that coset a least exponent by the fibre result; that exponent is least in the whole subset. Each exponent belongs to one fibre, where finite incidence is already known. Finally, a zero coordinate removes every term with a positive exponent in it, the surviving coefficients still satisfy (63), and the same proof applies to the remaining variables. $\square$

Finiteness of d is used twice, for the finiteness of bounded-degree index sets and for the finite basis of $\mathcal{L}$. Entireness is used in (66), not merely to obtain a lower bound at one point. Full Hahn fields license the support-controlled unit inverses and sums, and convexity licenses the ordered quotient. No root extraction, algebraic closedness, spherical completeness, real-valued norm or divisibility enters.

Corollary 11.9 (Independence of tails within the entire category). *Two members of $\mathcal{E}_{\Gamma,d}$ with the same ordinary support have the same whole-family strong evaluation domain in every K_{Δ}^d , and for either of them the domain depends only on the coarsened coordinate valuations, with zero coordinates treated by faces as above.*

Proof. The right side of (65) and its face variants contain no other data. $\square$

This corrects the guess recorded in question 16.5. Within the entire category, higher Hahn tails do *not* create an additional boundary obstruction; without entireness they can, by example 11.19 below, which is a rank-one analogue of the pair of section 14.2.

11.4 Consequences: one variable, cones and semilinear supports

Corollary 11.10 (The one-variable domain, recovered). *For a nonpolynomial $f \in \mathcal{E}_{\Gamma,1}$ the strong evaluation domain in K_{Δ} is exactly $\mathcal{D}_{\Gamma,\Delta}$; a polynomial is strongly evaluable everywhere. Hence a cofinal enlargement preserves every entire series in any finite number of variables.*

Proof. For an infinite $\Sigma \subseteq \mathbb{N}$ the set $\{n\bar{\delta} : n \in \Sigma\}$ is a singleton for $\bar{\delta} = 0$, lies in a well-ordered positive cyclic monoid for $\bar{\delta} > 0$, and has a strictly descending sequence for $\bar{\delta} < 0$. Apply theorem 11.8. A cofinal enlargement has $H = \Delta$, so every weight set is empty or a singleton. $\square$

This is the domain half of theorem 10.4, now without divisibility and over any trivially valued coefficient field. The value inclusion of theorem 10.4, zero conservation and the continuation theorems of section 10 are not part of it.

Definition 11.11 (Admissible weights). For $\Sigma \subseteq \mathbb{N}^d$ and an ordered abelian group G put $\text{Adm}_{\Sigma}(G) = \{\bar{\delta} \in G^d : \langle \Sigma, \bar{\delta} \rangle \text{ is well ordered}\}$. It depends on the whole support, not on an enumeration.

Proposition 11.12 (Cone and monotonicity properties). *$\text{Adm}_{\Sigma}(G)$ contains $G_{\geq 0}^d$, is closed under addition and under multiplication by nonnegative integers, and is upward closed in the coordinatewise order. If G is divisible it is closed under nonnegative rational scaling, and for $G = \mathbb{R}$ it is a convex*

cone, not necessarily closed. If $\Sigma_1 \subseteq \Sigma_2$ then $\text{Adm}_{\Sigma_2}(G) \subseteq \text{Adm}_{\Sigma_1}(G)$, and a finite change of Σ does not change $\text{Adm}_{\Sigma}(G)$.

Proof. For nonnegative weights every element of $\langle \Sigma, \bar{\delta} \rangle$ lies in the monoid generated by the positive $\bar{\delta}_j$, which is well ordered by lemma 2.1(b). Addition uses $\langle \Sigma, \bar{\delta} + \bar{\epsilon} \rangle \subseteq \langle \Sigma, \bar{\delta} \rangle + \langle \Sigma, \bar{\epsilon} \rangle$ and lemma 2.1(a). Positive integer, rational or real scaling is an order isomorphism of the weight set where it is defined, and zero scaling gives at most a singleton. Upward closure is addition of a nonnegative vector. The remaining assertions are immediate, since adding finitely many weights preserves well-ordering. $\square$

On the torus $(K_{\Delta}^{\times})^d$ the strong domain of a fixed entire series is therefore a multiplicative monoid, stable under coordinatewise multiplication by nonzero elements of $\mathcal{D}_{\Gamma, \Delta}$. Its values need not lie in $\mathcal{D}_{\Gamma, \Delta}$: a finite monomial prefactor can have negative quotient weight.

Proposition 11.13 (Evaluation algebras at a fixed point). *Fix $z \in K_{\Delta}^d$. The $F \in \mathcal{E}_{\Gamma, d}$ whose whole-family evaluation at z exists form a K_{Γ} -subalgebra, and evaluation is a K_{Γ} -algebra homomorphism from it to K_{Δ} .*

Proof. By lemma 11.2(b), the sum and the double product of two evaluation families at z are strongly summable. Regroup the product by the multi-index sum; each formal coefficient has finitely many decompositions in $\mathbb{N}^d$. $\square$

Definition 11.14 (Semilinear support). $\Sigma \subseteq \mathbb{N}^d$ is semilinear if $\Sigma = \bigcup_{\ell=1}^s (b_{\ell} + \mathbb{N}p_{\ell 1} + \cdots + \mathbb{N}p_{\ell r_{\ell}})$ with $b_{\ell}, p_{\ell j} \in \mathbb{N}^d$; empty unions and components without periods are allowed.

Theorem 11.15 (The period-inequality criterion). *For a support presented as in definition 11.14, $\bar{\delta} \in \text{Adm}_{\Sigma}(G)$ if and only if $\langle p_{\ell j}, \bar{\delta} \rangle \geq 0$ for every period vector. The offsets impose no inequalities, so the extension domain of an entire series with semilinear support is decided by finitely many ordered-group inequalities.*

Proof. A period of negative weight gives the strictly descending weights of $b_{\ell} + \mathbb{N}p_{\ell j}$. If every period weight is nonnegative, each component has as its weight set a translate of a monoid generated by finitely many nonnegative elements, well ordered by lemma 2.1(b), and a finite union of translates is well ordered. $\square$

The inequalities are weak. An infinite support ray of quotient weight zero is not an obstruction, because old coefficient growth controls the whole collapsed fibre; replacing ≥ 0 by > 0 would give a false boundary.

Example 11.16 (A coupled domain). Over $\mathbb{C}((t^{\mathbb{Q}}))$ the series $F_{\times}(X, Y) = \sum_{n \geq 0} t^{n^2} (XY)^n$ is entire, has the single period $(1, 1)$, and at a point with nonzero coordinates in any extension is strongly evaluable exactly when $\bar{\delta}_X + \bar{\delta}_Y \geq 0$. In particular $X = t^{-e_*}, Y = t^{e_*}$ is admissible although $X \notin \mathcal{D}_{\Gamma, \Delta}$: every quotient weight is zero and the family is $(t^{n^2})_n$. The domain is not a coordinatewise copy of the one-variable ring. By contrast $G_{+}(X, Y) = 1 + \sum_{n \geq 1} t^{n^2} X^n + \sum_{n \geq 1} t^{n^2} Y^n$ has torus domain $\bar{\delta}_X \geq 0, \bar{\delta}_Y \geq 0$, so two nonpolynomial series involving both coordinates can have different domain geometries.

Theorem 11.17 (A nonclosed domain in two variables). *Let $\Gamma = \mathbb{Q}$, $\Delta = \mathbb{R}_{e_*} \oplus \mathbb{Q}$ as above, and $F_{\text{par}}(X, Y) = \sum_{n \geq 1} t^{n^4} X^n Y^{n^2}$. This series is entire over $\mathbb{C}((t^{\mathbb{Q}}))$, and with Δ/H identified with $\mathbb{R}$,*

$$\text{Adm}_{\Sigma(F_{\text{par}})}(\mathbb{R}) = \{(u, w) : w > 0\} \cup \{(u, 0) : u \geq 0\}.$$

This set is convex but not closed, and it is not a finite intersection of weak linear half-spaces.

Proof. The total degree is $n + n^2$, and $n^4 + (n + n^2)\gamma \rightarrow +\infty$ for every rational γ . The weight set is $\{nu + n^2w : n \geq 1\}$. For $w > 0$ the successive differences $u + (2n + 1)w$ are eventually positive, so the set is well ordered; for $w < 0$ they are eventually negative; for $w = 0$ the set is well ordered exactly when $u \geq 0$. Apply [theorem 11.8](#). The admissible points $(-1, 1/m)$ tend in the real plane to $(-1, 0)$, which is not admissible, and a finite intersection of weak half-spaces is closed. $\square$

This concerns the Euclidean topology of the real weight parameter, not any surreal fine topology.

Example 11.18 (Bounded positive weights are not enough). Fix an irrational real $\theta > 0$, put $\ell_n = \lceil n\theta \rceil$, and let $F_\theta(X, Y) = \sum_{n \geq 1} t^{(n+\ell_n)^2} X^n Y^{\ell_n}$ over $\mathbb{C}((t^\mathbb{Q}))$, entire by the degree-squared damping. At the quotient weights $(-\theta, 1)$ its weights are $\ell_n - n\theta \in (0, 1)$, never 0, with infimum 0: for any N , two of the fractional parts of $0, \theta, \dots, N\theta$ lie within $1/N$, so some positive integer q has $q\theta$ within $1/N$ of an integer; if its fractional part exceeds $1 - 1/N$ this is a small positive gap, and if it is $\epsilon \in (0, 1/N)$ then, with $k = \lfloor 1/\epsilon \rfloor$, irrationality gives $0 < 1 - k\epsilon < \epsilon$ and kq has a gap below $1/N$. So the weights have no least element and the point is not in the strong domain, although every weight is positive. Well-ordering, not positivity or boundedness, is the criterion.

Example 11.19 (What fails without entireness). In $\mathbb{C}((t^\mathbb{Q}))[[X]]$ let $\tilde{a}_n = t^{1-1/(n+1)}$ and $\tilde{b}_n = t^{1-1/(n+1)} + t^{2+1/(n+1)}$. The series $A_b = \sum \tilde{a}_n X^n$ and $B_b = \sum \tilde{b}_n X^n$ have the same ordinary support and the same coefficient valuations. At $X = 1$ the family of A_b is strongly summable, its exponents strictly increasing and each occurring once, while that of B_b is not, its second exponents strictly decreasing. Neither series is entire: at $X = t^{-1}$ even the leading exponents strictly decrease. So [corollary 11.9](#) is not a general principle that leading data decide Hahn summability; entireness is doing real work.

11.5 Homogeneous blocks: a second, explicitly different convention

For a formal series F write $F_{(n)}(X) = \sum_{|\alpha|=n} a_\alpha X^\alpha$, a finite homogeneous polynomial over K_Γ , so that $F = \sum_n F_{(n)}$.

Definition 11.20 (Homogeneous-block evaluation). At $z \in K_\Delta^d$ the homogeneous-block evaluation of F exists if $(F_{(n)}(z))_{n \geq 0}$ is strongly summable; its value is written $F^{\text{hom}}(z)$ and its domain $\mathcal{D}^{\text{hom}}(F)$.

Whole-family evaluation implies homogeneous-block evaluation with the same value, by [lemma 11.2\(b\)](#). The reverse implication is not part of the definition and is false. Block evaluation is never called entire in this report.

Proposition 11.21 (Linear covariance). *For a matrix M over K_Γ defining the substitution $X = MY$, an entire F gives an entire formal series $F \circ M$ whose degree- n block is $F_{(n)} \circ M$. Consequently $y \in \mathcal{D}^{\text{hom}}(F \circ M)$ if and only if $My \in \mathcal{D}^{\text{hom}}(F)$, with equal values; this is a covariance law under $\text{GL}_d(K_\Gamma)$, and the same argument applies to non-square M , in particular to line restrictions.*

Proof. For $M \neq 0$ choose $\mu \in \Gamma$ not exceeding 0 or the valuation of any nonzero entry. Every new degree- n coefficient is a finite sum of old degree- n coefficients times products of n entries and integers, so its valuation is at least $b_n + n\mu$, where b_n is the least valuation of the nonzero coefficients of $F_{(n)}$; cancellation only raises it. Condition (63) at $\gamma + \mu$ gives the same condition for the new coefficients at γ . By homogeneity the degree- n block of $F \circ M$ at y is exactly $F_{(n)}(My)$, so the two block families coincide. $\square$

Example 11.22 (Cancellation changes the domain). Over $\mathbb{C}((t^{\mathbb{Q}}))$ let $F_{\text{can}}(X, Y) = \sum_{n \geq 0} t^{n^2} (X - Y)^n$. At (t^{-e_*}, t^{-e_*}) , with e_* above all old scales, every block of positive degree is zero, so the block evaluation exists and equals 1. The whole monomial family is not strongly summable: for every n it contains the term $t^{n^2} X^n$ of quotient weight $-n\pi_H(e_*)$, and these weights strictly decrease. In the coordinates $U = X - Y$, $V = Y$ the same series is $\sum t^{n^2} U^n$, whose whole family is summable at $(0, t^{-e_*})$. So whole-family domains depend on the chosen coordinate expansion, while block domains obey [proposition 11.21](#); the value 1 is a value of the block convention, not a sum assigned to the nonsummable expanded family.

Theorem 11.23 (Polynomial rays and new scales). *Let $F \in \mathcal{E}_{\Gamma, d}$ and $c \in K_{\Gamma}^d$, and let $F_c(T) = F(Tc) = \sum_n F_{(n)}(c)T^n$, an entire one-variable series over K_{Γ} . For every ordered-group enlargement,*

$$\{\lambda \in K_{\Delta} : \lambda c \in \mathcal{D}^{\text{hom}}(F)\} = \begin{cases} K_{\Delta}, & F_c \text{ is a polynomial,} \\ \mathcal{D}_{\Gamma, \Delta}, & F_c \text{ is not a polynomial.} \end{cases}$$

In particular, if $e_ > H$ then $t^{-e_*}c \in \mathcal{D}^{\text{hom}}(F)$ if and only if $F_{(n)}(c) = 0$ for all sufficiently large n .*

Proof. [Proposition 11.21](#) gives entireness of F_c . The block family at λc is exactly $(F_{(n)}(c)\lambda^n)_n$; apply [corollary 11.10](#). Directly: if infinitely many $F_{(n)}(c)$ are nonzero, the corresponding terms at $t^{-e_*}c$ have quotient valuations $-n\pi_H(e_*)$, strictly descending, and otherwise the evaluation is a finite sum; this needs no factorization or zero theorem. $\square$

The one-variable theorem is an input here, not a new priority claim. Directions not defined over K_{Γ} need a different analysis, and [theorem 11.23](#) does not classify block values at all points of K_{Δ}^d .

For nonempty $J \subseteq \{1, \dots, d\}$ put $\Sigma_J = \{\alpha \in \Sigma(F) : \alpha_j = 0 \text{ for } j \notin J\}$ and $\mathbb{P}_J = \{[c] \in \mathbb{P}^{d-1}(\mathbb{C}) : c_j = 0 \text{ for } j \notin J\}$.

Proposition 11.24 (Directions for the whole family). *Let $\mathbf{k} = \mathbb{C}$, $F \in \mathcal{E}_{\Gamma, d}$ and $e_* > H$. The ordinary complex directions $[c]$ at which the whole monomial family is strongly summable at $t^{-e_*}c$ are exactly $\bigcup \{\mathbb{P}_J : J \neq \emptyset, \Sigma_J \text{ finite}\}$, a finite union of coordinate projective subspaces. For nonpolynomial F it is a proper algebraic subset, and it is contained in the block-exceptional locus $\text{Pdir}(F)$ of [section 11.6](#).*

Proof. Let J be the set of nonzero coordinates of c . Every surviving term has quotient weight $-|\alpha|\pi_H(e_*)$, and these are well ordered exactly when the degrees in Σ_J are bounded, that is, when Σ_J is finite. The family of such J is downward closed, so the admissible directions are the union of the $\mathbb{P}_J$. For nonpolynomial F the full coordinate set is not admissible, so a direction with all coordinates nonzero lies outside. Inclusion in the block locus is regrouping. $\square$

Proposition 11.25 (Realizing the coordinate pattern). *Suppose $\text{cf}(\Gamma) = \aleph_0$. Every downward-closed family $\mathcal{J}$ of subsets of $\{1, \dots, d\}$ containing $\emptyset$ is the family of J with Σ_J finite, for some entire series; if the full set is not in $\mathcal{J}$, the series may be chosen nonpolynomial.*

Proof. If every subset lies in $\mathcal{J}$, take a polynomial. Otherwise let p_B be the indicator vector of each inclusion-minimal $B \notin \mathcal{J}$, and put $\Sigma = \bigcup_B \{np_B : n \geq 1\}$. Its face Σ_J is infinite exactly when some such B is contained in J , that is, exactly when $J \notin \mathcal{J}$, and otherwise it is empty. [Proposition 11.4](#) realizes Σ . $\square$

This finite coordinate classification contrasts with the much larger family of block-exceptional loci below, and it detects when a claimed multivariate extension theorem has silently changed its summation convention.

11.6 Exceptional projective directions

In this subsection $\mathbf{k} = \mathbb{C}$, and projective space means the ordinary set $\mathbb{P}^{d-1}(\mathbb{C})$.

Definition 11.26 (Polynomial-direction locus). For $F = \sum_n F_{(n)}$ over K_Γ put $\text{Pdir}(F) = \{[c] \in \mathbb{P}^{d-1}(\mathbb{C}) : F(Tc) \in K_\Gamma[T]\}$, which does not depend on the representative of $[c]$. For entire F , [theorem 11.23](#) identifies it with the ordinary directions admitting homogeneous-block evaluation at a new dominating scale.

The algebraic statements below hold for arbitrary formal F over K_Γ ; entireness is needed only for the radial domain formula and the analytic interpretation.

Theorem 11.27 (Necessary form and generic obstruction). *For every $F \in K_\Gamma[[X_1, \dots, X_d]]$ there are ascending projective algebraic sets $\mathcal{Z}_N \subseteq \mathbb{P}^{d-1}(\mathbb{C})$ with*

$$\text{Pdir}(F) = \bigcup_{N \geq 0} \mathcal{Z}_N, \quad \mathcal{Z}_N = \{[c] : F_{(n)}(c) = 0 \text{ for all } n > N\}.$$

If F is not a polynomial, every $\mathcal{Z}_N$ is proper, $\text{Pdir}(F)$ is not all of projective space, and its complement is dense in the ordinary complex topology. For $d = 2$ the set $\text{Pdir}(F)$ is at most countable.

Proof. Coefficientwise, $F_{(n)}(X) = \sum_{\eta \in S_n} t^\eta P_{n,\eta}(X)$ with ordinary complex homogeneous polynomials $P_{n,\eta}$ of degree n , S_n being the finite union of the Hahn supports of the finitely many coefficients of $F_{(n)}$. For a complex vector c , $F_{(n)}(c) = 0$ exactly when every $P_{n,\eta}(c) = 0$. So $\mathcal{Z}_N$ is the common zero set of a family of homogeneous complex polynomials; the ideal they generate is finitely generated by the Hilbert basis theorem [18], so $\mathcal{Z}_N$ is projective algebraic. The sets ascend, and their union is eventual vanishing.

If F is nonpolynomial, each N is followed by a nonzero block containing some nonzero $P_{n,\eta}$, which vanishes on $\mathcal{Z}_N$ and not identically; so $\mathcal{Z}_N$ is proper. Choose one nonzero homogeneous p_N vanishing on each $\mathcal{Z}_N$. Their coefficients generate a countable subfield $L \subseteq \mathbb{C}$. In the chart $c_1 = 1$ the polynomials $p_N(1, Y_2, \dots, Y_d)$ are nonzero; choosing $d - 1$ complex numbers algebraically independent over L , successively outside the algebraic closure of a countable field, gives a point outside every $\mathcal{Z}_N$. The same choices can be made inside any product of nonempty open disks, since only countably many values are forbidden at each step, which gives density. For $d = 1$ the locus of a nonpolynomial series is empty. Finally, in $\mathbb{P}^1(\mathbb{C})$ a proper algebraic set is finite, and a countable union of finite sets is at most countable. $\square$

The covering argument is the countable-field device of Lemma `hol:lem:generic` in the report `docs/surcomplex/holonomic-rigidity-for-entire-hahn-functions` [30], which chooses $c \in \mathbb{C}^d$ avoiding countably many nonzero polynomials over K_Γ ; here it is applied to complex polynomials and refined to disks for density. The device is elementary, and neither report claims it as new. No countability of Hahn supports is hidden: an individual coefficient may have uncountable support, the proof never enumerates exponents, and Noetherianity reduces each $\mathcal{Z}_N$ to finitely many equations while there are only countably many N .

Theorem 11.28 (Exact realization of the exceptional locus). *Suppose $\text{cf}(\Gamma) = \aleph_0$. For every countable union $A = \bigcup_{m \geq 1} Y_m \subseteq \mathbb{P}^{d-1}(\mathbb{C})$ of projective algebraic sets there is $F \in \mathcal{E}_{\Gamma,d}$ with $\text{Pdir}(F) = A$. If A is proper, F can be chosen nonpolynomial; if A is all of projective space, no nonpolynomial formal series has that locus, and a polynomial realizes it.*

Proof. The last case is $F = 0$ together with [theorem 11.27](#). Let A be proper and replace Y_m by $Y_1 \cup \dots \cup Y_m$, so that the sets ascend and are proper. For each m choose nonzero homogeneous polynomials $p_{m,1}, \dots, p_{m,r_m}$ with common projective zero set Y_m (the single polynomial 1 if Y_m is empty). To each triple (m, k, j) , $1 \leq k \leq r_m$, $1 \leq j \leq d$, assign a distinct integer $D_{m,k,j} > \deg p_{m,k}$, in increasing order block by block in m , and put $Q_{D_{m,k,j}} = X_j^{D_{m,k,j} - \deg p_{m,k}} p_{m,k}$, with $Q_n = 0$ at unassigned degrees. The block of index m has common projective zero set exactly Y_m : off Y_m some $p_{m,k}$ and some coordinate c_j are nonzero, and using every coordinate prevents a spurious exceptional hyperplane. Set $F = \sum_n t^{\varphi_n} Q_n$ with the damping of [proposition 11.4](#); each nonzero monomial coefficient of degree n has valuation φ_n , so F is entire, and it is nonpolynomial. If $[c] \in A$ it lies in some Y_M and all later blocks, so $F(Tc)$ is a polynomial. If $[c] \notin A$, every block has a polynomial not vanishing at c , and the assigned degrees are unbounded. $\square$

Corollary 11.29 (Sharp two-variable classification). *If $\text{cf}(\Gamma) = \aleph_0$, the ordinary projective directions admitting new-scale homogeneous-block evaluation of a nonpolynomial entire series in two variables are exactly the at most countable subsets of $\mathbb{P}^1(\mathbb{C})$.*

Proof. Necessity is [theorem 11.27](#); a countable set is a countable union of points, so [theorem 11.28](#) gives sufficiency; the identification with new-scale evaluation is [theorem 11.23](#). $\square$

For pairwise distinct $c_1, c_2, \dots \in \mathbb{C}$ the series

$$F_{\text{dir}}(X, Y) = 1 + \sum_{n \geq 1} t^{n^2} \prod_{j=1}^n (Y - c_j X)$$

over $\mathbb{C}((t^{\mathbb{Q}}))$ is entire, and its restriction to $[1 : c_m]$ is the polynomial $1 + \sum_{n < m} t^{n^2} \prod_{j \leq n} (c_m - c_j) T^n$ of degree $m - 1$, while every other slope and the direction $[0 : 1]$ give nonpolynomial restrictions. So $\text{Pdir}(F_{\text{dir}}) = \{[1 : c_m] : m \geq 1\}$; a projective linear change and [proposition 11.21](#) place a prescribed point at infinity. Enumerating $\mathbb{Q} + i\mathbb{Q}$ gives a countable dense exceptional locus: polynomial behaviour on a dense set of complex lines does not force a Hahn series to be polynomial, while polynomial behaviour on *every* complex line does, by [theorem 11.27](#).

The exceptional set is neither an ordinary convergence set nor a zero set of F : it asks for eventually vanishing coefficients on a line, which the enlargement turns into a summability condition because a negative new quotient valuation overwhelms every old coefficient. Ma and Neelon already realize countable unions of projective algebraic sets as convergence sets of divergent complex power series [[17](#), Corollary 4.19]. That realization idea is not claimed as new, their stronger analytic theorems are not reproved here, and convergence sets must not be conflated with polynomial-direction loci. What is proposed is the exact Hahn extension interpretation with the complete classification and an entire realization over every countably cofinal value group.

11.7 Transfer to surreal and surcomplex numbers

Conway's omega-map $x \mapsto \omega^x$ and his normal-form theorem identify surreal numbers with real generalized series with set-sized reverse well-ordered supports [[20](#), Theorem 10.2]; this is the map $(1), t^\gamma \mapsto \omega^{-\gamma}$, and it must not be confused with an exponential defined from Gonshor's exp. For a set-sized ordered subgroup $\Gamma \subseteq \mathbf{No}$ it embeds $\mathbb{R}((t^\Gamma))$ in $\mathbf{No}$ and $\mathbb{C}((t^\Gamma))$ in $\mathbf{No}[i]$, preserving and reflecting the support conditions that define strong sums.

Lemma 11.30 (Localization of the data). *Every set of surreal or surcomplex numbers lies in such an embedded workspace for a nonzero divisible set-sized ordered subgroup $\Gamma \subseteq \mathbf{No}$, and adjoining a finite tuple of arguments needs at most a set-sized enlargement $\Gamma \subseteq \Delta$.*

Proof. Take the union of the exponent supports of the real and imaginary parts of all given numbers, adjoin 1, and take the rational linear span in $\mathbf{No}$; this is a divisible set, unchanged by negating exponents. Repeat after adjoining the arguments. $\square$

This is the localization principle of the foundations report [29], already consumed by [corollary 10.12](#); the fifth source gives the short proof above. Localization says nothing about entireness, which remains a hypothesis on a specified old workspace.

Corollary 11.31 (Surcomplex support-domain theorem). *Let $\Gamma \subseteq \mathbf{No}$ be a set-sized ordered subgroup and F whole-family entire over the embedded field $\mathbb{C}((t^\Gamma))$. For $z \in \mathbf{No}[i]^d$ choose a set-sized Δ containing Γ and the exponent supports of z , and put $H = \text{conv}_\Delta \Gamma$. After discarding the terms killed by zero coordinates, the family defining $F(z)$ is strongly summable exactly when $\{\sum_j \alpha_j(v(z_j) + H) : \alpha_\alpha \neq 0\}$ is well ordered in Δ/H . The answer does not depend on the choice of Δ .*

Proof. Transport [theorem 11.8](#). For $\Delta \subseteq \Delta'$ the hull H' satisfies $H' \cap \Delta = H$, so Δ/H order-embeds in Δ'/H' and well-ordering of the same weight set is unchanged. $\square$

The real version follows with $\mathbb{R}$ in place of $\mathbb{C}$. Neither needs algebraic closedness of $\mathbf{No}[i]$, a derivation or a square root. The classification of [section 11.6](#) uses ordinary complex directions and is not asserted over an arbitrary residue field.

Two phenomena are explicit. With $\Gamma = \mathbb{Q}$ and $t = \omega^{-1}$, the series $\sum_{n \geq 0} \omega^{-n^2} (XY)^n$ is entire in that workspace; after enlarging by the dominant exponent $e_* = \omega$ it is strongly evaluable at $X = \omega^\omega$, $Y = \omega^{-\omega}$, both outside the old field and the first outside its one-variable extension domain, because the family is exactly $(\omega^{-n^2})_n$ and every weight is zero. Infinite fibres are therefore necessary in [theorem 11.8](#). And with pairwise distinct $c_j \in \mathbb{C}$, the series $1 + \sum_{n \geq 1} \omega^{-n^2} \prod_{j \leq n} (Y - c_j X)$ admits block evaluation at $\omega^\omega(1, c_m)$, where it is the finite polynomial above at $T = \omega^\omega$, while at every other ordinary slope and at the direction at infinity its nonzero blocks have strictly descending leading exponents: a prescribed countable set of directions crosses a genuinely surreal scale and every other direction fails.

Proposition 11.32 (Whole-class rigidity in both conventions). *Let $F \in \mathbf{No}[i][[X_1, \dots, X_d]]$ be one formal series in finitely many variables. If its whole monomial family is strongly summable at every surcomplex tuple, F is a polynomial. The same holds if instead its sequence of homogeneous blocks is strongly summable at every surcomplex tuple.*

Proof. Localize the coefficients in some K_Γ ([lemma 11.30](#)) and choose a positive surreal e_* above the set Γ . For a nonpolynomial whole family at the diagonal point $(\omega^{e_*}, \dots, \omega^{e_*})$, the leading exponents $v(a_\alpha) - |\alpha|e_*$ along strictly increasing degrees strictly decrease. For blocks, [theorem 11.27](#), which does not need entireness, gives a complex direction c with nonpolynomial restriction, and evaluating that restriction at ω^{e_*} again gives strictly descending leading exponents. $\square$

The whole-family half is already in this collection, in [corollary 10.12](#) for one variable and in the analysis report, and is not reclaimed; the block half shows that cancellation on exceptional lines does not evade global rigidity. Neither forbids a different analytic category of local germs,

non-power-series functions or another specified summation, and no strong sum is identified with a limit in the fine surreal topology.

11.8 What this section does not settle

The following limitations are the fifth source's and are retained without weakening.

- Only the scalar-extension clause of [question 16.5](#) is answered, for the whole-family convention. The preparation, interpolation, divisor and ideal-theoretic clauses are not settled; no several-variable GCD or Bézout classification is asserted, and a positive-dimensional zero set cannot be replaced by the one-variable finite root divisor of [section 7](#).
- The full homogeneous-block domain at an arbitrary tuple of new-field coordinates is not classified. [Theorem 11.23](#) covers rays whose direction is defined over the old field, and new-field direction ratios can contain new scales, so that the block coefficients leave the old field after substitution ([question 16.6](#)).
- The number of variables is finite, and this is used; all fields are full Hahn fields, and nothing is claimed for support-restricted or incomplete subfields, where an ambient strong sum need not lie in the smaller field.
- The semilinear test is a finite decision procedure only relative to a supplied semilinear presentation and exact ordered-group operations on Δ/H . It does not discover supports, recognize semilinearity, build the convex-hull quotient from a black-box exponent oracle or decide general well-ordering, and no finite sampling can certify the general criterion: every finite support has a well-ordered weight set at every point ([question 16.7](#)).
- [Section 11.6](#) depends on $\mathbf{k} = \mathbb{C}$ and ordinary complex directions.
- The finite-word well-quasi-order theorem behind [lemma 2.1\(b\)](#) and the Hilbert basis theorem are imported, and choice is used for lattice bases, coefficient bounds and algebraically independent coordinates.

12 Prime spectra above a one-node-per-shell product

This section is the sixth source's contribution. It describes every prime ideal of $\mathcal{E}_\Gamma$ that contains a canonical product with one simple node per shell, not only the maximal ones of [theorem 8.36](#); it decides when the quotient by such a product is zero-dimensional; it computes the contraction of these primes under a cofinal enlargement of the value group, with an explicit surreal example in which the prime spectrum refines while the function and its zeros do not change; and it locates the multiplicity threshold beyond which the reduced description is incomplete. It answers the “prime ideals” clause of [question 16.4](#) for one simple node per shell, and only for that case; the non-claim about prime ideals recorded at the end of [section 8.9](#) is kept there as written, now with a pointer to this section. Three warnings govern the section.

- (1) *What is inherited.* The analytic input is the quotient theorem [theorem 8.7](#) in its one-node form [corollary 8.13](#), over $\mathbb{R}((t^\Gamma))$ by [proposition 8.42](#). The sixth source reproved the simple-node case so as not to depend on this report's preparation theory; its argument is the first route of [theorem 8.6](#) specialized to one simple node per shell, and it is printed once, by

citation (remark 12.2). The maximal ideals, their residue fields and the maximal spectrum are theorems 8.33, 8.36 and 8.37; the sixth source recovers them from the full spectrum and does not claim them.

- (2) *What is classical.* The idempotent and ultraproduct mechanism for the primes of a product ring, including products of Prüfer domains described by ultrafilters, ultraproduct valuations and cuts in value groups, is due to Finocchiaro, Frisch and Windisch [15], a direct precedent for lemma 12.5 and proposition 12.6. The correspondence between the primes of a valuation domain and the convex subgroups of its value group is standard [19]. The Hahn support calculus is Neumann's [3], and the normal forms used in section 12.8 are Conway's and Gonshor's [1, 2]. The sixth source proposes as new only the exact analytic realization of these mechanisms, the order-unit dichotomy for the reduced divisor and the cofinal splitting construction, and it certifies none of them against the whole literature (section 12.10).
- (3) *Words and letters.* *Spectrum* means prime spectrum Spec throughout this section; the maximal spectrum is written Max , as in theorem 8.37. The sixth source's letters are translated by the last table of section 2.5. In particular its sequence ring $\mathcal{A}$ is written $\mathcal{R}_D$ here, because our $\mathcal{A}_\Gamma$ is the Hahn algebra of section 3; its convex subgroups C and D are written $\mathcal{C}$ and $\mathcal{C}'$, because our C_n are corrections and our D is a divisor; and its coefficient field k is written $\mathbf{k}$.

12.1 Setting and the inherited quotient

Convention 12.1 (Standing hypotheses of this section). Unless a second group is introduced explicitly, Γ is a nonzero set-sized divisible ordered abelian group with $\text{cf}(\Gamma) = \aleph_0$. The coefficient field $\mathbf{k}$ is $\mathbb{R}$ or $\mathbb{C}$, with the trivial valuation; in this section K_Γ denotes $\mathbf{k}((t^\Gamma))$ and $\mathcal{E}_\Gamma$ its ring of strongly entire series (for $\mathbf{k} = \mathbb{C}$ these are K_Γ and $\mathcal{E}_\Gamma$, for $\mathbf{k} = \mathbb{R}$ the real forms of proposition 8.42). In section 11 the letter $\mathbf{k}$ denoted an arbitrary trivially valued field; here it is $\mathbb{R}$ or $\mathbb{C}$, although the sixth source records that most of its proofs work over any trivially valued field. Fix a strictly increasing positive sequence $\gamma_1 < \gamma_2 < \dots$ cofinal in Γ and put

$$\rho_n = t^{-\gamma_n}, \quad D = \sum_{n \geq 1} [\rho_n], \quad P_D(Z) = \prod_{n \geq 1} (1 - t^{\gamma_n} Z) = \prod_{n \geq 1} \left(1 - \frac{Z}{\rho_n}\right). \quad (67)$$

The letters $H_n = H_{\gamma_n}$, $V_n = V_{\gamma_n}$ and $\mathfrak{q}_{\gamma_n}$ are those of definition 8.3 and lemma 8.4, and $\kappa_n = \mathbf{k}((t^{H_n}))$. Ultrafilters are ultrafilters on $\mathbb{N}_{>0}$; nonprincipal ones exist by the axiom of choice, which is used, and no particular one is computed. All rings, products and ultraproducts below are sets.

The node ρ_n is the shell normalizer τ_n of (18), and P_D is the canonical product of theorem 7.3. The product in (67) is not the product P of (85), which runs over $j \geq 0$; for $\Gamma = \Gamma_\infty$ and $\gamma_n = e_n$ it is the function f of (83). By theorem 8.7 and corollary 8.13 (and proposition 8.42 over $\mathbb{R}$), evaluation at the nodes induces a K_Γ -algebra isomorphism

$$\begin{aligned} \mathcal{E}_\Gamma/(P_D) &\xrightarrow{\sim} \mathcal{R}_D, & F &\mapsto (F(\rho_n))_n, \\ \mathcal{R}_D &= \{(b_n)_{n \geq 1} \in K_\Gamma^{\mathbb{N}_{>0}} : b_n \in V_n \text{ for all sufficiently large } n\}, \end{aligned} \quad (68)$$

the ring displayed in section 8.8; operations are coordinatewise. We write $\mathcal{R}_D^\Gamma$ when the group varies. From now on $\mathcal{E}_\Gamma/(P_D)$ is identified with $\mathcal{R}_D$. A prime of $\mathcal{R}_D$ is the same as a prime of $\mathcal{E}_\Gamma$ containing P_D , through its inverse image; in particular a *minimal* prime of $\mathcal{R}_D$ is *minimal above* (P_D) , not a minimal prime of the domain $\mathcal{E}_\Gamma$.

Remark 12.2 (The sixth source's analytic route, printed once). The sixth source proves, for the monomial nodes (67), a coefficient criterion, Gauss multiplicativity, the canonical product with a weighted bound on the formal inverse of its tails, division by a linear factor, translation and additivity of orders, the principal kernel and the exact value image. These are [theorem 3.1](#), [proposition 3.5](#) with (5), [theorem 7.3](#) with (26), [lemma 6.4](#), [proposition 3.5](#) with [corollary 7.6](#), [proposition 8.8](#) and [corollary 8.13](#); its weighted value $\min_j \{v(a_j) - jr\}$ is m_{-r} of (14). Its sufficiency construction fits finitely many values by a polynomial with coefficients in some V_L and then adds

$$C_n(Z) = y_n \prod_{j < n} \frac{Z - \rho_j}{\rho_n - \rho_j} \left(\frac{Z}{\rho_n} \right)^{d_n}, \quad d_n \geq \max\{n, 2c_n + 2\},$$

where y_n is the residual value and $v(y_n) \geq -c_n \gamma_n$. This is the correction (32) of the first route for one simple node per shell: with $\tau_n = \rho_n$ the normalized node is 1, the shell algebra is K_Γ , $\Psi_{n,d}$ is the constant $y_n/Q_n(1)$, and $(Z/\tau_n)^d Q_n(Z/\tau_n)/Q_n(1)$ is the displayed Lagrange factor times $(Z/\rho_n)^d$. The initialization over V_L and the damping bound (34) are the same. Its kernel proof, like [proposition 8.8](#), divides by the finite product $P_{\leq N}$ and multiplies by the formal inverse of the tail; it controls the quotients through the stabilized Gauss value $m_{-r}(F/P_{\leq N}) = m_{-r}(F) - \sum_{n \leq N} \min\{0, \gamma_n - r\}$ rather than through a coefficient bound uniform in N . Both are therefore printed once, here by citation; the kernel argument differs from [proposition 8.8](#) only in that bookkeeping and is not kept as a separate route. Like the first route, neither uses preparation or algebraic closedness.

Lemma 12.3 (Indicator idempotents). *For every $S \subseteq \mathbb{N}_{>0}$ the indicator sequence $e_S = (\mathbf{1}_S(n))_n$ lies in $\mathcal{R}_D$, and $e_{S \cap T} = e_S e_T$, $e_S + e_{\mathbb{N}_{>0} \setminus S} = 1$. Some $F \in \mathcal{E}_\Gamma$ interpolates it. The ring $\mathcal{R}_D$ is reduced, so (P_D) is a radical ideal of $\mathcal{E}_\Gamma$.*

Proof. The values 0 and 1 lie in every V_n , so [corollary 8.13](#) supplies an interpolant; it need not be idempotent in $\mathcal{E}_\Gamma$ and becomes idempotent only modulo (P_D) . The identities are coordinatewise. The ring $\mathcal{R}_D$ is a subring of a product of fields, hence reduced. $\square$

The subscript of e_S is a set; an integer subscript, as in e_j , denotes a basis vector of Γ_∞ , as elsewhere in this report. The same idempotents were used in the proof of [theorem 8.36](#).

Remark 12.4 (Other nodes on the same radii). The sixth source takes the monomial nodes $\rho_n = t^{-\gamma_n}$. The ring $\mathcal{R}_D$ of (68) depends only on the radii: by [theorem 8.7](#) and [corollary 8.13](#) it is also the quotient of $\mathcal{E}_\Gamma$ by the canonical product of any divisor with one simple node ρ'_n of valuation $-\gamma_n$ on each shell and empty finite part, through $F \mapsto (F(\rho'_n))_n$. Consequently [sections 12.2 to 12.6](#), which are statements about $\mathcal{R}_D$ and about inverse images under that evaluation map, hold verbatim for such nodes. This observation is the merge's. [Sections 12.7 to 12.9](#) are stated only in the source's setting.

12.2 The ultrafilter fibres

For an ultrafilter $\mathcal{U}$ put

$$\mathfrak{n}_\mathcal{U} = \{a \in \mathcal{R}_D : \{n : a_n = 0\} \in \mathcal{U}\}. \quad (69)$$

For the principal ultrafilter at n this is the coordinate ideal

$$\mathfrak{e}_n = \{a \in \mathcal{R}_D : a_n = 0\}, \quad \mathcal{R}_D/\mathfrak{e}_n \cong K_\Gamma,$$

whose inverse image in $\mathcal{E}_\Gamma$ is the evaluation ideal $(Z - \rho_n)\mathcal{E}_\Gamma$ of [corollary 7.6](#).

Lemma 12.5 (Idempotents select exactly one ultrafilter). *For a prime ideal $\mathfrak{p}$ of $\mathcal{R}_D$ the set $\mathcal{U}_{\mathfrak{p}} = \{S \subseteq \mathbb{N}_{>0} : e_S \notin \mathfrak{p}\}$ is an ultrafilter, and $\mathfrak{p} \supseteq \mathfrak{n}_{\mathcal{U}_{\mathfrak{p}}}$. Primes with different selected ultrafilters are incomparable.*

Proof. In the domain $\mathcal{R}_D/\mathfrak{p}$ every idempotent is 0 or 1, and the identities of lemma 12.3 give the ultrafilter axioms. If a vanishes on some $S \in \mathcal{U}_{\mathfrak{p}}$, then $ae_S = 0$ with $e_S \notin \mathfrak{p}$, so $a \in \mathfrak{p}$. If $\mathfrak{p} \subseteq \mathfrak{p}'$ and $e_S \notin \mathfrak{p}'$, then $e_S \notin \mathfrak{p}$; thus $\mathcal{U}_{\mathfrak{p}'} \subseteq \mathcal{U}_{\mathfrak{p}}$, and two ultrafilters one of which contains the other are equal. $\square$

Proposition 12.6 (The fibre ring). *Let $\mathcal{U}$ be nonprincipal. Replacing the finitely many coordinates of $a \in \mathcal{R}_D$ that lie outside V_n by zero and passing to the class modulo $\mathcal{U}$ defines a surjective ring homomorphism*

$$\text{red}_{\mathcal{U}} : \mathcal{R}_D \longrightarrow \mathcal{V}_{\mathcal{U}} = \prod_{\mathcal{U}} V_n$$

with kernel $\mathfrak{n}_{\mathcal{U}}$, so $\mathcal{R}_D/\mathfrak{n}_{\mathcal{U}} \cong \mathcal{V}_{\mathcal{U}}$. Here an ultraproduct is a quotient by agreement on a member of $\mathcal{U}$, not a space of convergent sequences. The ring $\mathcal{V}_{\mathcal{U}}$ is a valuation domain with fraction field $\mathcal{K}_{\mathcal{U}} = \prod_{\mathcal{U}} K_{\Gamma}$ and value group

$$\Gamma_{\mathcal{U}} = \prod_{\mathcal{U}} (\Gamma/H_n),$$

the valuation of a nonzero class being

$$w_{\mathcal{U}}([a_n]_{\mathcal{U}}) = [v(a_n) + H_n]_{\mathcal{U}}, \quad (70)$$

where zero coordinates on a set outside $\mathcal{U}$ may be changed arbitrarily.

Proof. Changing finitely many coordinates does not change a class at a nonprincipal ultrafilter, which gives well-definedness and compatibility with the ring operations. Every sequence in $\prod_n V_n$ already lies in $\mathcal{R}_D$, so $\text{red}_{\mathcal{U}}$ is onto, and its kernel is $\mathfrak{n}_{\mathcal{U}}$.

The ultraproduct $\mathcal{K}_{\mathcal{U}}$ is a field: a nonzero class has a representative that is nonzero on a member of $\mathcal{U}$, and reciprocals there invert it. Every class in $\mathcal{K}_{\mathcal{U}}^{\times}$ lies in $\mathcal{V}_{\mathcal{U}}$ or has its inverse there, because the alternative holds in each coordinate and $\mathcal{U}$ contains one of the two index sets. So $\mathcal{V}_{\mathcal{U}}$ is a valuation domain with fraction field $\mathcal{K}_{\mathcal{U}}$. Formula (70) respects agreement modulo $\mathcal{U}$, multiplication and the ultrametric inequality coordinatewise. Its range is the whole ordered group ultraproduct: choose an exponent in each coordinate coset and take the corresponding monomials. The classes of nonnegative value are exactly the nonzero classes in $\mathcal{V}_{\mathcal{U}}$. No model-theoretic transfer theorem is used. $\square$

Composing $\text{red}_{\mathcal{U}}$ with the coordinatewise coarsened residue maps of lemma 8.4 gives $\Phi_{\mathcal{U}}$ of (53) on $\mathcal{R}_D$: the kernel of $[a_n]_{\mathcal{U}} \mapsto [\text{res}_n(a_n)]_{\mathcal{U}}$ on $\mathcal{V}_{\mathcal{U}}$ consists of the classes with $a_n \in \mathfrak{q}_{\gamma_n}$ on a member of $\mathcal{U}$, which is the maximal ideal $\{w_{\mathcal{U}} > 0\} \cup \{0\}$, and the map is onto $L_{\mathcal{U}} = \prod_{\mathcal{U}} \kappa_n$. This remark is the merge's bridge between the two notations.

Remark 12.7 (Relation to the product-ring literature). Lemma 12.5 and proposition 12.6 are the familiar idempotent and ultraproduct mechanisms of product-ring theory; compare [15]. What the analytic theory supplies is the particular restricted sequence ring $\mathcal{R}_D$, with coarsenings that change with n . This is why a statement about product rings becomes a statement about the values of entire functions at the nodes.

Lemma 12.8 (Primes of a valuation domain). *Let $\mathcal{V}$ be a valuation domain with surjective valuation $w : \text{Frac}(\mathcal{V})^\times \rightarrow \mathcal{G}$. For a convex subgroup $\mathcal{C} \leq \mathcal{G}$ put*

$$\mathfrak{p}_{\mathcal{C}}(\mathcal{V}) = \{0\} \cup \{x \in \mathcal{V} \setminus \{0\} : w(x) > \mathcal{C}\},$$

where $g > \mathcal{C}$ means that g exceeds every element of $\mathcal{C}$. Every prime of $\mathcal{V}$ is $\mathfrak{p}_{\mathcal{C}}(\mathcal{V})$ for exactly one $\mathcal{C}$, and $\mathcal{C} \subseteq \mathcal{C}_1$ if and only if $\mathfrak{p}_{\mathcal{C}_1}(\mathcal{V}) \subseteq \mathfrak{p}_{\mathcal{C}}(\mathcal{V})$. The endpoints are $\mathfrak{p}_{\mathcal{G}}(\mathcal{V}) = 0$ and $\mathfrak{p}_0(\mathcal{V})$, the maximal ideal.

Proof. For $g \geq 0$, lying outside a convex subgroup is the same as exceeding all of it. Hence $\mathfrak{p}_{\mathcal{C}}(\mathcal{V})$ is zero together with the elements of positive value for the coarsening $w + \mathcal{C}$; it is an ideal by the valuation inequalities, and prime because two nonnegative values in $\mathcal{C}$ have their sum in $\mathcal{C}$.

Conversely let $\mathfrak{p}$ be prime. The values of the nonzero elements of $\mathcal{V} \setminus \mathfrak{p}$ form an additive submonoid S of $\mathcal{G}_{\geq 0}$ that is closed downwards: if $0 \leq g \leq w(s)$ with $s \notin \mathfrak{p}$, choose x of value g ; then $s = xu$ with $u \in \mathcal{V}$, so $x \notin \mathfrak{p}$. The set $S \cup (-S)$ is a convex subgroup whose nonnegative part is S , and $\mathfrak{p} = \mathfrak{p}_{S \cup (-S)}(\mathcal{V})$. Surjectivity of w separates distinct convex subgroups and gives the inclusion rule. This is the standard correspondence [19]. $\square$

12.3 The complete classification

Theorem 12.9 (All primes above the reduced product). *The prime ideals of $\mathcal{R}_D \cong \mathcal{E}_\Gamma/(P_D)$ are exactly the following.*

- (a) *For each $n \geq 1$, the maximal ideal $\mathfrak{e}_n$, with inverse image $(Z - \rho_n)\mathcal{E}_\Gamma$.*
- (b) *For each nonprincipal ultrafilter $\mathcal{U}$ and each convex subgroup $\mathcal{C} \leq \Gamma_{\mathcal{U}}$, the prime*

$$\mathfrak{p}_{\mathcal{U}, \mathcal{C}} = \text{red}_{\mathcal{U}}^{-1}(\mathfrak{p}_{\mathcal{C}}(\mathcal{V}_{\mathcal{U}})). \quad (71)$$

The parameters are unique. Primes over different ultrafilters are incomparable; within one fibre, inclusion of primes reverses inclusion of convex subgroups. The fibre over $\mathcal{U}$ has the unique minimal prime $\mathfrak{n}_{\mathcal{U}} = \mathfrak{p}_{\mathcal{U}, \Gamma_{\mathcal{U}}}$ and the unique maximal prime

$$\overline{\mathfrak{m}}_{\mathcal{U}} = \mathfrak{p}_{\mathcal{U}, 0},$$

whose inverse image in $\mathcal{E}_\Gamma$ is the ideal $\mathfrak{M}_{\mathcal{U}}$ of theorem 8.33. For a principal ultrafilter at n both endpoints are $\mathfrak{e}_n$.

Proof. Let $\mathfrak{p}$ be prime and $\mathcal{U} = \mathcal{U}_{\mathfrak{p}}$. If $\mathcal{U}$ is principal at n , then $1 - e_{\{n\}} \in \mathfrak{p}$, and evaluation identifies $\mathcal{R}_D/(1 - e_{\{n\}})\mathcal{R}_D$ with the field K_Γ ; hence $\mathfrak{p} = \mathfrak{e}_n$. If $\mathcal{U}$ is nonprincipal, lemma 12.5 makes $\mathfrak{p}$ the inverse image of a prime of $\mathcal{R}_D/\mathfrak{n}_{\mathcal{U}} \cong \mathcal{V}_{\mathcal{U}}$ (proposition 12.6), and lemma 12.8 applies. The converse and uniqueness follow from the same correspondence and from the idempotents, which recover $\mathcal{U}$. Convex subgroups of an ordered group are linearly ordered by inclusion, which gives the uniqueness of the endpoints. The kernel of $\Phi_{\mathcal{U}}$ on $\mathcal{R}_D$ is $\mathfrak{p}_{\mathcal{U}, 0}$ by the remark after proposition 12.6, and its inverse image in $\mathcal{E}_\Gamma$ is $\mathfrak{M}_{\mathcal{U}}$ by (54). $\square$

The maximal ideals among these are the $\mathfrak{e}_n$ and the $\overline{\mathfrak{m}}_{\mathcal{U}}$. For simple nodes this is theorem 8.36, which the sixth source recovers and does not claim; the new content is the nonmaximal part of each fibre.

Corollary 12.10 (Membership test for entire functions). *The inverse image of $\mathfrak{p}_{\mathcal{U}, \mathcal{C}}$ in $\mathcal{E}_\Gamma$ consists of the F with $\{n : F(\rho_n) = 0\} \in \mathcal{U}$ and, in the complementary case, of the F whose ultraproduct value satisfies*

$$[v(F(\rho_n)) + H_n]_{\mathcal{U}} > \mathcal{C}.$$

Thus an entire function can lie in a non-evaluation prime even when none of its node values vanishes. The condition measures an asymptotic scale of the values, not an additional zero.

Theorem 12.11 (Local rings and residue towers). *For nonprincipal $\mathcal{U}$ and convex $\mathcal{C} \leq \Gamma_{\mathcal{U}}$ there are canonical isomorphisms*

$$(\mathcal{R}_D)_{\mathfrak{p}_{\mathcal{U}, \mathcal{C}}} \cong (\mathcal{V}_{\mathcal{U}})_{\mathfrak{p}_{\mathcal{C}}(\mathcal{V}_{\mathcal{U}})} = \{0\} \cup \{x \in \mathcal{K}_{\mathcal{U}}^\times : w_{\mathcal{U}}(x) + \mathcal{C} \geq 0\}.$$

This is a valuation domain with value group $\Gamma_{\mathcal{U}}/\mathcal{C}$. Its residue field carries a residual valuation with value group $\mathcal{C}$, whose residue field is

$$L_{\mathcal{U}} = \prod_{\mathcal{U}} \mathbf{k}((t^{H_n})). \quad (72)$$

At a principal prime $\mathfrak{e}_n$ the local ring and the residue field are K_Γ .

Proof. The quotient map to $\mathcal{V}_{\mathcal{U}}$ localizes to a surjection, and its kernel dies after localization: if $a \in \mathfrak{n}_{\mathcal{U}}$ vanishes on $S \in \mathcal{U}$, then $ae_S = 0$ and $e_S \notin \mathfrak{p}_{\mathcal{U}, \mathcal{C}}$.

For a valuation domain $\mathcal{V}$ with value group $\mathcal{G}$, localizing at $\mathfrak{p}_{\mathcal{C}}(\mathcal{V})$ inverts exactly the elements whose nonnegative values lie in $\mathcal{C}$, so the localization is the ring of the coarsening $w + \mathcal{C}$. In the one case needing an argument, if $w(x) \in \mathcal{C}$ is negative, choose $h \in \mathcal{C}_{\geq 0}$ with $h \geq -w(x)$ and $s \in \mathcal{V}$ of value h ; then $s \notin \mathfrak{p}_{\mathcal{C}}(\mathcal{V})$ and $sx \in \mathcal{V}$. This gives the description and the value group. A nonzero residue of a unit for $w + \mathcal{C}$ has a well-defined residual value in $\mathcal{C}$, every element of $\mathcal{C}$ occurs by surjectivity of w , and taking residues once more gives the residue field of $\mathcal{V}$. For $\mathcal{V}_{\mathcal{U}}$ the coordinatewise residue maps and [lemma 8.4](#) identify that field with (72). The principal case follows by inverting $e_{\{n\}}$. $\square$

Remark 12.12 (What is and is not new about the residue fields). The fields (72) are the residue fields $L_{\mathcal{U}}$ of [theorems 8.33](#) and [8.36](#), algebraically closed for $\mathbf{k} = \mathbb{C}$ and real closed for $\mathbf{k} = \mathbb{R}$ by [corollary 8.39](#). At intermediate primes the residual valuation is additional structure. No ultraproduct field is asserted to be canonically a Hahn field on its ultraproduct value group; no such identification is needed.

12.4 Topology: a Stone space of valuation fibres

Give $\text{Spec } \mathcal{R}_D$ the Zariski topology, with basic open sets $D(a) = \{\mathfrak{p} : a \notin \mathfrak{p}\}$, and give $\beta\mathbb{N}_{>0}$ the basic clopen sets $\hat{S} = \{\mathcal{U} : S \in \mathcal{U}\}$, as in [theorem 8.37](#).

Theorem 12.13 (The spectrum over the ultrafilter space). *The map $\pi : \text{Spec } \mathcal{R}_D \rightarrow \beta\mathbb{N}_{>0}$, $\mathfrak{p} \mapsto \mathcal{U}_{\mathfrak{p}}$, is continuous and onto, with $\pi^{-1}(\hat{S}) = D(e_S)$. Its fibre at a nonprincipal $\mathcal{U}$ is the closed irreducible subspace $V(\mathfrak{n}_{\mathcal{U}}) \cong \text{Spec } \mathcal{V}_{\mathcal{U}}$, with generic point $\mathfrak{n}_{\mathcal{U}}$ and unique closed point $\overline{\mathfrak{m}}_{\mathcal{U}}$. Principal fibres are open points. Both endpoint maps are homeomorphisms*

$$\text{Min } \mathcal{R}_D \cong \beta\mathbb{N}_{>0}, \quad \text{Max } \mathcal{R}_D \cong \beta\mathbb{N}_{>0}$$

for the subspace topologies, and the evaluation primes $\mathfrak{e}_n$ form a discrete dense subset of the whole prime spectrum.

Proof. The formula for $\pi^{-1}(\widehat{S})$ is the definition of $\mathcal{U}_{\mathfrak{p}}$, and π is onto by [theorem 12.9](#). A prime contains $\mathfrak{n}_{\mathcal{U}}$ exactly when it selects $\mathcal{U}$: if $S \notin \mathcal{U}$ then $e_S \in \mathfrak{n}_{\mathcal{U}}$, and if $S \in \mathcal{U}$ the complementary idempotent lies there. So the fibre is $V(\mathfrak{n}_{\mathcal{U}})$, homeomorphic to $\text{Spec } \mathcal{V}_{\mathcal{U}}$; the spectrum of a domain is irreducible with generic point 0, and a valuation domain has one maximal ideal. Principal fibres are $D(e_{\{n\}}) = \{\mathfrak{e}_n\}$.

For the minimal primes put $S_a = \{n : a_n \neq 0\}$; then $a \notin \mathfrak{n}_{\mathcal{U}}$ exactly when $S_a \in \mathcal{U}$, so $D(a)$ meets the minimal primes in $\widehat{S}_a$, and the idempotents give every $\widehat{S}$. For the maximal primes put $T_a = \{n : a_n \in V_n^\times\} \subseteq S_a$. At a nonprincipal $\mathcal{U}$, $a \notin \mathfrak{M}_{\mathcal{U}}$ exactly when $T_a \in \mathcal{U}$; at the principal point n , exactly when $n \in S_a$. So $D(a)$ meets the maximal primes in $\widehat{T}_a \cup \{\text{principal point at } n : n \in S_a \setminus T_a\}$, which is open, and the idempotents again give continuity of the inverse. This recovers [theorem 8.37](#) for simple nodes. Finally a nonempty $D(a)$ has $a \neq 0$, hence some $a_n \neq 0$ and $\mathfrak{e}_n \in D(a)$; together with the open points this gives density and discreteness. $\square$

Remark 12.14 (Not a disjoint union). The fibres partition the primes but are not open in general, so the spectrum is not a topological disjoint union of valuation spectra. A nonprincipal fibre is closed, and its position relative to the principal points is governed by all indicator idempotents at once.

Corollary 12.15 (Semiprimitive, and Jacobson only in the degenerate case). *The Jacobson radical of $\mathcal{R}_D$ is zero. If some nonprincipal fibre has more than one point, then $\mathcal{R}_D$ is not a Jacobson ring.*

Proof. The intersection of the $\mathfrak{e}_n$ is zero, hence so is the intersection of all maximal ideals. A nonmaximal prime of a nonprincipal fibre lies in exactly one maximal ideal, by [theorem 12.9](#), so it is not the intersection of the maximal ideals containing it; that is the failure of the Jacobson property. $\square$

12.5 The order-unit dichotomy and continuum prime chains

Order unit is the notion of [definition 1.3](#); countable cofinality does not imply that one exists ([remark 1.4](#)). In this subsection θ denotes a real number in $(0, 1)$, and $\lfloor \cdot \rfloor$ the integer part.

Lemma 12.16 (Uniform Archimedean gaps in an ultraproduct). *Let $\mathcal{G}_n$ be nonzero ordered abelian groups, choose $\bar{\eta}_n > 0$ in $\mathcal{G}_n$, and let $\mathcal{U}$ be nonprincipal. For $0 < \theta < 1$ put*

$$\zeta_\theta = [\lfloor n^\theta \rfloor \bar{\eta}_n]_{\mathcal{U}} \in \prod_{\mathcal{U}} \mathcal{G}_n, \quad \mathcal{C}_\theta = \text{conv}(\mathbb{Z}\zeta_\theta),$$

the convex subgroup generated by ζ_θ . If $\theta < \theta'$ then $m\zeta_\theta < \zeta_{\theta'}$ for every positive integer m , and hence $\mathcal{C}_\theta \subsetneq \mathcal{C}_{\theta'}$.

Proof. For fixed $\theta < \theta'$ and m , the integer inequality $m\lfloor n^\theta \rfloor < \lfloor n^{\theta'} \rfloor$ holds for all large n ; multiplying by $\bar{\eta}_n > 0$ preserves it, and a nonprincipal ultrafilter contains every cofinite set. So $\zeta_{\theta'}$ exceeds every ordinary multiple of ζ_θ . The convex subgroup generated by a positive ζ consists of the elements bounded in absolute value by an ordinary finite multiple of ζ , which gives the strict inclusion. $\square$

Theorem 12.17 (Reduced-divisor dichotomy). *Under [convention 12.1](#) the following are equivalent.*

- (a) Γ has an order unit.
- (b) $V_n = K_\Gamma$ for all sufficiently large n , and $\mathcal{R}_D = K_\Gamma^{\mathbb{N}_{>0}}$.
- (c) $\mathcal{R}_D$ is von Neumann regular: for every $a \in \mathcal{R}_D$ there is $b \in \mathcal{R}_D$ with $a^2b = a$.

- (d) Every prime of $\mathcal{R}_D$ is maximal; equivalently $\dim \mathcal{R}_D = 0$.
- (e) $\mathcal{R}_D$ is a Jacobson ring.

If they fail, then for every nonprincipal $\mathcal{U}$ the fibre over $\mathcal{U}$ contains a strict chain of primes indexed by $(0, 1)$ with its order reversed. In particular $\mathcal{E}_\Gamma/(P_D)$ has infinite Krull dimension, although every zero of P_D is simple.

Proof. If h is an order unit, cofinality gives $\gamma_n \geq h$ eventually, so $H_n = \Gamma$ and $V_n = K_\Gamma$. Conversely $V_n = K_\Gamma$ forces $H_n = \Gamma$, by testing monomials of arbitrarily negative value, so γ_n is an order unit. This is (a) $\Leftrightarrow$ (b), which is [corollary 8.15](#) together with [corollary 8.16](#). In a product of fields put $b_n = a_n^{-1}$ where $a_n \neq 0$ and $b_n = 0$ elsewhere; this gives (c). Under (b) every Γ/H_n is zero, so every $\Gamma_{\mathcal{U}}$ is zero and [theorem 12.9](#) gives (d); and (d) implies (e).

Suppose there is no order unit. Every H_n is proper; choose $\eta_n > H_n$ and put $\bar{\eta}_n = \eta_n + H_n > 0$ in Γ/H_n . [Lemma 12.16](#) gives strictly increasing convex subgroups $\mathcal{C}_\theta \leq \Gamma_{\mathcal{U}}$, and the primes $\mathfrak{p}_{\mathcal{U}, \mathcal{C}_\theta}$ strictly decrease by [theorem 12.9](#), so (d) fails. Each nonprincipal fibre then has a nonmaximal prime, so (e) fails by [corollary 12.15](#). For (c) take $a_n = t^{\eta_n}$, which lies in $\mathcal{R}_D$ and has no zero coordinate: $a^2b = a$ would force $b_n = t^{-\eta_n}$, which lies in no V_n , so $b \notin \mathcal{R}_D$. A real interval contains finite strict chains of every length, which gives the assertion about Krull dimension. $\square$

The assertion concerns the quotient $\mathcal{E}_\Gamma/(P_D)$. It does *not* say that $\mathcal{E}_\Gamma$ itself is zero-dimensional in the order-unit case.

Proposition 12.18 (Entire witnesses for every strict inclusion). *Without an order unit, choose $\eta_n > H_n$ as above. For each $0 < \theta < 1$ there is $F_\theta \in \mathcal{E}_\Gamma$ with $F_\theta(\rho_n) = t^{\lfloor n^\theta \rfloor \eta_n}$ for all n . For the subgroups $\mathcal{C}_{\theta'}$ of the proof of [theorem 12.17](#), the class of F_θ lies in $\mathfrak{p}_{\mathcal{U}, \mathcal{C}_{\theta'}}$ when $\theta > \theta'$ and not when $\theta \leq \theta'$. None of these node values is zero.*

Proof. All prescribed values lie in V_n , so [corollary 8.13](#) supplies F_θ , whose ultraproduct value is ζ_θ . If $\theta > \theta'$ it exceeds every element of $\mathcal{C}_{\theta'}$ by [lemma 12.16](#); if $\theta \leq \theta'$ it lies in $\mathcal{C}_{\theta'}$. Apply [corollary 12.10](#). $\square$

Remark 12.19 (Cardinality and order type). The theorem asserts a chain with continuum many distinct members. It assigns no cardinal-valued Krull dimension, does not assert that the chain is well ordered, and does not bound the number of convex subgroups of an ultraproduct; those are different set-theoretic questions.

Remark 12.20 (Rank is not the dividing invariant). The sixth source's example is [section 14.4](#) together with [section 14.1](#), printed once there. Every nonzero Archimedean divisible group, and $\mathbb{Q}^d$ in lexicographic order, has an order unit, so for them $\mathcal{R}_D = K_\Gamma^{\mathbb{N}_{>0}}$ and every prime above P_D is maximal. The group Γ_∞ has countable cofinality and no order unit, so every nonprincipal fibre carries a continuum chain. What divides the two regimes is one cofinal cyclic scale, not rank greater than one; this is the same dividing line as case (ii) versus case (iii) of [theorem 1.5](#).

12.6 Finite algebra in an infinite-dimensional spectrum

The quotient can have a very deep spectrum and still admit simple finite linear algebra: every coordinate ring V_n is a valuation ring, and every subset of the index set has an indicator idempotent.

Theorem 12.21 (Finitely generated ideals and annihilators). *Every finitely generated ideal of $\mathcal{R}_D$ is principal. For $a \in \mathcal{R}_D$,*

$$\text{Ann}(a) = e_{\{n:a_n=0\}} \mathcal{R}_D, \quad a\mathcal{R}_D \cong e_{\{n:a_n \neq 0\}} \mathcal{R}_D.$$

Hence every finitely generated ideal is projective and $\mathcal{R}_D$ is semihereditary. It is a Bézout ring with zero divisors, not a Bézout domain.

Proof. Let $a^{(1)}, \dots, a^{(s)} \in \mathcal{R}_D$. Outside a finite exceptional set all their coordinates lie in V_n . At such a coordinate with some nonzero value choose the first index j whose coarsened valuation $v + H_n$ is least among the nonzero values; at an exceptional coordinate choose any index with a nonzero value, if there is one. This partitions the coordinates with some nonzero value into sets $S_1, \dots, S_s$; put $g = \sum_j e_{S_j} a^{(j)}$, an element of the ideal. Define $b_n^{(j)} = a_n^{(j)} / g_n$ where $g_n \neq 0$ and 0 elsewhere. The minimality of the chosen valuation puts these ratios in V_n at every nonexceptional coordinate, so $b^{(j)} \in \mathcal{R}_D$ and $a^{(j)} = gb^{(j)}$; the ideal is $g\mathcal{R}_D$. The annihilator of a consists of the sequences supported on its zero set, and multiplication by a maps $e_{\{a_n \neq 0\}} \mathcal{R}_D$ isomorphically onto $a\mathcal{R}_D$, being onto by definition and injective because each surviving coordinate is a nonzero element of a field. A direct summand of the free module $\mathcal{R}_D$ is projective. $\square$

Remark 12.22 (Translation to $\mathcal{E}_\Gamma$). Every finitely generated ideal I of $\mathcal{E}_\Gamma$ that contains P_D is (P_D, G) for a single $G \in \mathcal{E}_\Gamma$: the image of I in $\mathcal{R}_D$ is finitely generated, hence $g\mathcal{R}_D$, and any lift G of g works. Membership of an arbitrary element in (P_D, G) is decided by [corollary 8.28](#). This translation is the merge's; it bears on [question 16.3](#) only for ideals containing this product.

Theorem 12.23 (Smith forms for finite matrices). *For every $r \times s$ matrix M over $\mathcal{R}_D$ there are $U \in \text{GL}_r(\mathcal{R}_D)$ and $V \in \text{GL}_s(\mathcal{R}_D)$ with*

$$UMV = \text{diag}(d_1, \dots, d_\ell, 0, \dots), \quad \ell = \min(r, s), \quad d_i \mid d_{i+1},$$

the diagonal taken in the rectangular sense and zero entries allowed. Thus $\mathcal{R}_D$ is an elementary divisor ring, and every finitely presented $\mathcal{R}_D$ -module is a direct sum of finitely many cyclic modules and a finite free module.

Proof. Choose a finite exceptional set outside which every entry of M lies in V_n . Over a valuation domain a finite matrix has the usual pivot reduction: a nonzero entry of least valuation divides every entry; swaps move it to the corner, elementary operations clear its row and column, the remaining entries stay divisible by it, and one inducts on the smaller block. A zero matrix takes identity transformations. Perform this reduction over V_n at each nonexceptional coordinate and over K_Γ at the exceptional ones. The row and column operations and their inverses have entries in V_n eventually, so their coordinate sequences define invertible matrices U, V over $\mathcal{R}_D$. The diagonal entries lie in $\mathcal{R}_D$, and the coordinatewise quotients d_{i+1}/d_i lie in V_n eventually, so $d_i \mid d_{i+1}$ in $\mathcal{R}_D$; a zero pivot is followed only by zero pivots at that coordinate, and the quotient there is taken to be 0. Applying this to a presentation matrix gives the module decomposition. No uniqueness of the diagonal representatives is claimed. $\square$

In this subsection U and V are invertible matrices, not an ultrafilter or a coarsened valuation ring.

Proposition 12.24 (Total quotient ring and integral closedness). *An element of $\mathcal{R}_D$ is a non-zero-divisor exactly when none of its coordinates is zero, and the total quotient ring is $Q(\mathcal{R}_D) = K_\Gamma^{\mathbb{N}_{>0}}$. The ring $\mathcal{R}_D$ is integrally closed in it, and it is not Noetherian.*

Proof. A zero coordinate gives a nonzero annihilating indicator, and a sequence with no zero coordinate acts injectively; so the total quotient ring embeds in $K_\Gamma^{\mathbb{N}_{>0}}$. Given $b \in K_\Gamma^{\mathbb{N}_{>0}}$ put $s_n = 1$ if $b_n \in V_n$ and $s_n = b_n^{-1}$ otherwise; then s is a non-zero-divisor of $\mathcal{R}_D$, $sb \in \mathcal{R}_D$ and $b = (sb)/s$. If b satisfies a monic equation over $\mathcal{R}_D$, all its coefficients lie in V_n for large n , and a valuation ring is integrally closed: an element of negative valuation cannot satisfy a monic equation, since its leading power would have strictly smaller valuation than every other term. So $b_n \in V_n$ eventually and $b \in \mathcal{R}_D$. The ideals generated by $e_{\{1,\dots,N\}}$ increase strictly with N . $\square$

Remark 12.25 (Two Bézout statements that do not conflict). Without an order unit the full ring $\mathcal{E}_\Gamma$ is not Bézout ([theorem 1.5\(iii\)](#)); [theorem 12.21](#) says that its quotient by this particular simple one-node-per-shell product *is* a Bézout ring. There is no contradiction: the quotient acquires indicator idempotents, which permit coordinatewise choices of generator.

12.7 Cofinal scalar extension and contraction of primes

Let $\Gamma \subseteq \Delta$ be an ordered-group inclusion of divisible ordered abelian groups with Γ cofinal in Δ , and keep the radii $\gamma_n \in \Gamma$ and the nodes $\rho_n = t^{-\gamma_n}$. Superscripts record the group:

$$H_n^\Gamma = H_{\gamma_n} \subseteq \Gamma, \quad H_n^\Delta = \{\delta \in \Delta : |\delta| \leq m\gamma_n \text{ for some } m \in \mathbb{N}\},$$

and similarly $V_n^\Gamma, V_n^\Delta, \mathcal{R}_D^\Gamma, \mathcal{R}_D^\Delta$ and $\Gamma_\mathcal{U} = \prod_\mathcal{U}(\Gamma/H_n^\Gamma), \Delta_\mathcal{U} = \prod_\mathcal{U}(\Delta/H_n^\Delta)$. The notation Δ is that of [section 10](#); the convex hull there is all of Δ , because Γ is cofinal.

Lemma 12.26 (The comparison maps). *Coefficientwise inclusion gives $K_\Gamma \subseteq K_\Delta, \mathcal{E}_\Gamma \subseteq \mathcal{E}_\Delta$ and $\mathcal{R}_D^\Gamma \subseteq \mathcal{R}_D^\Delta$, the last induced on the quotients by the same series P_D . Moreover*

$$H_n^\Delta \cap \Gamma = H_n^\Gamma, \quad V_n^\Delta \cap K_\Gamma = V_n^\Gamma,$$

and the zero set of P_D in either field is $\{\rho_n : n \geq 1\}$, every zero being simple.

Proof. An ordered-group inclusion preserves well-ordered supports and the Hahn operations. Coefficient slopes cofinal in Γ are cofinal in Δ , so [theorem 3.1](#) gives the second inclusion; for $\mathbf{k} = \mathbb{C}$ this is [corollary 10.5](#). The equality of convex subgroups is the common inequality $|g| \leq m\gamma_n$ for $g \in \Gamma$, and the equality of valuation rings follows by testing the leading exponent. The evaluation maps commute with the inclusions, and both kernels are (P_D) by [proposition 8.8](#), so the induced map of quotients is injective and is the displayed inclusion of sequence rings. The radii remain cofinal in Δ . At $z \neq 0$ choose N with $\gamma_{N+1} + v(z) > 0$: the tail $\prod_{n>N}(1 - t^{\gamma_n}z)$ is 1 plus an element of positive valuation, hence nonzero, so only the listed nodes are zeros; removing one factor leaves a cofactor that does not vanish at its node, so each zero is simple. $\square$

Remark 12.27 (What scalar extension means here). Rings are compared by coefficientwise inclusion and the induced map of quotients. It is *not* asserted that $\mathcal{E}_\Gamma \otimes_{K_\Gamma} K_\Delta = \mathcal{E}_\Delta$, or that the corresponding tensor product of quotients is the larger quotient; these surjectivity assertions are neither needed nor proved. Cofinality cannot simply be dropped: without it a nonpolynomial series entire over K_Γ is not entire over K_Δ ([theorem 10.1](#)).

For nonprincipal $\mathcal{U}$ the coordinate maps $\Gamma/H_n^\Gamma \rightarrow \Delta/H_n^\Delta$ reflect order and equality, so they induce an order embedding $\Gamma_\mathcal{U} \hookrightarrow \Delta_\mathcal{U}$, by which $\Gamma_\mathcal{U}$ is identified with a subgroup of $\Delta_\mathcal{U}$.

Lemma 12.28 (All convex extensions of a convex subgroup). *Let $\mathcal{G} \subseteq \mathcal{G}'$ be an inclusion of ordered abelian groups with $\mathcal{G}$ divisible, and let $\mathcal{C}$ be a convex subgroup of $\mathcal{G}$. Put*

$$\mathcal{C}_{\min} = \{x \in \mathcal{G}' : |x| \leq c \text{ for some } c \in \mathcal{C}_{\geq 0}\}, \quad (73)$$

$$\mathcal{C}_{\max} = \{x \in \mathcal{G}' : |x| < g \text{ for every } g \in \mathcal{G}_{>0} \setminus \mathcal{C}\}, \quad (74)$$

with $\mathcal{C}_{\max} = \mathcal{G}'$ when the condition is empty. Both are convex subgroups of $\mathcal{G}'$ with trace $\mathcal{C}$ on $\mathcal{G}$, and the convex subgroups $\mathcal{C}' \leq \mathcal{G}'$ with $\mathcal{C}' \cap \mathcal{G} = \mathcal{C}$ are exactly those with

$$\mathcal{C}_{\min} \subseteq \mathcal{C}' \subseteq \mathcal{C}_{\max}. \quad (75)$$

Proof. $\mathcal{C}_{\min}$ is the convex subgroup of $\mathcal{G}'$ generated by $\mathcal{C}$, and its trace is $\mathcal{C}$ by convexity of $\mathcal{C}$ in $\mathcal{G}$. The set $\mathcal{C}_{\max}$ is symmetric and convex. If $\mathcal{C} = \mathcal{G}$ there is nothing more to prove; otherwise, for $g > 0$ outside $\mathcal{C}$ also $g/2$ is outside $\mathcal{C}$, by divisibility, and $|x + y| < g/2 + g/2 = g$ for $x, y \in \mathcal{C}_{\max}$, so $\mathcal{C}_{\max}$ is a subgroup. A positive element of $\mathcal{G}$ outside $\mathcal{C}$ exceeds all of $\mathcal{C}$, so $\mathcal{C} \subseteq \mathcal{C}_{\max}$, while no $g \in \mathcal{G} \setminus \mathcal{C}$ lies in $\mathcal{C}_{\max}$; its trace is $\mathcal{C}$, and it contains $\mathcal{C}_{\min}$. If $\mathcal{C}'$ is convex with trace $\mathcal{C}$, it contains $\mathcal{C}_{\min}$; and for $x \in \mathcal{C}'$ and $g \in \mathcal{G}_{>0} \setminus \mathcal{C}$, the inequality $g \leq |x|$ would put g in $\mathcal{C}' \cap \mathcal{G} = \mathcal{C}$, so $\mathcal{C}' \subseteq \mathcal{C}_{\max}$. Conversely every convex subgroup in the interval has trace $\mathcal{C}$. $\square$

Theorem 12.29 (Exact contraction and all extensions of a prime). *Contraction of primes along $\mathcal{R}_D^\Gamma \subseteq \mathcal{R}_D^\Delta$ preserves the selected ultrafilter. At a principal point it sends $\mathfrak{e}_n^\Delta$ to $\mathfrak{e}_n^\Gamma$. At a nonprincipal $\mathcal{U}$,*

$$\mathfrak{p}_{\mathcal{U}, \mathcal{C}'}^\Delta \cap \mathcal{R}_D^\Gamma = \mathfrak{p}_{\mathcal{U}, \mathcal{C}' \cap \Gamma_{\mathcal{U}}}^\Gamma. \quad (76)$$

Every prime of $\mathcal{R}_D^\Gamma$ extends to $\mathcal{R}_D^\Delta$. The primes contracting to $\mathfrak{p}_{\mathcal{U}, \mathcal{C}}^\Gamma$ are exactly the $\mathfrak{p}_{\mathcal{U}, \mathcal{C}'}^\Delta$ with $\mathcal{C}'$ in the interval (75) for $\mathcal{G} = \Gamma_{\mathcal{U}}$ and $\mathcal{G}' = \Delta_{\mathcal{U}}$, so each fibre of the contraction map is anti-isomorphic, as an ordered set, to that interval.

Proof. Each indicator e_S maps to the same indicator, so contraction preserves the zero-or-one test and hence the ultrafilter; the principal case is coordinate evaluation. For nonprincipal $\mathcal{U}$ the maps of ultraproduct valuation rings are injective and compatible with $\Gamma_{\mathcal{U}} \hookrightarrow \Delta_{\mathcal{U}}$. A nonzero element of $\mathcal{V}_{\mathcal{U}}^\Gamma$ of value $g \geq 0$ maps into the prime indexed by $\mathcal{C}'$ exactly when $g \notin \mathcal{C}'$, that is, when $g \notin \mathcal{C}' \cap \Gamma_{\mathcal{U}}$; zero is preserved. This is (76). The group $\Gamma_{\mathcal{U}}$ is divisible, since division by a positive integer is performed coordinatewise, so lemma 12.28 supplies at least one convex extension and describes all of them, and theorem 12.9 translates this into primes with the order reversed. $\square$

Corollary 12.30 (Uniqueness of extension). *The prime $\mathfrak{p}_{\mathcal{U}, \mathcal{C}}^\Gamma$ has a unique extension exactly when $\mathcal{C}_{\min} = \mathcal{C}_{\max}$ in $\Delta_{\mathcal{U}}$. If $\Gamma_{\mathcal{U}} = \Delta_{\mathcal{U}}$, every prime of that fibre extends uniquely. If Γ has an order unit, the spectral map $\text{Spec } \mathcal{R}_D^\Delta \rightarrow \text{Spec } \mathcal{R}_D^\Gamma$ is the identity of the common ultrafilter space $\beta\mathbb{N}_{>0}$.*

Proof. Both endpoints of the interval are admissible, so uniqueness means that they coincide; if the groups coincide, both endpoints are $\mathcal{C}$. An order unit of Γ remains one in Δ : first bound a given element by an element of Γ , then by a multiple of the order unit. Both quotients are then products of fields (theorem 12.17), whose spectra are the ultrafilter spaces, and indicators are preserved. $\square$

Corollary 12.31 (Stable endpoint spectra). *For every cofinal extension of this subsection, contraction restricts to homeomorphisms of the minimal and of the maximal spectra, both identified with the identity of $\beta\mathbb{N}_{>0}$. Every minimal prime of $\mathcal{R}_D^\Gamma$ has a unique prime extension to $\mathcal{R}_D^\Delta$. Every maximal ideal has a unique maximal extension, but it can have further nonmaximal extensions.*

Proof. The embedding $\Gamma_{\mathcal{U}} \subseteq \Delta_{\mathcal{U}}$ is cofinal: dominate a representative of each new coordinate by an old exponent and pass to the ultraproduct. A convex subgroup of $\Delta_{\mathcal{U}}$ containing all of $\Gamma_{\mathcal{U}}$ is therefore all of $\Delta_{\mathcal{U}}$, which by [theorem 12.29](#) gives unique extension at the minimal endpoint $\mathcal{C} = \Gamma_{\mathcal{U}}$. At the maximal endpoint $\mathcal{C}' = 0$ contracts to $\mathcal{C} = 0$, and [theorem 12.9](#) gives exactly one maximal prime over the selected ultrafilter. Principal points behave in the same way by coordinate evaluation, and [theorem 12.13](#) makes both endpoint maps the identity of $\beta\mathbb{N}_{>0}$. $\square$

12.8 An explicit continuum splitting with no new zeros

The sixth source's main construction separates new cuts in the value group from new zeros, inside the surreal numbers. It enlarges the group Γ_{∞} of [section 14.1](#) by rational indices:

$$\Gamma_{\infty} = \bigoplus_{j \in \mathbb{Z}_{\geq 0}} \mathbb{Q}e_j \subseteq \Delta_{\mathbb{Q}} = \bigoplus_{q \in \mathbb{Q}_{\geq 0}} \mathbb{Q}e_q, \quad (77)$$

finitely supported rational combinations, a nonzero one being positive exactly when its coefficient at the largest occupied index is positive. The integer-indexed basis vectors of $\Delta_{\mathbb{Q}}$ are those of Γ_{∞} ; the sixth source writes f_q for the basis of $\Delta_{\mathbb{Q}}$ and $f_j = e_j$, and the single letter e_q is used here because our f is the function of [\(83\)](#). Take $\gamma_n = e_n$ for $n \geq 1$. Then P_D is that function f , and $\rho_n = t^{-e_n}$. In this subsection q and u denote rational numbers, not the node count q_n or a unit.

Lemma 12.32 (The two groups). *Both groups in (77) are divisible and of countable cofinality, neither has an order unit, and Γ_{∞} is cofinal in $\Delta_{\mathbb{Q}}$. Their radius subgroups are*

$$H_n^{\Gamma_{\infty}} = \bigoplus_{0 \leq j \leq n} \mathbb{Q}e_j, \quad H_n^{\Delta_{\mathbb{Q}}} = \bigoplus_{q \in \mathbb{Q} \cap [0, n]} \mathbb{Q}e_q,$$

and the quotients are identified with the finite combinations supported on indices strictly greater than n .

Proof. Every nonzero finite support has a largest index, and a basis vector with a larger integer index dominates every ordinary multiple of the given vector; this gives the absence of an order unit, countable cofinality and cofinality of Γ_{∞} in $\Delta_{\mathbb{Q}}$. A vector supported on indices at most n is bounded by a multiple of e_n , and a vector with a larger leading index is not; this gives the radius subgroups, the first of which is [corollary 8.35](#). Discarding the coordinates of index at most n gives the quotient identifications. $\square$

Fix a nonprincipal $\mathcal{U}$ and write $\Gamma_{\infty, \mathcal{U}}$ and $\Delta_{\mathbb{Q}, \mathcal{U}}$ for the two ultraproduct value groups. Define the convex subgroup

$$\mathcal{C}_{\text{int}} = \prod_{\mathcal{U}} (\mathbb{Q}e_{n+1}) \subseteq \Gamma_{\infty, \mathcal{U}}, \quad (78)$$

the internal ultraproduct of the coordinate subgroups $\mathbb{Q}e_{n+1}$ under the identifications of [lemma 12.32](#), and for real $0 < \theta < 1$

$$\mathcal{C}'_{\theta} = \prod_{\mathcal{U}} \left(\bigoplus_{\substack{q \in \mathbb{Q} \\ n < q \leq n+1+\theta}} \mathbb{Q}e_q \right) \subseteq \Delta_{\mathbb{Q}, \mathcal{U}}. \quad (79)$$

These are convex subgroups: each coordinate subgroup is convex, and an order interval bounded by an element of one of them stays inside it on a member of $\mathcal{U}$.

Theorem 12.33 (Cofinal prime splitting without new zeros). *For the inclusion (77), the function $P_D(Z) = \prod_{n \geq 1} (1 - t^{e_n} Z)$ is entire over both K_{Γ_∞} and $K_{\Delta_{\mathbb{Q}}}$ and has exactly the same simple zeros t^{-e_n} in both fields. Nevertheless, for every nonprincipal $\mathcal{U}$ the primes $\mathfrak{p}_{\mathcal{U}, \mathcal{C}'_\theta}^{\Delta_{\mathbb{Q}}}$, $0 < \theta < 1$, form a strictly decreasing chain with continuum many members, all of which contract to the single intermediate prime $\mathfrak{p}_{\mathcal{U}, \mathcal{C}_{\text{int}}}^{\Gamma_\infty}$. Intermediate means strictly between the unique minimal and the unique maximal prime of its fibre.*

Proof. Entireness and the zeros are lemma 12.26. At coordinate n the only integer index in $(n, n+1+\theta]$ is $n+1$, so $\mathcal{C}'_\theta \cap \Gamma_{\infty, \mathcal{U}} = \mathcal{C}_{\text{int}}$; the equality holds for ultraproduct classes, because membership on a member of $\mathcal{U}$ can be represented coordinatewise there and changes off it are irrelevant. If $\theta < \theta'$, choose a rational u with $\theta < u < \theta'$; then $b_u = [e_{n+1+u} + H_n^{\Delta_{\mathbb{Q}}}]_{\mathcal{U}}$ lies in $\mathcal{C}'_{\theta'}$ but not in $\mathcal{C}'_\theta$, so $\mathcal{C}'_\theta \subsetneq \mathcal{C}'_{\theta'}$. The subgroup $\mathcal{C}_{\text{int}}$ is nonzero, since it contains $[e_{n+1}]_{\mathcal{U}}$, and proper, since it omits $[e_{n+2}]_{\mathcal{U}}$, so its prime is intermediate. The inclusion rule of theorem 12.9 and (76) finish the proof. $\square$

Proposition 12.34 (Entire functions separating the new primes). *For each rational $u \in (0, 1)$ there is $G_u \in \mathcal{E}_{\Delta_{\mathbb{Q}}}$ with $G_u(t^{-e_n}) = t^{e_{n+1+u}}$ for all $n \geq 1$. Whenever $\theta < u < \theta'$, its class lies in $\mathfrak{p}_{\mathcal{U}, \mathcal{C}'_\theta}^{\Delta_{\mathbb{Q}}}$ and not in $\mathfrak{p}_{\mathcal{U}, \mathcal{C}'_{\theta'}}^{\Delta_{\mathbb{Q}}}$. So the distinctions of theorem 12.33 are witnessed by entire functions over the enlarged workspace, not only by elements of an ordered group.*

Proof. The prescribed values have positive coarsened valuation for every n , so corollary 8.13 over $K_{\Delta_{\mathbb{Q}}}$ realizes them. Their ultraproduct value is b_u , whose leading index $n+1+u$ exceeds every index allowed in the coordinate subgroups of $\mathcal{C}'_\theta$, so $b_u > \mathcal{C}'_\theta$, while $b_u \in \mathcal{C}'_{\theta'}$. Apply corollary 12.10. $\square$

Corollary 12.35 (A maximal ideal with continuum many prime extensions). *In the same example, for every nonprincipal $\mathcal{U}$ the old maximal ideal $\overline{\mathfrak{M}}_{\mathcal{U}}^{\Gamma_\infty}$ is the contraction of a strictly decreasing chain of continuum many nonmaximal primes of $\mathcal{R}_D^{\Delta_{\mathbb{Q}}}$, all strictly below $\overline{\mathfrak{M}}_{\mathcal{U}}^{\Delta_{\mathbb{Q}}}$. Unchanged zero sets and homeomorphic maximal spectra therefore do not make the full spectral contraction map injective.*

Proof. Replace (79) by $\mathcal{C}'_\theta{}^0 = \prod_{\mathcal{U}} (\bigoplus_{q \in \mathbb{Q} \cap (n, n+\theta]} \mathbb{Q} e_q)$. There is no integer strictly between n and $n+1$, so these convex subgroups meet $\Gamma_{\infty, \mathcal{U}}$ in 0; they are nonzero, and they increase strictly with θ by a rational witness between two cutoffs. Their primes are nonmaximal, strictly decreasing, and contract to $\mathfrak{p}_{\mathcal{U}, 0}^{\Gamma_\infty} = \overline{\mathfrak{M}}_{\mathcal{U}}^{\Gamma_\infty}$. An analytic witness at a rational cutoff u interpolates the values $t^{e_{n+u}}$ at the nodes. $\square$

Remark 12.36 (Why the internal subgroup is needed). The subgroup $\mathcal{C}_{\text{int}}$ is not the convex subgroup generated by the single class $[e_{n+1}]_{\mathcal{U}}$: it contains $[q_n e_{n+1}]_{\mathcal{U}}$ for arbitrary rational sequences (q_n) , unbounded ones included, and the intersection in the proof of theorem 12.33 keeps every such sequence. Replacing $\mathcal{C}_{\text{int}}$ by that smaller convex hull would give a false contraction statement.

What the splitting does and does not mean. Every prime of the chain lies over the same ultrafilter of ordinary nodes, and P_D vanishes only at the same node coordinates. The new distinctions come from scales of the enlarged value group lying between consecutive old integer-indexed levels. They are invisible to the values of old-field functions after contraction and are detected by the G_u . The theorem is a failure of the inference “no new zeros under cofinal extension, hence no new primes above the divisor”. It is not a failure of injectivity of the coefficient map, nor a claim that a polynomial acquires more roots than its degree: the divisor is infinite, the primes

are algebraic objects of its quotient, and the new points are non-evaluation points. The cofinal extension of [section 14.3](#), which adjoins $\omega^{1/2}$ to Γ_∞ , lies between the two groups of (77); nothing is asserted here about its own prime spectrum.

The surreal realization. Under the normal-form embedding (1), $t^g \mapsto \omega^{-g}$, take $e_q = \omega^q$ for $q \in \mathbb{Q}_{\geq 0}$ as elements of the additive group of $\mathbf{No}$; finite rational combinations of distinct such monomials are ordered by their largest exponent, as (77) requires, and $e_j = \omega^j$ is the choice of (81). The real workspaces $\mathbb{R}((\omega^{-\Gamma_\infty})) \subset \mathbb{R}((\omega^{-\Delta_\mathbb{Q}})) \subset \mathbf{No}$ and their complexifications in $\mathbf{No}[i]$ are set-sized subfields, and

$$P_D(Z) = \prod_{n \geq 1} (1 - \omega^{-\omega^n} Z), \quad \rho_n = \omega^{\omega^n}, \quad G_u(\omega^{\omega^n}) = \omega^{-\omega^{n+1+u}} \quad (u \in \mathbb{Q} \cap (0, 1)),$$

the first being (83) and the nodes being (82). These formulas specify values of genuine entire power series over the larger fixed Hahn workspace.

Corollary 12.37 (Surreal-workspace prime splitting). *There are explicit set-sized subfields $\mathbb{R}((\omega^{-\Gamma_\infty})) \subset \mathbb{R}((\omega^{-\Delta_\mathbb{Q}})) \subset \mathbf{No}$, and their complexifications in $\mathbf{No}[i]$, such that the entire divisor $\prod_{n \geq 1} (1 - \omega^{-\omega^n} Z)$ has the same simple surreal zeros ω^{ω^n} in both workspaces, while a single prime of the smaller divisor quotient is the contraction of a chain of continuum many primes of the larger one. This holds over real and over complex coefficients.*

Proof. Apply [theorem 12.33](#) through (1). The interpolation, classification and contraction arguments do not use algebraic closedness and hold for either coefficient field. $\square$

Remark 12.38 (Workspace-relative, not class-wide). “The same zeros” means the zeros in the two specified fields. No strongly summable value of this product is asserted at every point of $\mathbf{No}[i]$: the ambient class contains scales beyond any set-sized cofinal sequence, and at such a scale strong summability fails by [theorem 10.1](#). The ultraproduct residue fields are not identified canonically with subfields of the surreal field. The construction shows a dependence on the chosen set-sized universe of scales: an analytic object keeps its formula and every zero under an enlargement while its prime spectrum refines, the refinement being encoded by convex subgroups of an ultraproduct of quotient value groups. All carriers are sets and all embeddings are explicit; no paradox is involved. The Boolean layer is stable — every stage has the same indicators and the same ultrafilter space — and what changes inside a fibre is the ordered hierarchy of value cuts. Ordinary zero coordinates, ultrafilter choices of infinitely many shells, and convex subgroups of their asymptotic value group are three separate pieces of information.

12.9 Multiplicity: the limit of the reduced description

The results so far concern one *simple* node per shell. Multiplicity is another, logically separate, source of nonmaximal primes. For positive integers m_n put

$$D_{\mathbf{m}} = \sum_{n \geq 1} m_n [\rho_n], \quad P_{D_{\mathbf{m}}}(Z) = \prod_{n \geq 1} \left(1 - \frac{Z}{\rho_n}\right)^{m_n},$$

the canonical product of [theorem 7.3](#) and the setting of [section 8.9](#); the sixth source calls it Q , a letter reserved here for (29). Write $\text{ord}_{\rho_n}(F)$ for the order of a nonzero F at ρ_n , zero when $F(\rho_n) \neq 0$.

Lemma 12.39 (Divisibility with arbitrary finite multiplicities). *P_{D_m} is entire, with zeros exactly the ρ_n of multiplicities m_n , and $F \in \mathcal{E}_\Gamma$ is divisible by P_{D_m} if and only if $\text{ord}_{\rho_n}(F) \geq m_n$ for every n .*

Proof. The first assertion is [theorem 7.3](#); over $\mathbb{R}$, a zero in $\mathbb{R}((t^\Gamma))$ is a zero of the same order in $\mathbb{C}((t^\Gamma))$. The divisibility criterion is [proposition 8.8](#), since $\text{ord}_{\rho_n}(F) \geq m_n$ says that the prescribed zero jets vanish; over $\mathbb{R}$ it holds by [proposition 8.42](#). The sixth source reproves this lemma along the lines of [remark 12.2](#), with each index repeated m_n times. $\square$

Theorem 12.40 (Radicals and the bounded-multiplicity threshold). *For every $F \in \mathcal{E}_\Gamma$,*

$$F \in \sqrt{P_{D_m}\mathcal{E}_\Gamma} \iff \text{some integer } d \geq 1 \text{ has } d \text{ord}_{\rho_n}(F) \geq m_n \text{ for every } n. \quad (80)$$

In particular $\sqrt{P_{D_m}\mathcal{E}_\Gamma} = P_D\mathcal{E}_\Gamma$ if and only if $\sup_n m_n < \infty$. If $m_n \leq M$ for all n , the natural map $\mathcal{E}_\Gamma/(P_{D_m}) \rightarrow \mathcal{E}_\Gamma/(P_D)$ has nilpotent kernel of exponent at most M and induces a homeomorphism of prime spectra, so the prime and dimension statements of [theorems 12.9](#) and [12.17](#) persist for uniformly bounded multiplicities. This transfers the prime spaces and dimensions, not the reduced-ring conclusions such as von Neumann regularity or semiheredity. If (m_n) is unbounded, some prime ideal of $\mathcal{E}_\Gamma$ contains P_{D_m} but not P_D , so the reduced description cannot account for all primes above such a divisor.

Proof. $F \in \sqrt{P_{D_m}\mathcal{E}_\Gamma}$ means $F^d \in P_{D_m}\mathcal{E}_\Gamma$ for some $d \geq 1$; orders multiply by d , and [lemma 12.39](#) gives (80). If $m_n \leq M$, then P_{D_m} divides P_D^M and P_D divides P_{D_m} , so $P_D^M\mathcal{E}_\Gamma \subseteq P_{D_m}\mathcal{E}_\Gamma \subseteq P_D\mathcal{E}_\Gamma$. Since $P_D\mathcal{E}_\Gamma$ is radical ([lemma 12.3](#)), the radicals agree. The kernel $P_D\mathcal{E}_\Gamma/P_{D_m}\mathcal{E}_\Gamma$ has vanishing M th power, every prime contains a nilpotent ideal, and the quotient correspondence gives the homeomorphism. Conversely, if $P_D \in \sqrt{P_{D_m}\mathcal{E}_\Gamma}$, some P_D^M is divisible by P_{D_m} , and its order M at each ρ_n forces $m_n \leq M$. When (m_n) is unbounded, the class of P_D in $\mathcal{E}_\Gamma/(P_{D_m})$ is therefore not nilpotent; localizing at its powers gives a nonzero ring, a maximal ideal of which contracts to a prime avoiding that class, and its inverse image in $\mathcal{E}_\Gamma$ is the required prime. $\square$

Example 12.41 (Pointwise nilpotence is not uniform nilpotence). Take $m_n = n$. In the finite jet algebra at each ρ_n the class of P_D is nilpotent, since P_D has a simple zero there. Its class in $\mathcal{E}_\Gamma/(P_{D_m})$ is not nilpotent, because no single power has order at least n at every node. Unbounded multiplicity cannot be removed by observing that every jet variable is nilpotent; the exponent must be uniform over the whole sequence.

Remark 12.42 (What the threshold does not do). [Theorem 12.40](#) does not classify the primes that avoid P_D when the multiplicities are unbounded. It identifies the exact threshold at which the reduced spectrum suffices and proves that beyond it there are additional primes; no assertion about the complete spectrum in that regime is hidden in the other results. By [theorem 8.36](#), every maximal ideal containing P_{D_m} is an evaluation ideal or some $\mathfrak{M}_\mathcal{U}$, and each of these contains P_D ; so the additional primes are not maximal. That consequence is the merge's.

12.10 Sensitive steps and what this section does not establish

Logically sensitive steps. The sixth source lists the following as parts of its proofs, not as qualifications. The coarsened ultrafilter values are read in the quotient groups Γ/H_n ; the scalar field is not embedded as a constant sequence into the residue fields κ_n , and a scalar may have support in no single H_n although its leading exponent lies in H_n eventually (compare [remark 8.34](#)). The

convex-subgroup parameters of [theorem 12.9](#) need not be internal ultraproduct subgroups; every convex subgroup of $\Gamma_{\mathcal{U}}$ is allowed, and only the subgroups of the splitting example are internal by construction. Every localization of the reduced quotient is a domain although the quotient has many zero divisors; the annihilating indicator kills the kernel of the fibre map after localization. Prime-chain length is not inferred from the number of zeros: the witnesses of [propositions 12.18](#) and [12.34](#) have no zero node values and distinguish only asymptotic valuation cuts. The analytic steps it lists — a second evaluation radius in the coefficient criterion, a coefficient bound for a formal inverse that is not itself entire, and a finite initialization before damping — are those of [theorems 3.1](#) and [8.6](#) and [proposition 8.8](#). Finally, the quantifiers differ in statements that look alike: $x_n \in V_n$ means $v(x_n) \geq -m\gamma_n$ for some m depending on n ; a class of $\Gamma_{\mathcal{U}}$ lies in the convex subgroup generated by ζ when $|g| \leq m\zeta$ for one ordinary integer m outside the ultraproduct, whereas an internal subgroup of sequences of rational multiples is in general larger; $g > \mathcal{C}$ quantifies over every element of $\mathcal{C}$; a family of locally nilpotent elements need not be globally nilpotent in the product; and a chain with continuum many members is not thereby well ordered.

Non-claims of the sixth source. The following are retained without weakening.

- “Entire” is strong summability at every point of one fixed set-sized Hahn field, not entireness on $\mathbf{No}[i]$; nothing is claimed for the proper class.
- The exact value image and quotient, the canonical product and divisibility, the nonprincipal maximal ideals with their residue fields, the identification of the maximal spectrum with $\beta\mathbb{N}_{>0}$, and the preservation of entireness and zeros under cofinal extension are inherited from this report and are not new; the specialized simple-node proof is supplied only for dependency control ([remark 12.2](#)).
- The ultraproduct and valuation-domain mechanism for primes is classical [[15](#), [19](#)]; the Hahn support machinery is classical [[3](#)]; the normal forms are classical [[1](#), [2](#)]; the valuation-ring elimination behind the Smith forms is standard. No new abstract theory of product rings is claimed.
- The proposed contributions — the analytic prime fibres, the reduced-divisor dichotomy and the cofinal splitting construction — were not found in the sources it compared. That is not proof of their absence from the literature; the search was targeted, not exhaustive, and no priority is certified.
- No named published conjecture is solved. The manuscript has not been refereed; no Lean file is supplied and no machine-checked proof is claimed; its suggested Lean dependency order (Hahn-series, analytic, commutative-algebra, ordered-group and ultraproduct interfaces) is a suggestion, not code or a claim about present Mathlib coverage. It does not claim to have verified this repository completely: its comparison was concentrated on this report at its pin, and it states that the formalization status of those reports is not inferred from the presence of their LaTeX statements.
- The complete classification is for one simple node per occupied shell, extended to bounded multiplicities only for prime spaces and dimension. Shells carrying several nodes, unbounded multiplicities and the proper class $\mathbf{No}[i]$ are not classified.
- In the order-unit case the quotient is zero-dimensional; the full ring $\mathcal{E}_{\Gamma}$ is not claimed to be. The bounded-multiplicity transfer covers prime spaces and dimension, not von Neumann regularity or semiheredity.

- No assertion is made that zeros change under cofinal extension; the point of [theorem 12.33](#) is that they do not.
- The finite checks (Suite F of [section B](#)) do not establish strong summability, construct ultrafilters, prove continuum chains or certify novelty.

13 Assembly of the classifications

Proof of [theorem 1.5](#). The ring is an integral domain because it is a subring of $K_\Gamma[[Z]]$. Its units are exactly the nonzero constants by [corollary 6.3](#): a unit is zero-free, and nonzero constants are invertible. GCDs exist by [corollary 7.5](#).

If Γ has no countable cofinal subset, [corollary 3.3](#) gives $\mathcal{E}_\Gamma = K_\Gamma[Z]$, which is a PID over a field. If Γ has an order unit, its positive integer multiples form a countable cofinal subset, so nonpolynomial functions exist, and [theorem 9.3](#) supplies the Bézout property. If Γ has countable cofinality and no order unit, [theorem 8.29](#) supplies a proper ideal generated by two coprime functions. Such an ideal cannot be principal, since any principal generator would be a common divisor and hence a unit. Non-Noetherianity in both nonpolynomial cases is [proposition 7.9](#).

These cases are mutually exclusive and exhaustive. For $\Gamma = 0$, strong summability at 1 says that only finitely many coefficient supports may contain 0, hence only finitely many coefficients are nonzero. $\square$

Proof of [theorem 1.7](#). Part (a) is [theorem 10.4](#) together with [lemma 10.3](#); part (b) is [corollary 10.5](#), whose coarse form is [theorem 10.1](#); part (c) is [theorem 10.7](#); part (d) is [theorem 10.9](#); part (e) is [corollary 10.10](#). The residue field statement is [lemma 10.3](#). $\square$

Value-group condition	Entire series	Ideal structure
No countable cofinal subset	Only polynomials	PID
An order unit exists	Nonpolynomial series exist	Bézout, non-Noetherian
Countable cofinality, no order unit	Nonpolynomial series exist	GCD, not Bézout; two-generated proper ideal with no common zero

14 Explicit higher-rank and surreal examples

14.1 An explicit rank-infinite subgroup of the surreal field

Let

$$\Gamma_\infty = \bigoplus_{j \geq 0} \mathbb{Q}e_j, \quad e_j = \omega^j \in \mathbf{No}, \quad (81)$$

with the order inherited from $\mathbf{No}$: every element has finite support, and the largest index with a nonzero coefficient determines its sign. Thus $e_{j+1} \gg_A e_j$ and (e_j) is cofinal, so $\text{cf}(\Gamma_\infty) = \aleph_0$. No element is an order unit: the largest index in its finite support can always be exceeded. This is case (iii) of [theorem 1.5](#), and by [corollary 4.2](#) also $\mathcal{T}_{\Gamma_\infty}^\times = K_{\Gamma_\infty}^\times$.

The Bézout counterexample realized

Take $\gamma_n = e_n$ for $n \geq 1$, a positive cofinal sequence with $\gamma_{n+1} \gg_A \gamma_n$, as [lemma 8.1](#) demands. Using $t^\gamma = \omega^{-\gamma}$, [theorem 8.29](#) gives roots

$$\rho_n = \omega^{\omega^n}, \quad \sigma_n = \omega^{\omega^n} + \omega^{-\omega^{n+1}} \quad (n \geq 1), \quad (82)$$

and the functions

$$f(Z) = \prod_{n \geq 1} (1 - \omega^{-\omega^n} Z) = \prod_{n \geq 1} (1 - t^{e_n} Z), \quad (83)$$

$$g(Z) = \prod_{n \geq 1} \left(1 - \frac{Z}{\omega^{\omega^n} + \omega^{-\omega^{n+1}}} \right) \quad (84)$$

are entire on the fixed surcomplex Hahn field K_{Γ_∞} , not on all of $\mathbf{No}[z]$. They have real Hahn coefficients and all the displayed roots are real surreal numbers; by [corollary 7.8](#) this is visible from the conjugation-invariance of their divisors. The failure of a Bézout identity remains true even when complex Hahn coefficients are allowed in A, B , and consequently also when A, B are restricted to real Hahn coefficients. By [corollary 8.38](#), the maximal ideals of $\mathcal{E}_{\Gamma_\infty}$ containing f and g are exactly the $\mathfrak{M}_{\mathcal{U}}$ for nonprincipal $\mathcal{U}$, with the residue fields of [corollary 8.35](#). The fourth source records the same instance in its group $\Gamma_* = \bigoplus_{j \geq 1} \mathbb{Q}e_j$, which omits e_0 ; there the residue fields are built on $\bigoplus_{1 \leq j \leq n} \mathbb{Q}e_j$, of rank n . The values $t^{-n!e_n} = \omega^{n!\omega^n}$ at the nodes ω^{ω^n} are interpolated by the explicit series of [remark 8.21](#), and the values $t^{-e_{n+1}} = \omega^{\omega^{n+1}}$ are not interpolable.

The valuation formula (49) becomes

$$v(g(\rho_n)) = e_{n+1} + (2 - n)e_n + \sum_{1 \leq j < n} e_j,$$

whose first values are

$$\begin{aligned} \lambda_1 &= e_2 + e_1, & \lambda_2 &= e_3 + e_1, \\ \lambda_3 &= e_4 - e_3 + e_2 + e_1, & \lambda_4 &= e_5 - 2e_4 + e_3 + e_2 + e_1. \end{aligned}$$

The negative lower-level coefficients do not affect domination: the coefficient of the highest basis element e_{n+1} is 1.

For additional coefficient information, the k th coefficient of f is

$$[Z^k]f = (-1)^k \sum_{1 \leq j_1 < \dots < j_k} t^{e_{j_1} + \dots + e_{j_k}}, \quad v([Z^k]f) = e_1 + \dots + e_k,$$

the minimum coming from the unique subset $\{1, \dots, k\}$, whose coefficient is $(-1)^k$ and therefore cannot cancel. Division by the finite integer k still leaves a cofinal sequence of Archimedean classes, which independently exhibits the coefficient growth required by [theorem 3.1](#). The reciprocal $1/\sigma_j$ has the explicit support-certified form

$$\frac{1}{\sigma_j} = t^{e_j} \sum_{r \geq 0} (-1)^r t^{r(e_j + e_{j+1})},$$

a valid positive-support geometric identity by [lemma 2.1\(c\)](#). Its support need not itself be cofinal in the rank-infinite group; that is irrelevant to the legitimacy of this single coefficient. Entirety of the product is supplied by the cofinal escape of the factor indices j , not by a false claim about $r(e_j + e_{j+1})$.

The same workspace seen from its canonical product

The natural product over the *whole* basis,

$$P(Z) = \prod_{j \geq 0} (1 - t^{e_j} Z), \quad (85)$$

differs from (83) by the single linear factor of index zero:

$$P(Z) = (1 - t^{e_0} Z) f(Z) = (1 - \omega^{-1} Z) f(Z). \quad (86)$$

It is therefore an equally explicit member of the same ring, with one extra zero at ω .

Theorem 14.1 (A genuine infinite-rank surcomplex entire function). *The product (85) belongs to $\mathcal{E}_{\Gamma_\infty}$ and is nonpolynomial. Its zeros are exactly*

$$r_j = t^{-e_j} = \omega^{\omega^j} \quad (j \geq 0), \quad (87)$$

all simple and positive surreal. Its coefficient of Z^k is

$$c_k = (-1)^k \sum_{j_1 < \dots < j_k} t^{e_{j_1} + \dots + e_{j_k}}, \quad v(c_k) = e_0 + e_1 + \dots + e_{k-1} \quad (88)$$

for $k \geq 1$, with $c_0 = 1$. At a scale $s \in \Gamma_\infty$ put

$$N_s = \#\{j : e_j + s < 0\}, \quad M_s = \#\{j : e_j + s = 0\}.$$

These are finite, $M_s \in \{0, 1\}$, and

$$m_s(P) = \sum_{e_j + s < 0} (e_j + s), \quad (89)$$

$$I_s(P)(X) = (-1)^{N_s} X^{N_s} (1 - X)^{M_s}. \quad (90)$$

Finally

$$v(P'(r_j)) = e_j + \sum_{k < j} (e_k - e_j). \quad (91)$$

Proof. Since $e_j \rightarrow +\infty$ cofinally, the proposed zero divisor is radially finite, and [theorem 7.3](#) gives the product's existence, entireness and exact simple zero set. Expanding over finite subsets gives (88); among the k -element subsets, the sum of the first k basis elements is uniquely least in the largest-index order and its coefficient is $(-1)^k$, so it cannot cancel and every c_k is nonzero.

The slope sequence $(e_0 + \dots + e_{k-1})/k$ is cofinal: once $k-1$ exceeds the largest index occurring in a prescribed group element, its positive e_{k-1} coordinate dominates that element. This is a second, direct verification of entireness via [theorem 3.1](#).

At scale s only finitely many factors have $e_j + s \leq 0$. A factor of negative value contributes that value to m_s and contributes $-X$ to the normalized initial polynomial; a factor of value zero contributes $1 - X$; all positive factors contribute 1. This proves (89)–(90).

Differentiating the factor of index j and evaluating at r_j kills all other product-rule terms, giving

$$P'(r_j) = -t^{e_j} \prod_{k \neq j} (1 - t^{e_k - e_j}).$$

The factors with $k > j$ form a positive-support unit, and the finitely many factors with $k < j$ have valuations $e_k - e_j$; adding these proves (91). $\square$

This example is confined to no single Archimedean scale, yet its entire theory is nontrivial and its zero divisor completely explicit. Together with [remark 4.5](#) it shows that nonpolynomial entire functions and nonconstant restricted units are genuinely different existence questions.

14.2 Two series with the same leading valuations and different boundary behaviour

Let $\Gamma = \mathbb{Q}\epsilon \oplus \mathbb{Q}\eta$ with $0 < \epsilon \ll \eta$ and define

$$A(Z) = \sum_{n \geq 1} t^{n^2\epsilon} Z^n, \quad (92)$$

$$B(Z) = \sum_{n \geq 1} \left(t^{n^2\epsilon} + t^{\eta - n^2\epsilon} \right) Z^n. \quad (93)$$

Their nonzero coefficients have exactly the same valuations $n^2\epsilon$. Nevertheless their *monomial* strong evaluation domains differ. For $s = a\eta + b\epsilon$ with $a, b \in \mathbb{Q}$,

	$a < 0$	$a = 0$	$a > 0$	
$A(t^s)$	not strong	strong	strong	
$B(t^s)$	not strong	not strong	strong	(94)

For $a > 0$ the leading valuations have η coordinate $na \rightarrow +\infty$, and [lemma 2.2](#) proves both positive assertions. For $a < 0$ those leading valuations strictly decrease. For $a = 0$ the exponents of $A(t^s)$ are $(n^2 + nb)\epsilon$, which after finitely many terms strictly increase with only finite repetition at any exponent, so they form a well-ordered support. The extra exponents of $B(t^s)$ are $\eta + (nb - n^2)\epsilon$, which strictly decrease for all sufficiently large n ; they destroy strong summability without affecting a single leading coefficient valuation.

Neither series is entire over the whole rank-two field: their coefficient slopes $n\epsilon$ remain below η , so [theorem 3.1](#) correctly excludes both. There is no tension between this example and the valuation-only criterion. Leading valuations can fail to decide strong summability at a *boundary* scale, while global summability at *every* scale removes the ambiguity. Formula (94) is stated for monomial arguments only; it does not silently assert a classification for arbitrary nonmonomial arguments of B .

14.3 One extension preserves the function; another destroys entireness

Adjoin $\omega^{1/2}$ to Γ_∞ and take the divisible additive span. This is an ordered-group extension in which Γ_∞ remains cofinal, since a sufficiently large integer power of ω dominates every finite combination of the added and old generators. The series P , and likewise the pair f, g , remain entire by [corollaries 10.5](#) and [10.11](#); in particular the Bézout failure persists.

Instead adjoin the new generator $e_* = \omega^\omega > \Gamma_\infty$ and set $\Delta = \Gamma_\infty \oplus \mathbb{Q}e_*$. At the point $Z = t^{-e_*}$ the nonzero summands of P have leading exponents $(e_0 + \dots + e_{k-1}) - ke_*$, whose consecutive differences are $e_k - e_* < 0$; the defining family is therefore not strongly summable, exactly as [theorem 10.1](#) predicts. [Theorem 10.9](#) rules out any alternative entire power series over K_Δ agreeing with P on the old field, or even on $\mathbb{C}$. On the still-defined domain $\mathcal{D}_{\Gamma_\infty, \Delta}$, however, P is unchanged and its zeros remain exactly (87), by [theorems 10.4](#) and [10.7](#). This concrete pair is the clearest illustration of [remark 10.6](#).

14.4 Why finite rank does not give the counterexample

For $\Gamma = \mathbb{Q}^d$ in lexicographic order let h lie in the largest Archimedean class. Integer multiples of h are cofinal, so h is an order unit and the entire ring is Bézout by [theorem 9.3](#), even when $d > 1$. The same holds for $\mathbb{Q}e_1 \oplus \mathbb{Q}e_2$ with $e_2 \gg_A e_1$: take $h = e_2$. It is *absence of a greatest Archimedean class*, together with countable cofinality, that forces the negative case — not high rank.

14.5 An uncountable hierarchy gives polynomial rigidity

Let

$$\Gamma_{\omega_1} = \bigoplus_{\alpha < \omega_1} \mathbb{Q}e_\alpha, \quad e_\alpha = \omega^\alpha, \quad (95)$$

again ordered by the largest nonzero index. Its cofinality is $\aleph_1$: the basis is cofinal, whereas any countable family of finite supports has its indices bounded below ω_1 and a larger basis element bounds the whole family. Therefore

$$\mathcal{E}_{\Gamma_{\omega_1}} = \mathcal{T}_{\Gamma_{\omega_1}} = K_{\Gamma_{\omega_1}}[Z], \quad (96)$$

the middle equality because a nonpolynomial restricted series would itself supply a countable sequence of coefficient valuations tending cofinally to infinity. This example shows that infinite rank by itself does not even guarantee nonpolynomial entire functions: rank is not the driver, the order type above the scales is.

14.6 The separation table

Value group	cf	Has an order unit	Nonpolynomial entire series	Nonconstant restricted units
$\mathbb{Q}$ or $\mathbb{R}$	$\aleph_0$	Every positive element	Yes	Yes
$\mathbb{Q}\epsilon \oplus \mathbb{Q}\eta$, $\epsilon \ll \eta$	$\aleph_0$	Yes; not every positive element	Yes	Yes, with the sharper gap test
Γ_∞	$\aleph_0$	No	Yes	No
Γ_{ω_1}	$\aleph_1$	No	No	No

The second and third rows are the two nonpolynomial cases of [theorem 1.5](#); the third is the non-Bézout one. Note that $\mathbb{R}$ is uncountable of rank one with $\text{cf} = \aleph_0$, while Γ_∞ has infinite rank with $\text{cf} = \aleph_0$: neither cardinality nor rank is the invariant at work.

15 Relation to other reports in this collection

This report is a merge of six manuscripts, each prepared against a pinned snapshot of the same repository, and each states explicitly which of its neighbours' results it does *not* claim. Those statements are preserved here, and the cross-references are reciprocal.

Rank-one Berkovich geometry — this report extends it. The report *From Surcomplex Hahn Fields to Berkovich Geometry* [25] fixes $K = \mathbb{C}((t^{\mathbb{R}}))$: the value group is $\mathbb{R}$, the field is spherically complete, and $|a|_v = \exp(-v(a))$ is a genuine real-valued non-Archimedean absolute value. It defines intrinsic convergence using that real-valued absolute value. Its entire-function section proves zero-free entire rigidity and a convergent canonical product *there*, and none of the source manuscripts merged here claims to originate either point. The present report is the arbitrary-rank generalization. Three specific relations should be recorded.

- *Where rank one sits in the classification.* For $\Gamma = \mathbb{R}$ we have $\text{cf}(\mathbb{R}) = \aleph_0$ and 1 is an order unit, so rank one falls squarely in case (ii) of [theorem 1.5](#): nonpolynomial entire functions exist, $\mathcal{E}_{\mathbb{R}}$ is a non-Noetherian Bézout domain, and arbitrary radially discrete Hermite interpolation is available. The pathology of case (iii) is invisible at rank one.

- *Why rank one is not general.* That report’s `prop:rankobstruction` proves that $\mathbb{Q} + \mathbb{Q}\omega$ admits no order-preserving *injective* additive map to $\mathbb{R}$. That is exactly why [lemma 9.1](#) constructs a *coarsening* with a possibly nonzero kernel H rather than an embedding, and why it is available only when an order unit exists. The two statements are complementary: one forbids faithful scalarization in general, the other supplies the best available non-faithful substitute in the order-unit case.
- *The ring chains resemble each other and are not equal.* That report’s strict chain $K[Z] \subsetneq \mathcal{T} \subsetneq \mathcal{P} \subsetneq \mathcal{A} \subsetneq \mathcal{F} \subsetneq \mathcal{E}^-$ locates its polynomial-coefficient and formal algebras. Our chain (4) resembles it but is not the same chain, and the symbols $\mathcal{T}$, $\mathcal{A}$, $\mathcal{E}$ do not denote the same rings in the two reports. [Convention 1.2](#) forbids transferring a statement across that boundary.

Conversely, the rank-one report’s own entire-function section should be read with the knowledge that the arbitrary-rank classification now exists and that its rank-one instance lands in the Bézout case.

Analysis and Foundations — this report is published beside them. [Corollary 10.12](#) re-derives a conclusion the collection already contains twice: the *Analysis* report proves all-scale polynomial rigidity directly (its `Theorem f:thm-rigidity`), and the *Foundations* report records the exact boundary of the phenomenon with a pair of series over $\mathbb{C}((t^{\mathbb{Q}}))$ that differ only in the sign of an exponent, noting that $\sum t^{+n^2} z^n$ is strongly summable at every point of that field and is *nonpolynomial* while remaining coherent at every radius, and that this does not refute the all-scale theorem but shows that the all-scale theorem does not transfer to a *fixed* value group. That example is precisely [remark 3.2](#) of this report, and the trichotomy of [theorem 1.5](#) explains why it works over $\mathbb{Q}$ — $\text{cf}(\mathbb{Q}) = \aleph_0$ and 1 is an order unit, so $\mathbb{Q}$ is a case (ii) group — and what it would take to break it, namely replacing $\mathbb{Q}$ by a group of uncountable cofinality. Conversely, our reduction in [corollary 10.12](#) consumes the foundations report’s *localization principle*: every set of surcomplex numbers lies in one divisible set-sized workspace. That principle is what makes the reduction legitimate, and it is cited here, not reproved.

Global divisors and analytic geometry — different rings, and two restricted products that restrict different things. The global-divisor report [26] works in $\mathcal{O}(U)((t^{\Gamma}))$, with ordinary holomorphic coefficient functions on a common complex domain; its divisor, interpolation and Picard obstructions live in a different function ring from the strongly entire ring studied here, and nothing in this report contradicts or restates them. The analytic-geometry report states the ring discipline that [convention 1.2](#) invokes by name, for its own four coefficient rings. Neither the objects nor the theorems transfer.

That report and [theorem 1.6](#) nevertheless have the same moral, and the resemblance is close enough that the difference has to be stated rather than left to be inferred. Its `global:thm:interpolation` and `global:cor:CRT` prove a *support-restricted* Chinese remainder theorem: the correct target of global interpolation is a restricted subalgebra $\mathcal{B}$ of the product of the local finite algebras, not the full product, and its `global:ex:selector` shows the failure already for prescribed values in $\{0, 1\}$ — a 0/1 selector on clusters of growing size whose intrinsic quotient coordinates acquire an infinite strictly decreasing sequence of exponents. [Theorem 8.7](#) has the same shape, and [theorem 8.31](#) is likewise a 0/1 prescription that is not interpolable. The two restricted products are not the same construction and neither implies the other:

- *What is restricted.* There, the condition is that the *union of the coefficient supports* of the local remainders be *well ordered*. Here, the condition is *eventual membership* of the shell remainder's coefficients in a *coarsened valuation ring* V_n . Well-ordering of a support is not a valuation bound, and a valuation bound is not a well-ordering condition.
- *What the index set is.* There, the index set is a family of clusters of *infinitesimally close moving points*, each cluster bounded by a polynomial P_c , and the obstruction comes from negative powers of the separations *inside* one cluster. Here, the index set is the sequence of *valuation shells escaping through the Archimedean hierarchy*, and the obstruction comes from how far out in that hierarchy the data reaches. A single shell may itself contain close nodes, and then [theorem 8.31](#) shows the same divided-difference mechanism acting inside the shell — but the shellwise condition is a tail condition on n , which has no counterpart there.
- *Which ring.* That theorem is stated in $\mathcal{O}(U)((t^\Gamma))$, which [convention 1.2](#) and [section 16.3](#) explicitly disclaim for every statement in this report, and which the third source manuscript disclaims for its own. Its value group may be $\mathbb{Q}$, with an order unit, where [corollary 8.15](#) says our restriction disappears entirely; its obstruction survives there, because it is not our obstruction.

Both reports therefore say “the image of an infinite Chinese remainder map is a proper restricted subproduct, and the restriction is forced, not chosen”. They compute different restrictions, for different reasons, in different rings. The fourth source's restricted product $\prod'_n(A_n, B_n)$ is this report's $\mathcal{R}_D$, and the distinction applies to it verbatim; that source, too, disclaims every ring other than the strongly entire one.

A related collision of words concerns the analytic-geometry report. There an ideal is called *free* when its elements have no common geometric zero, and `analytic:thm:fixed-structure` produces, in $\mathcal{O}(D)((t^\mathbb{R}))$ on a fixed disk, maximal ideals containing no evaluation kernel and two-generated ideals with empty zero set; “boundary” there refers to a fixed disk. The fourth source calls a nonprincipal ultrafilter *free* and calls the non-evaluation maximal ideals of [section 8.9](#) *boundary ideals*. This report writes “nonprincipal” and “non-evaluation” throughout. The objects live in different rings and no statement transfers, although the moral — a proper finitely generated ideal with no common zero — is shared, and both reports credit Henriksen [13].

The same report also has a prime spectrum with continuum chains, and the resemblance to [section 12](#) is again close enough to state the difference. Its section on the prime spectrum of the fixed-polydisk ring $\mathcal{O}(D)((t^\mathbb{R}))$ (`analytic:sec:fixed-spectrum`, `analytic:thm:fixed-gauge-prime`, `analytic:thm:fixed-fibre`) builds, from a nonprincipal ultrafilter and the gauges $w(n) = n^r$, a continuum-sized chain of primes in one fibre over a classical maximal ideal; [lemma 12.16](#) uses the same power gauges $[n^\theta]$. The rings are different — ordinary holomorphic coefficients on a fixed disk and value group $\mathbb{R}$ there [27], strongly entire series over a fixed Hahn field here — and so are the reasons: its chain comes from the growth of Hahn valuations of values at a fixed simple divisor inside one classical fibre, ours from the quotient value groups Γ/H_n , which are zero when Γ has an order unit, as $\mathbb{R}$ does ([theorem 12.17](#)). No statement transfers. That report credits Henriksen's work on prime ideals of the ring of classical entire functions, which attaches primes to the growth of zero multiplicities [14]; the sixth source does not cite it, and we record it as context for [theorem 12.40](#), not as a checked comparison.

Holonomic rigidity and several variables — one device, two uses. The report *Holonomic Rigidity for Entire Hahn Functions* [30] works with the same whole-family entire functions of several variables. Its Lemma `hol:lem:generic` chooses a complex point avoiding countably many nonzero

polynomials over K_Γ by picking coordinates algebraically independent over a countable field; the covering step of [theorem 11.27](#) is the same device, applied to complex polynomials and refined to open disks. Its several-variable theorem uses the whole-family convention of [definition 11.1](#) and groups a family into total-degree blocks only as a step inside a proof, which is consistent with the separation of conventions in [section 11.5](#). No theorem is shared: that report proves D-finite rigidity, not an extension domain. The locus $\text{Pdir}(F)$ is written so to avoid the nonabelian-support report's Pol , which denotes a normalized polar factor.

16 What the results do and do not establish

16.1 Three independent distinctions

The first distinction is between strong Hahn summation and intrinsic valuation convergence. Positive well-ordered support suffices for the former; cofinal progress through every group threshold is needed for the latter. On a single disk they differ. For a series strongly evaluable at *every* point, [theorem 3.1](#) shows the distinction disappears within its original field.

The second is between countable cofinality and an order unit. Countable cofinality supplies a sequence of ever larger scales, possibly moving through infinitely many Archimedean classes; an order unit supplies all those bounds by the integer multiples of one element. Entire functions need only the first; a nonconstant restricted inverse requires the second. The group Γ_∞ separates them, and that separation is the whole content of the gap between cases (ii) and (iii) of [theorem 1.5](#).

The third is between an intrinsic topology in one workspace and the fine topology tested against every surreal scale. The same countable sequence may converge intrinsically in K_Γ and fail to converge after a noncofinal enlargement. None of our proofs identifies these topologies. See [section 2.3](#).

16.2 Established, and no more

The results isolate three logically different phenomena. Coefficient slopes determine whether an ordinary countable power series can be evaluated at every scale in a fixed field. Support-controlled preparation and canonical products completely describe zeros and divisibility within that category. The order structure above those scales determines whether arbitrary values and jets can be prescribed and whether coprime functions generate the unit ideal.

In particular the factorization monoid alone cannot decide the Bézout property. In both nonpolynomial cases a function up to a scalar is exactly a radially finite zero divisor, and every effective subdivisor is again principal; nevertheless only one of those cases admits unrestricted discrete interpolation. The additive structure of the ring, not merely its multiplicative divisibility relation, is essential.

The non-Bézout example is not a delicate choice of a transcendental ordinary complex function or of an uncomputable coefficient. Its nodes, displacements and factor coefficients are explicit Hahn expressions over rational data. What cannot be replaced by a finite calculation is the quantification over all coefficient indices and all Archimedean scales.

16.3 Scope that must not be silently enlarged

The following restrictions are stated by the source manuscripts and are retained without weakening.

Hypotheses that cannot be relaxed. All supports and index sets are sets; there is no proper-class summation. The field is a *full* Hahn field with complex residue coefficients, and Γ is divisible. Algebraic closedness is used essentially, to split the prepared finite polynomial; it is the sole import of [theorem 10.7](#). A nondivisible group, or a support-restricted Hahn subfield, may lose the roots needed to split the prepared polynomial, so the factorization theorem cannot simply be copied there. The Hahn–Neumann support lemmas ([lemma 2.1](#)) and algebraic closedness [4] are imported inputs, not proved here. [Section 11](#) alone works without divisibility and over an arbitrary trivially valued coefficient field, under its own statement of hypotheses, and none of the divisible-group results of the other sections is transferred to it.

Categories excluded. Outside [section 11](#) the ring is one-variable and strong-summation entire. It is *not* the ring of ordinary-holomorphic-coefficient Hahn families on one common complex domain, not a ring of arbitrary fine-local surcomplex germs, and not a several-variable analytic algebra; [section 11](#) treats finitely many variables only for coefficient and unit criteria, extension domains and exceptional directions, with no preparation, divisor or ideal theory. The arbitrary-rank preparation theorem is proved in $\mathbb{C}[X]((t^\Gamma))$, not asserted in a larger power-series coefficient ring. There is no contour integral here, no global surcomplex exponential at unrestricted imaginary arguments, and nothing about the Berarducci–Mantova derivation; the statement about primitives in [proposition 3.5](#) concerns termwise Z -integration with coefficients fixed and says nothing about differential equations in the coefficient field. All entire functions here are represented by one ordinary formal power series at the origin: translation preserves this class, but does not identify it with every locally analytic or surreal-differentiable function. Piecewise functions on fine-topological domains, Gonshor’s exponential, locally represented classes and other support or summation conventions lie outside the hypotheses, so [corollary 10.12](#) is *not* a general no-go theorem for surreal analysis.

Limits of the coarsening and of the extension theorem. The rank-one coarsening of [lemma 9.1](#) changes residue information even though it preserves the topology and the entire ring; residue-direction data is not identified across the two valuations. The scalar-extension results keep the coefficient field $\mathbb{C}$ fixed and use canonical ordered-group inclusions of full Hahn fields only: they are not theorems about arbitrary extensions of incomplete subfields, about changing residue fields, or about analytic sheaves. The hidden-tail table (94) is stated for monomial arguments and does not classify arbitrary nonmonomial arguments of B . The single-variable uniform-domain statement is *false* in several variables — a series independent of one coordinate imposes no constraint in that coordinate. Its exact replacement for whole-family summation is now [theorem 11.8](#), whose limits are recorded in [section 11.8](#); what remains open is recorded in [questions 16.5 to 16.7](#). No density of $\mathbb{C}$ in any fine surreal topology is asserted, and strong Hahn summability, intrinsic valuation convergence and the fine surreal topology are deliberately never identified.

Limits of the exact image theorem. [Theorem 1.6](#) concerns ordinary power series in *one* coordinate and radially finite divisors in a full Hahn field. It is not a theorem about $\mathcal{O}(U)((t^\Gamma))$, about common-domain holomorphic coefficient germs, or about functions continuous for the fine surreal topology; the third source manuscript disclaims that ring by name, as [convention 1.2](#) does. It does not extend automatically to several variables, where a shell need not carry one finite monic quotient controlling all the prescribed data, and it supplies no interpolation criterion for an infinite node set confined to a single valuation ball, where [lemma 8.2](#) gives no shell decomposition at all. The phrase “restricted product” in [definition 8.5](#) names an algebraic union and asserts no locally compact topology, no adelic structure and no Banach norm. [Theorem 8.33](#) constructs a specified

family of maximal ideals with a field presentation for each; it does *not* classify all maximal ideals of $\mathcal{E}_\Gamma$ and does not assert that every non-evaluation maximal ideal arises this way, and its nonprincipal ultrafilter is an ordinary set-theoretic choice, not an algorithmically computed object. The fourth source’s [theorem 8.36](#) shows that, for one node per shell, these and the evaluation ideals are all the maximal ideals containing P_D ; it too does not classify the maximal ideals of $\mathcal{E}_\Gamma$ that contain no such product, does not treat shells with several nodes, and does not classify prime ideals. The sixth source’s [theorem 12.9](#) classifies the prime ideals containing such a product when every node is simple, and [theorem 12.40](#) extends that to uniformly bounded multiplicities for prime spaces and dimension; primes above products with unbounded multiplicities or with several nodes on a shell, and primes containing no such product, remain unclassified.

Limits recorded by the fourth source. Its unit-ideal criteria are not a corona theorem: no bounded function algebra and no norm lower bound is specified. Its ultrafilter characters are ring homomorphisms and are not asserted to preserve strong Hahn sums, so $\Phi_{\mathcal{U}}(P_D)$ must not be computed termwise ([remark 8.41](#)); they are not evaluation maps on a larger surreal domain. Its word “entire” has fixed-field scope, and its nonpolynomial interpolants make no claim against all-scale polynomial rigidity. Its restricted product is algebraic, with no adelic or locally compact topology. Its jets are formal derivatives in Z , not the Berarducci–Mantova derivation. Its construction is a finite-step procedure relative to exact field operations and to bounds c_n , cofactor classes and remainders supplied as certificates; it does not make arbitrary Hahn coefficients computable and does not decide membership in H_n from an unspecified representation of Γ . It claims nothing about surreal exponential automorphisms, birthdays, integration or the proper-class analytic theory. It reproved the support, product and division ingredients it needed so that its interpolation theorem would not depend on this report’s unrefereed preparation and factorization arguments; [section A](#) records that the second route indeed uses neither.

The construction behind [theorem 8.6](#) is constructive only in a limited and stated sense. Given explicit finite representations of a shell’s nodes and jets, (31) is a finite computation in a quotient of $K_\Gamma[T]$, and a certificate $v([T^j]\Theta_n) \geq -c\gamma_n$ yields the explicit damping exponent $d_n \geq \max\{n, 2c+2\}$; each coefficient of the assembled F receives only finitely many corrections. This is not a decision procedure for arbitrary input Hahn series: deciding their equality, computing a valuation, or deciding membership in V_n may require information absent from a finite encoding. The theorem is an existence theorem with a finite-step construction *relative to* exact field operations and valuation-bound certificates, and the maximal ideals of [theorem 8.33](#) are not constructive even in that sense.

Finally, [proposition 8.42](#) states exactly which results survive over $\mathbb{R}((t^\Gamma))$, and no more. Algebraic closedness remains essential to [theorems 6.2](#) and [7.4](#) and hence to [corollary 7.5](#) and [theorem 10.7](#), as stated above; the real form is available for the image, quotient and Bézout statements only because their proofs split the relevant polynomials at the given nodes by hypothesis and because [proposition 8.8](#) supplies the kernel without preparation. Nothing in [proposition 8.42](#) should be read as weakening the algebraic-closedness hypothesis of the factorization theory.

Limits recorded by the sixth source. They are listed in full in [section 12.10](#). In summary: its “entire” has fixed-field scope and nothing is claimed for $\mathbf{No}[i]$; the quotient, the maximal ideals, their residue fields, the maximal spectrum and the cofinal-extension results on entireness and zeros are inherited, not new; the ultraproduct and valuation-domain mechanism, the Hahn support calculus, the normal forms and the elimination behind the Smith forms are classical; its classification covers one simple node per shell, and bounded multiplicities only for prime spaces and dimension; in the

order-unit case only the quotient, not $\mathcal{E}_\Gamma$, is claimed zero-dimensional; and no change of zeros under extension is asserted.

What is and is not claimed about priority and verification. The results of this report are self-declared proposed original contributions, not a solution of any named published conjecture. The first two source manuscripts state that priority is *not* certified and that the literature checks were targeted rather than exhaustive; in one of them several general web searches returned irrelevant results and were explicitly not treated as evidence of absence, and in the other a connector keyword search returning nothing was likewise not treated as proof. The rank-one factorization specialization is stated to be classical; standard Hahn algebra, algebraic closedness and support lemmas are not claimed; Cherry himself calls his factorization results well known; the rank-two distinction between positive valuation and topological nilpotence is established [8, Section 6.2]; and the all-surcomplex polynomial corollary is credited to the repository, as recorded in [remark 10.13](#). One source’s audit further notes that Lazard was checked bibliographically only and that Conway and Gonshor were not re-audited in full; its repository review was a targeted read of four files, not a line-by-line proof audit or an independent build. None of the six manuscripts was independently refereed and none was machine-checked: one records `formal_verification false`, `independent_peer_review false` and `priority_certified false` in its build record, the second states flatly that no Lean verification is claimed, and the third states that it has not been independently refereed or verified in Lean, that its literature comparison was targeted and not a certification of priority, and that its repository review was a targeted read of repository files at a pinned commit rather than a successful clone, a line-by-line audit of every report, or a Lean build. The third also states that its main image theorem is proved independently of this repository’s unrefered general factorization and preparation claims, and [section A](#) records where that independence does and does not hold in the merged text. The fourth states that its proofs have not been independently refereed or checked in a proof assistant, that its targeted literature search is not an exhaustive priority determination, that irrelevant search results were not used as evidence, that its repository review was a targeted read through a connector at its pinned commit and not an audit of every theorem, that no changes were made to the repository, and that it credits the general ultrafilter method and the finite algebra as classical; its package manifest records `independent_peer_review` and `proof_assistant_verification` as false. The fifth records `priority_certified`, `formal_verification` and `independent_peer_review` as false; states that its comparison was targeted, that no search returning nothing was treated as proof of novelty, and that binary repository archives, retired historical manuscripts and the Lean modules were not audited; and states that a successful build shows only that its source typesets. The sixth states that no named published conjecture is declared solved, that it has not been refereed or verified in Lean and supplies no Lean file, that its novelty language means only “not found in the compared sources” after a targeted comparison, not a priority certification, that its review concentrated on this report and the catalogue at its pin and is not a complete verification of the repository; its build record gives the verification boundary as written proofs and finite sanity checks, with no peer review or Lean verification claimed. No theorem below depends for its validity on the correctness of an unrefered repository report. The repository’s Lean modules on finite algebra and Hahn supports are plausible *prerequisites* for a future formalization, not a certificate that these proofs have been formalized; missing implementation is not treated here as an open mathematical problem. No repository files under `code/` or `data/` have been modified.

What the finite checks do not do. See [section B](#) for the full statement. In summary: the six delivered suites verify 831, 664, 1958, 1876, 1072 and 10889 finite exact assertions respectively,

and they prove *no* infinite theorem. The first three articles and their READMEs state that the finite truncations of the counterexample pair *are* coprime polynomials and *do* satisfy polynomial Bézout identities, so no finite computation can witness [theorem 8.29](#). The preparation checks run modulo t^9 in an integer exponent subgroup, not an arbitrary-rank group, and the finite dominance samples test only $k \in \{1, 2, 10, 100, 10000\}$, not all integer multiples. The checks cannot verify arbitrary well-ordered supports, the cofinality dichotomy, the necessity direction of [theorem 4.1](#), or any infinite product identity. The third suite additionally cannot verify infinite strong summability, the existence or nonexistence of an infinite entire interpolant, the restricted-product quotient, or the ultrafilter construction; its rational polynomial specializations verify algebraic identities only and are not offered as examples of positive fine valuation in $\mathbb{C}$. The fourth suite likewise uses rational proxies for the finite shell algebras and finite rational vectors for the valuation inequalities; it verifies neither infinite summability nor cofinality nor the theorems, and is not evidence of priority. The fifth verifies finite identities and example certificates; it cannot prove that an infinite support is well ordered, that a series is entire, that a general quotient fibre has finite incidence, or that every countable union of algebraic sets is realized, and it does not test the irrational example of [example 11.18](#) through a rational approximation. The sixth checks finite identities, ordered-group inequalities in finite-support rational vectors, finite multiplicity tests and Smith reductions over $\mathbb{Z}_{(2)}$; it does not establish strong summability, construct a nonprincipal ultrafilter, enumerate convex subgroups, prove the continuum-chain assertions, treat a proper-class construction, certify entirety from finite data or certify novelty.

16.4 Questions opened by the classification

The following are research questions generated by this work, not claims that any particular publication lists them as open.

Question 16.1 (Exact interpolation image). When Γ has countable cofinality but no order unit, characterize the image of

$$\mathcal{E}_\Gamma \longrightarrow \prod_{x \in \text{supp } D} K_\Gamma[U]/(U^{D(x)})$$

for a prescribed radially finite divisor D .

Answered, by [theorem 8.6](#). The question is retained in the form in which the two earlier manuscripts posed it, together with the note they attached to it, because the note turned out to be the right guess and locating the answer against it is the clearest statement of what was added. That note read: the growth barrier gives explicit necessary inequalities and proves that the image can be proper, but sufficiency for arbitrary nodes and jets is not established, and a satisfactory answer should encode the compatibility of divided differences across convex subgroups rather than only the size of individual values. The shell Hermite polynomial R_n of (19) is exactly that record of divided differences, and V_n is exactly the convex-subgroup datum; [theorem 8.31](#) is the instance in which individual values are harmless and the divided difference is not. What remains open after [theorem 8.6](#) is narrower and is recorded in [question 16.2](#).

Question 16.2 (What the image theorem does not decide). [Theorem 8.6](#) concerns radially finite divisors in a full Hahn field and one ordinary variable. Is there a criterion for interpolation on an infinite node set confined to a single valuation ball, where no shell decomposition exists? Is there a several-variable replacement, where a shell need not carry a single finite monic quotient controlling all the prescribed data? Does the membership condition (25) admit a decision procedure for any natural class of finitely encoded inputs?

The first two are genuinely open here; lemma 8.2 is what fails in each case. The several-variable results of section 11 concern extension domains, not interpolation, and do not bear on the second. The third is discussed in section 16.3: the construction behind theorem 8.6 is finite-step relative to exact field operations and valuation-bound certificates, and is not by itself an algorithm for arbitrary Hahn input.

Question 16.3 (Finitely generated ideals beyond their common divisor). Can one describe the ideal (f, g) by its common zero divisor together with an explicit obstruction in an interpolation quotient?

Answered for the case in which one generator is a canonical product, by theorem 8.26. The exact reduction the earlier manuscripts recorded — after dividing out the GCD, the Bézout problem is whether the inverse jets of one function along the other’s zero divisor lie in the interpolation image — is still the right reduction, and theorem 8.6 now supplies the missing intrinsic description of that image, while lemmas 8.24 and 8.25 turn it into the pointwise test (46). Theorem 8.29 still proves that a zero-divisor description alone is insufficient; the criterion says what must be added, namely an upper valuation bound on the values of the second function along the shells. Two parts of the question were left unsettled by the criterion: it decides membership of 1 in (P_D, F) , not the structure of (f, g) as an ideal for arbitrary f ; and theorem 8.33 constructs many maximal ideals containing such a proper ideal without classifying the maximal spectrum of $\mathcal{E}_\Gamma$, or the relations between restricted products attached to different divisors.

Partly answered further by the fourth source. Theorem 8.27 decides membership of 1 in $(P_D, G_1, \dots, G_s)$ for any finite number of generators, and corollary 8.28 decides membership of an arbitrary element in (P_D, G) when D has one simple node per shell; since every nonzero entire function is a scalar multiple of the canonical product of its divisor (theorem 7.4), so that $(f, g) = (P_{\text{Div } f}, g)$, this decides membership in (f, g) whenever the zeros of f are simple, of negative valuation, and one on each shell. Theorem 8.36 lists every maximal ideal containing such an ideal. Still open: membership in (P_D, G) when shells carry several nodes or multiplicities, and a description of (f, g) as an ideal beyond membership tests. See also question 16.4.

A related structural fact from the sixth source. For one simple node per shell the quotient $\mathcal{E}_\Gamma/(P_D)$ is a Bézout ring (theorem 12.21), so every finitely generated ideal of $\mathcal{E}_\Gamma$ containing P_D equals (P_D, G) for a single G (remark 12.22), and corollary 12.10 lists every prime ideal containing it. This does not describe (f, g) for arbitrary f , and the open parts listed above remain open.

Question 16.4 (The maximal spectrum). Do all non-evaluation maximal ideals of $\mathcal{E}_\Gamma$ arise as $\mathfrak{M}_\mathcal{U}$ of theorem 8.33 for some divisor and some nonprincipal ultrafilter, and how do the restricted products attached to different divisors relate?

Partly answered by the fourth source. Above a canonical product with one node per shell, with arbitrary multiplicities and empty finite part, the answer is yes, exactly: the maximal ideals containing P_D are the evaluation ideals and the $\mathfrak{M}_\mathcal{U}$ (theorem 8.36), and they form a copy of $\beta\mathbb{N}_{>0}$ (theorem 8.37). Still open, and not a hidden part of theorem 8.33 or theorem 8.36: maximal ideals containing only canonical products whose shells carry several nodes, where the finite residue algebras have several maximal ideals; whether every non-evaluation maximal ideal of $\mathcal{E}_\Gamma$ contains a product of the treated kind; the prime ideals; and the relations between the restricted products of different divisors.

The “prime ideals” clause is partly answered by the sixth source. Above a product with one simple node per shell and empty finite part, every prime ideal is an evaluation ideal or one of the primes

$\mathfrak{p}_{\mathcal{U},\mathcal{C}}$, indexed by a nonprincipal ultrafilter and a convex subgroup of $\prod_{\mathcal{U}}(\Gamma/H_n)$ (theorem 12.9), with local rings, topology and Krull dimension described in theorems 12.11, 12.13 and 12.17 and contraction under cofinal extension in theorem 12.29. For uniformly bounded multiplicities the prime spectrum is the same (theorem 12.40). Still open after the sixth source: the primes above products whose shells carry several nodes (question 16.9); the primes above products with unbounded multiplicities, where only the existence of primes not containing the reduced product is known (question 16.10); primes and maximal ideals of $\mathcal{E}_\Gamma$ containing no product of the treated kind; and the relations between the restricted products of different divisors. The first clause of the question is not affected.

Question 16.5 (Several variables). Which parts of the cofinality criterion and of the Bézout obstruction extend to several ordinary variables, and which local preparation statements retain explicit Hahn support certificates? In particular, since the single-variable uniform extension domain is false in several variables, what is the correct replacement — presumably one depending on the unbounded degree directions of the support, including the boundary where higher Hahn tails matter?

One must separate the cofinal growth of total degree from the geometry of positive-dimensional zero sets. Cherry’s several-variable work is an important rank-one antecedent [6]; no arbitrary-rank multivariable GCD theorem is established here. Section 11.2 proves only the finite-variable convergence and unit criteria, and it does not extend the one-variable divisor classification to higher-dimensional zero sets.

The “In particular” clause is answered by the fifth source, for whole-family summation: theorem 11.8 is the exact replacement. It depends only on the ordinary support, through its weight image in Δ/H , and for semilinear supports only the period vectors, the unbounded degree directions, enter (theorem 11.15), in line with the question’s guess. The guess about the boundary was wrong in one respect: within the entire category, higher Hahn tails do *not* matter (corollary 11.9); they matter only for non-entire series (example 11.19, and in rank two section 14.2). The other clauses remain open: whether the Bézout obstruction and the ideal theory extend, and which local preparation statements retain support certificates, in several variables. The cofinality criterion does extend (theorem 11.3).

The fifth source adds two questions, suggested by its results; like the others, they are not claims that any publication lists them as open.

Question 16.6 (The full block domain). Can the homogeneous-block domain $\mathcal{D}^{\text{hom}}(F)$ at arbitrary tuples of new-field coordinates be described by a finite hierarchy of quotient valuations and initial homogeneous forms, with a precise condition for indefinite cancellation? Such a description should be invariant under linear changes of coordinates and recover theorem 11.23 on old-field rays.

Question 16.7 (Effective support classes). For which finitely encoded supports does the well-ordering test of theorem 11.8 admit an effective exact procedure over specified finite-rank ordered groups? Polynomially parametrized supports already give nonclosed domains (theorem 11.17), and example 11.18 shows that lower bounds on the weights are not enough for arbitrary supports.

Question 16.8 (Restricted and incomplete subfields). How much of the divisor theory survives for incomplete surreal subfields or for support-restricted Hahn subfields? Our proofs isolate two requirements: closure under the positive-support preparation construction, and enough algebraic roots to split the finite prepared polynomials. A classification of natural surreal subfields meeting both would turn these full-Hahn results into a more computationally selective theory.

The sixth source adds three questions, suggested by section 12; like the others, they are not claims that any publication lists them as open.

Question 16.9 (Several nodes on a shell). [Theorem 8.6](#) uses finite shell algebras when several nodes share a radius. Which additional prime fibres appear when distinct normalized nodes collide after coarsening? The one-node valuation-domain description of [theorem 12.9](#) does not apply directly to those finite algebras.

Question 16.10 (Unbounded multiplicities). For $P_{D_m} = \prod_n (1 - Z/\rho_n)^{m_n}$ with (m_n) unbounded, can the primes not containing P_D be described directly from asymptotic zero orders and coarsened coefficient values? [Theorem 12.40](#) gives an exact radical test but not that classification.

Question 16.11 (When is the prime spectrum invariant?). [Theorem 12.29](#) reduces unique extension of a prime to the equality $\mathcal{C}_{\min} = \mathcal{C}_{\max}$. Which convenient conditions on a cofinal inclusion $\Gamma \subseteq \Delta$ make these equalities hold for every nonprincipal ultrafilter and every convex subgroup? The order-unit case is one sufficient condition ([corollary 12.30](#)), but it does not exhaust the conceivable cases.

16.5 A proof-assistant route

The research statement does not require a formalization of the entire proper class **No**. A first formal development can work over a set-sized divisible linearly ordered abelian group and a Hahn field. The most useful first target is the global criterion [theorem 3.1](#), whose nontrivial implication is a short ordered-group argument plus the definition of strong summability; the next is the coarsened polynomial argument for [theorem 4.1](#). Support preparation should be formalized only after the finite-word Neumann lemma and arbitrary regrouping are available. The explicit obstruction then requires only ordered-group inequalities, finite product valuations and evaluation homomorphisms. This is a dependency plan, not a formal-verification claim. The current repository ledger `docs/FORMALIZATION.md` distinguishes checked Hahn support, summability and algebraic-closure prerequisites from the still-pending statements of this report. No Lean code is included in this report package. For [section 12](#) the sixth source suggests five interfaces in dependency order: Hahn series (strong summability, the coefficient criterion, weighted Gauss multiplication, linear division); the analytic layer (canonical product, divisibility, exact evaluation quotient); commutative algebra of the eventual-integrality sequence ring (idempotents, annihilators, finite generators, localization); ordered groups (the convex-subgroup prime correspondence and the interval [\(73\)](#)–[\(74\)](#)); and ultraproducts, combining these and specializing the explicit example. It is a suggestion, not code or a claim about Mathlib coverage.

17 Conclusion

The threshold for nonpolynomial entire behaviour in a Hahn–surcomplex workspace is countable cofinality, not rank one. Among workspaces admitting nonpolynomial entire series, the universal Bézout property and full radially finite Hermite interpolation require the stronger condition that an order unit exist. This is also the threshold for existence of a nonconstant restricted unit; for a particular restricted series, its least nonconstant coefficient gap relative to the constant term must itself be a positive order unit. Individual coprime pairs may admit a Bézout identity even when the universal property fails. The two existence conditions separate in an explicit infinite-rank subgroup of the surreal numbers, and the separation is witnessed by two entire functions with disjoint simple zero sets whose ideal is proper.

After enlarging the value group, every nonpolynomial entire series has the same exact strong domain: the valuation ring of the convex hull of its original scales. Cofinality is precisely what

preserves entireness, while support preparation preserves every existing zero and prevents new ones throughout the surviving domain. The canonical product theorem then shows that coefficients, zeros and the extension boundary encode the same cofinal scale structure. Together this gives one explanation of both nonpolynomial fixed-workspace functions and polynomial rigidity on the whole surcomplex class, without confusing Hahn identities with convergence in the fine surreal topology.

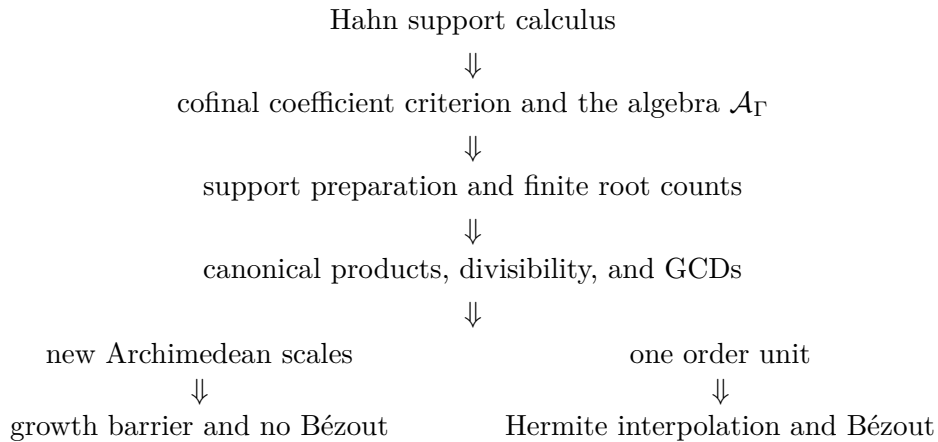
Between these thresholds the obstruction is measured exactly. An infinite Hermite prescription is realizable precisely when its shell data are eventually integral for the coarsening that kills the Archimedean scale of each shell, and two independent constructions, sequential and cardinal, realize it. Above a product with one node per shell, the maximal ideals are the evaluations and one ideal for each nonprincipal ultrafilter on the shells, with algebraically closed residue fields assembled from lower-rank Hahn fields. In finitely many variables the uniform extension domain gives way to a well-ordering condition on the quotient weights of the ordinary support, while for nonpolynomial series homogeneous-block summation along complex directions sees exactly the proper countable unions of projective algebraic sets, all of which occur when the value group has countable cofinality.

Below the maximal ideals, a product with one simple node per shell carries two kinds of boundary data. Its idempotents record which infinite collections of shells an ultrafilter selects, and produce the Stone space $\beta\mathbb{N}_{>0}$ common to its minimal and maximal primes; its ultraproduct value groups record how small the other functions become along those shells, and produce the prime chains inside each fibre. With an order unit the second layer disappears from the quotient; without one it contains a continuum chain in every nonprincipal fibre. Under a cofinal enlargement the nodes and the Stone base stay fixed while the second layer can refine, and the rational-indexed enlargement of Γ_∞ realizes this inside surreal and surcomplex Hahn workspaces: preserving zeros is not the same as preserving the prime spectrum of an entire divisor.

A Proof-dependency and hypothesis audit

A.1 The dependency chain

The main proof can be read along the following chain.



The arrow from preparation to root counting additionally uses algebraic closedness of the Hahn field. The arrow to interpolation uses completeness of the real-valued coarsening, proved directly in [lemma 9.1](#). No proof of the negative result depends on the positive interpolation theorem.

The scalar-extension results of [section 10](#) sit on a deliberately separate branch. The restricted-unit theorem ([theorem 4.1](#)) uses Gauss multiplicativity (5) and a convex coarsening only; it uses

neither zero counting nor canonical products. The universal-domain theorem (theorem 10.4) uses the global criterion and integral evaluation; it does not use factorization. *Only* zero conservation (theorem 10.7) imports both algebraic closedness and preparation. The divisor classification is downstream of the local tools and the elementary entire algebra. Thus the principal classifications are not consequences of an unproved global factorization assertion, and conversely the factorization proof never assumes that an arbitrary zero-free restricted series is restricted-invertible — by example 4.3 that is exactly the assertion that can fail in higher rank.

The fourth source’s cardinal route to the image theorem (section 8.5) uses the canonical product (theorem 7.3), Gauss multiplicativity including its equality case, the integral inverse of the cofactor class and the summation lemma lemma 8.17. Like the first route it uses neither preparation nor algebraic closedness, and it does not use the first route. The fourth source’s ideal-theoretic results (theorems 8.27 and 8.30 and corollary 8.28 and section 8.9) use the quotient isomorphism, and hence either route together with proposition 8.8; algebraic closedness enters only the statement about the type of the residue fields, corollary 8.39, through the Hahn-field theorem for $\mathbb{C}((t^{H_n}))$.

Section 11 sits on a third separate branch. It uses the Hahn–Neumann support calculus, the finite lattice estimate, unit twisting, the Hilbert basis theorem and countable-field avoidance. It uses no divisibility, preparation, root count, factorization or algebraic closedness. Of the other sections it uses only lemma 2.1, the definitions (61) and (62) and the normal-form map (1), none of which needs them; theorem 11.5 alone keeps the report’s standing hypotheses and cites theorem 4.1.

Section 12 sits downstream of the quotient isomorphism. Of the analytic theory it uses theorem 8.7 in the form corollary 8.13, hence either route to the image together with proposition 8.8, and, for the canonical product with multiplicities, theorem 7.3; for the comparison of two workspaces it uses theorem 3.1 and proposition 8.8 over the larger group. It uses no preparation, root count, factorization or algebraic closedness, and it does not use zero conservation: the zeros of its product in the larger field are located directly (lemma 12.26). The rest is commutative algebra of the sequence ring, ultraproducts and the valuation-domain prime correspondence. Algebraic closedness enters only through the type of the residue fields $L_{\mathcal{U}}$, quoted from corollary 8.39.

A.2 Where an attained minimum is load-bearing

Attained minima. The proof of theorem 3.1 uses a minimum attained at an ordinary *finite* coefficient index, not a termwise inequality. The proof of lemma 8.22 uses an attained minimum in the same way: termwise strict inequalities alone would not justify a strict inequality for an arbitrary limiting process. These are the two places in the article where the distinction is load-bearing, and neither can be replaced by an estimate valid only term by term.

Two independent infinities. The finite index k in a power series is not a Hahn exponent. Integer multiples of a positive exponent need not be cofinal. The support lemma lemma 2.1(b) handles geometric-series expansions without that property; the positive interpolation proof uses it only after constructing a legitimate real-valued coarsening, which is available only under the order-unit hypothesis.

Divisibility is not assumed in its own proof. The quotient F/G in theorem 7.4 is first formed, after removing the shared zero at the origin, in $K_{\Gamma}[[Z]]$. Only preparation at every scale proves it entire. This avoids assuming the global divisibility theorem in its own proof.

The counterexample is irreducibly infinite. No finite truncation of the pair (48) can witness non-Bézout behaviour: the finite truncations are polynomials with disjoint root sets and therefore generate the unit ideal. The contradiction requires a single candidate coefficient lower bound to fail at cofinally many successively dominating scales.

Why preparation must be a support recursion. [Corollary 5.3](#) — the fact that the prepared factorization is unchanged by enlarging the value group — is available precisely because preparation was proved by well-founded recursion over S^* . An iteration whose convergence had been asserted in K_Γ would have to have that convergence re-established in K_Δ , and [example 4.3](#) shows it can genuinely fail.

B Reproducible finite checks

Six independent exact-arithmetic suites accompany this report, one from each source manuscript, preserved unmodified in `code/` and `data/` under the file names given below. All six use only the Python standard library, with exact integer and `fractions.Fraction` arithmetic and no floating point. The first four represent group elements as tuples of rational coordinates ordered by the largest nonzero coordinate; the fifth needs no ordered-group tuples beyond integer relation vectors; the sixth uses finitely supported rational vectors with rational indices, ordered in the same way.

Suite A (831 checks). The source program `verify_examples.py` is shipped as

`code/06-cofinality-bezout-trichotomy-verify-examples.py`

It checks finite instances of the reverse-lexicographic exponent inequalities, the valuation formula (49) in the first nine coordinates of Γ_∞ , leading exponents of finite canonical products, a finite-truncation implementation of the preparation recursion (11)–(12), and the local inverse-jet identity underlying (57). The delivered run passed all **831** checks, with category counts 72 paired-root checks, 40 finite dominance samples, 88 product-coefficient checks, 456 preparation checks and 175 inverse-jet checks; the record is the delivered verification JSON. The script takes a *required* output path argument and exits nonzero without one; a fresh output path preserves the historical record. A review run on a temporary copy passed 831 of 831 again.

Suite B (664 checks). The source program `verify.py` is shipped as

`code/03-cofinality-and-scalar-extension-verify.py`

It checks finite truncations of (85) for $N = 1, \dots, 8$ — each coefficient’s leading exponent, its binomial subset count and its signs — exactness and simplicity of each root t^{-e_j} and the simple-root derivative (91), closed- and open-ball Newton counts at sampled scales including half-shifts by e_0 , descent of the valuations after adjoining the dominant scale e_9 , the rank-two hidden-tail comparison of [section 14.2](#) for $n = 1, \dots, 100$ and $\lambda \in \{-7, -\frac{1}{2}, 0, \frac{3}{2}, 10\}$, and the preparation recursion for (13) through perturbation order eight. The delivered run reports **664** passing checks. A maintained review on a temporary copy passed all 664 checks, leaving historical records unchanged. The delivered results are recorded in `data/03-cofinality-and-scalar-extension-verification.json` and `.txt`. The script takes no arguments and writes `verification.json` and `verification.txt` into the `data/` directory next to its own folder, so it should be run on a copy of the directory.

Suite C (1958 checks). `code/01-exact-jet-image-verify.py` accompanies the third source and the material of [sections 8.2 to 8.4](#). It checks, over 24 finite polynomial cases, 5 damping exponents and rank vectors of dimension 12, the finite Hermite jet identities and degree bounds, preservation of earlier jets under a shell correction, the shell correction congruence and its low-degree vanishing, the finite quotient inverse and the multiplication-determinant formula, the paired-root valuation formula and its coarse maximal ideal, and the close-pair corrected value and forbidden coefficient. The delivered record `data/01-exact-jet-image-checks.json` reports **1958** passing checks in 17 categories, with random seed 20260922. The script takes a required output path and refuses to overwrite an existing file. Its own record lists what it does not verify: infinite strong summability or cofinality, the existence or nonexistence of an infinite entire interpolant, the infinite quotient and ultrafilter theorems, and independent refereeing or Lean verification.

Suite D (1876 checks). `code/07-scale-moderate-interpolation-verify.py` accompanies the fourth source, chiefly [section 8.5](#) and the paired threshold. Over 18 random cases with three shells each, at the rational scales 2, 17 and 103 and with near-coincident rational roots 1 and $\frac{102}{101}$ of multiplicity one or two, it checks exact division of the product by each shell factor (54), the cofactor inverse and the inverse of T in the finite algebra (54 each), the cardinal shell jets — the prescribed class on its own shell and zero on the others (162) — the multiplication-determinant formula (27) (54), and simultaneous interpolation by the sum of the cardinal corrections (54), together with the two values of the close-pair polynomial (2). In nine-dimensional rational vectors ordered by the largest coordinate it checks the scale condition $\gamma \geq -\gamma_n/2$ and the suppression inequality behind (43) (700 each), that the paired value (50) lies in H_n exactly when the displacement does (21) and is positive when it does not (14), and that the gain $\varkappa_n$ is nonnegative (7). Its random seeds are 22092026 and 20260922. The delivered record `data/07-scale-moderate-interpolation-verification.json` reports **1876** passing assertions; it lists as not verified infinite Hahn summability, the full theorems, novelty and Lean formalization. The file `data/07-scale-moderate-interpolation-package_manifest.json` records the source package’s pin, page count, assertion count, the two verification flags, and SHA-256 hashes of the six files of that package; of those, the script, the record and the source audit (shipped as `07-scale-moderate-interpolation-SOURCE_AUDIT.md`) are in this directory and match their hashes, while the manuscript’s own `article.tex`, `article.pdf` and `README.md` are not shipped. The `-output` argument is optional; without it the script only prints. Its documented command writes to `data/verification.json`, so a re-run should name a scratch path instead.

Suite E (1072 checks). `code/08-multivariate-extension-verify_examples.py` accompanies the fifth source and [section 11](#). With bivariate polynomials over $\mathbb{Q}$ it checks the homogeneous products of the explicit direction example and their value at infinity (20), the prescribed polynomial directions and the nonvanishing products off them (210), binomial block cancellation in $(X - Y)^n$ (80), the quadratic weight-difference identity and eventual-sign certificates of [theorem 11.17](#) (100 and 25), samples approaching its nonclosed boundary (60), semilinear period-weight identities (225), a two-generator relation lattice and its residual bound (243), coordinate-face realization from minimal excluded faces (57), and degree-separated projective block zero loci (52). The delivered record `data/08-multivariate-extension-verification.json` reports **1072** passing assertions and sets `formal_verification`, `independent_peer_review` and `priority_certified` to false. `data/08-multivariate-extension-research-audit.json` records the pin, the blob of this article at that pin, the answered clause, the items not claimed solved and the audit scope, and `data/08-multivariate-extension-build.json` records the source’s own build, including SHA-256 hashes of its `article.tex` and `article.pdf`, which are not shipped here. The `-output`

argument is optional; the documented command replaces `data/verification.json`, so a re-run should name a scratch path.

For this merge both new suites were re-run with `-output` pointing outside the repository; each reproduced its delivered record exactly, up to platform line endings.

Suite F (10889 checks). `code/09-prime-spectra-verify_finite.py` accompanies the sixth source and [section 12](#). It checks the node values of the simple-node Lagrange corrections of [remark 12.2](#) at the rational nodes $2, 4, \dots, 2^n$ (153; the source calls them cardinal corrections, which are not the entire cardinal corrections of [section 8.5](#)) and their damping inequalities in rational vectors (390); the intersection of old integer scales with new rational cutoffs, at an intermediate subgroup and at the maximal endpoint (3600 each), with 192 strict-containment witnesses, in finite-support rational vectors; 30 representative power-gap inequalities $mn^\theta < n^{\theta'}$ at the single value $n = 2^{60}$, with $\theta, \theta' \in \{\frac{1}{5}, \dots, \frac{4}{5}\}$, behind [lemma 12.16](#); 200 Smith diagonalizations of random matrices of sizes up to 5×5 over the valuation ring $\mathbb{Z}_{(2)}$, checking the diagonal shape (200), the identity $UMV = \text{diag}$ (200), the integrality of the transformations and their inverses (800), the pivot ratios (1190) and the divisibility chain (240); and the finite multiplicity tests behind [theorem 12.40](#) (196 uniform-power and 98 radical-order checks). Its random seeds are 23092026, 10050 and 31. The delivered record `data/09-prime-spectra-verification_results.json` reports **10889** passing assertions in 13 categories and states that the checks are finite sanity checks, not proofs of infinite or ultrafilter theorems; `data/09-prime-spectra-BUILD_REPORT.json` records the source’s own build of its 28-page article, which is not shipped. The `-json` argument is optional; without it the script only prints, and the documented command rewrites the delivered record in place, so a re-run should name a scratch path. For this merge it was re-run with `-json` pointing outside the repository and reproduced the delivered record exactly, up to platform line endings.

What the checks cannot do. All six suites print their own scope disclaimers, and all six are correct to do so. The counts are counts of finite assertions, not confidence scores for any infinite theorem. No suite can prove cofinality statements, Hahn support well-ordering for every permitted group, any infinite product identity, the necessity direction of [theorem 4.1](#), or the nonexistence of a global Bézout solution. In the preparation checks the t -adic truncation is explicitly finite, so higher coefficients are discarded only after the claimed congruence has been specified, and the exponent subgroup used is an integer one rather than an arbitrary-rank group. The finite dominance samples test only $k \in \{1, 2, 10, 100, 10000\}$. Most importantly, the finite truncations of (48) are coprime polynomials and *do* satisfy polynomial Bézout identities, so no finite computation can witness [theorem 8.29](#). What the checks can detect is an indexing error, a sign error in a paired root valuation, a reversed order convention, a wrong root-count endpoint or a wrong finite preparation coefficient. Everything else rests on the proofs in the body.

C Source and novelty ledger

The first two of this report’s six source manuscripts are both dated 21 September 2026, and each pinned the repository at its own snapshot: one at commit `4896a2808ce30e01b1c86ae3ba2295a64762d246` and the other at `aa846271b4dcae2c055b216126a87210292ec19b`. The first re-read four files at its pinned commit — the root README, the documentation catalogue, and the READMEs of the rank-one and global-divisor reports — after its working branch changed during preparation; the second cited the documentation catalogue and the rank-one article. Both

reviews were targeted reads, not line-by-line proof audits or independent builds. Neither reinterprets missing Lean implementation as an unsolved mathematical problem.

The primary literature checked across the two manuscripts comprises Poonen’s Hahn-field construction and algebraic-closedness corollary, Neumann’s ordered division-ring paper for the support calculus, Cherry’s complete real-valued non-Archimedean setting together with his one-variable factorization statement and his lectures, Conrad’s higher-rank valuation examples, Lazard’s bibliographic record as the historical reference for one-variable zeros, van der Hoeven’s generalized-series operator framework, and the Conway–Gonshor surreal normal-form background with cross-checks against [10, 11]. Lazard was checked bibliographically only, and Conway and Gonshor were not re-audited in full. General web searches were attempted; several returned irrelevant results and were not treated as evidence that a theorem is absent from the literature.

A third manuscript, dated 22 September 2026, was written later against the merged report itself, at the snapshot `a3124af79f66b8b9c196d76b4cbc5ac3938907c4`. Its decisive comparison was that snapshot’s subsection recording the missing image of the infinite jet map, now answered by [theorem 1.6](#). It was a targeted source inspection through repository file reads, not a local clone, a line-by-line audit of every report, or a Lean build. Its literature comparison covered Lazard, Cherry and van der Hoeven and located no matching statement of the shellwise restricted-product theorem; that is a reason to propose it as new, not a proof that it occurs nowhere. The ultrafilter construction is standard algebra; what is proposed there is the concrete analytic surjection and the explicitly identified residue fields. It credits the coefficient criterion, the canonical product, the order-unit interpolation theorem and the explicit non-Bézout pair to the first two sources and reproves them only for dependency control.

A fourth manuscript, *Scale-Moderate Interpolation and Boundary Ideals in Surcomplex Hahn Fields*, dated 22 September 2026, pinned the repository at `0097304c7ae9d5de46d2ea342e2494f8fc126303`. At that snapshot the merged report still contained the subsection “The missing image of the infinite jet map” and recorded [question 16.1](#) as open, so the manuscript’s statement that it resolves “the repository’s explicitly unidentified infinite jet image” was accurate for its pin. It is stale for this tree: the third source, pinned earlier at `a3124af`, was merged at commit `de84d8b` about an hour after `0097304` and had already answered the question. The fourth source is therefore recorded as an independent, near-simultaneous second proof of [theorem 8.6](#), [theorem 8.7](#) and [theorem 8.26](#), under the same hypotheses (Γ divisible and set-sized, coefficients $\mathbb{C}$ or $\mathbb{R}$), and not as their source. Its review read the documentation index, the formalization ledger and the relevant parts of this report through a connector at the pin; it was targeted, not an audit of every theorem. Its literature comparison covered Poonen, both Cherry references, Helmer (metadata only, a historical antecedent and not a proof input), Bruno’s dissertation (whose value group is explicitly a subgroup of $\mathbb{R}$), Henriksen and Gonshor. Where the two texts prove the same statement it is printed once: its coefficient criterion and Gauss estimates, canonical product, kernel theorem, exact sequence, one-node criterion, close-pair example, factorial example, order-unit check and unit test for shell algebras are [theorem 3.1](#), [proposition 3.5](#) with (5), [theorem 7.3](#) with (26), [proposition 8.8](#), [theorem 8.7](#), [corollary 8.13](#), [theorem 8.31](#), [corollary 8.23](#), [corollary 8.15](#) and [lemma 8.24](#). Its sufficiency construction is genuinely different and is kept as the second route of [section 8.5](#); its finite-prefix corollary is the completion step there. Its new material is [corollaries 8.16](#) and [8.20](#) and [lemmas 8.17](#) to [8.19](#), [remark 8.21](#), [theorems 8.27](#) and [8.30](#) and [corollary 8.28](#), all of [section 8.9](#), and [remark 8.43](#). Where the merge had to choose: its letters were translated into this report’s ([section 2.5](#)); its surreal instance was restated in Γ_∞ , which contains e_0 , instead of its Γ_* ; and its finite-generator and membership results are stated, as in the source, for divisors with empty finite part.

A fifth manuscript, *Multivariate Entire Hahn Series across New Surreal Scales*, dated 22 September 2026, pinned the repository at `4cf691c7d951e037739d32d9f5c387dcce724f3c`, after the third source had been merged, and recorded the blob `f034bdb` of this article at that pin; the body it read is unchanged in the present base, which differs only in the ledger and checks appendices. It read this report’s introduction, extension section, questions, scope discussion and finite-variable appendix, together with the documentation inventory and some neighbouring summaries; binary archives, retired manuscripts and the Lean modules were not audited. Its claims about this report — that [question 16.5](#) asks for the several-variable replacement, that a series independent of one variable cannot constrain it, and that the appendix already proved the finite-variable coefficient criterion and cofinality dichotomy — were checked and hold; that appendix is now [section 11.2](#). Its literature comparison covered van der Hoeven’s operators, Achinger’s notes on rank-one non-Archimedean geometry [21], Joswig and Smith on convergent Hahn series and higher-rank tropical geometry [22], Ma and Neelon, the Stacks Project for the Hilbert basis theorem, and Berarducci for Conway normal forms. Its growth criterion replaces the earlier appendix’s ratio-form proof as the statement of [theorem 11.3](#), under weaker hypotheses; the earlier [theorem 11.5](#) is kept with its original hypotheses; everything else in [section 11](#) is its new material. Where the merge had to choose: its homogeneous block, admissible set, support, quotient, scale and polynomial-direction symbols were renamed ([section 2.5](#)); its standalone localization proposition is printed as [lemma 11.30](#) with credit to the foundations report; and none of this report’s divisible-group or algebraically closed results was attached to its weaker hypotheses.

A sixth manuscript, *Prime Spectra at Surreal Infinity: Reduced entire divisors, valuation fibres, and splitting under cofinal scalar extension*, dated September 2026, pinned the repository at `465a54b479a1ee842cbf7db1689a7d2f6bfe25e1`, after the fifth source had been merged; this report’s directory is identical at that pin and at the base of the sixth merge, and the seven labels its audit cites (`ent:thm:image`, `ent:thm:quotient`, `ent:cor:single-node`, `ent:thm:product`, `ent:thm:maximal`, `ent:sm:thm:maximal-all`, `ent:sm:thm:beta`) exist with the meaning it gives them. Its statements about this report — that it identifies the quotient by a one-simple-node product with the eventually integral sequences, describes all maximal ideals of that quotient with ultraproduct residue fields, identifies the maximal spectrum with $\beta\mathbb{N}_{>0}$, proves that the full entire ring fails to be Bézout without an order unit, and uses finite shell algebras for several nodes on a shell — were checked and hold; no stale statement was found. Its review concentrated on this report, its README and the documentation catalogue; it compared the literature of Finocchiaro, Frisch and Windisch [15], the Stacks Project on valuation rings [19], Neumann, and Conway and Gonshor (the last two not reviewed page by page). It is [section 12](#), all of which is its material except where stated: its analytic section is printed once by citation ([remark 12.2](#)); its recovered maximal ideals, residue fields and maximal spectrum are [theorems 8.33](#), [8.36](#) and [8.37](#); its example “rank is not the dividing invariant” is [section 14.4](#) with [section 14.1](#); and its divisibility lemma for arbitrary multiplicities is [theorem 7.3](#) with [proposition 8.8](#). Its three questions are [questions 16.9](#) to [16.11](#), and its suggested Lean dependency order is summarized in [section 16.4](#). Where the merge had to choose: its letters were renamed ([section 2.5](#)), including e_q for its f_q ; its kernel argument was not kept as a separate route, being a bookkeeping variant of [proposition 8.8](#); and four statements are the merge’s, not the source’s, and are marked where they occur — that the ring-theoretic results hold for any nodes on the same radii ([remark 12.4](#)), that $\Phi_{\mathcal{U}}$ is $\text{red}_{\mathcal{U}}$ followed by residues (after [proposition 12.6](#)), that finitely generated ideals containing the product are (P_D, G) ([remark 12.22](#)), and that the additional primes for unbounded multiplicities are not maximal (the remark after [theorem 12.40](#)). Its finite checks are Suite F of [section B](#).

The strongest proposed original statements of the first two sources are [theorem 8.29](#) and its use in [theorem 1.5](#), together with [theorems 10.4](#), [10.7](#) and [10.9](#) and their assembly in [theorem 1.7](#). The sharp workspace criterion and the persistence result are [theorem 10.1](#) and [corollary 10.11](#); the restricted-unit criterion [theorem 4.1](#) is closely related to established topological nilpotence distinctions, and it is its precise necessity-and-sufficiency formulation for (3), including groups with no order unit, that is proposed here. The rank-one factorization formula is explicitly classical. The all-surcomplex polynomial corollary is credited to the pre-existing all-scale theme in the repository ([remark 10.13](#)). A later discovery of an antecedent would change the priority discussion, not the content or the hypotheses of the displayed proofs.

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